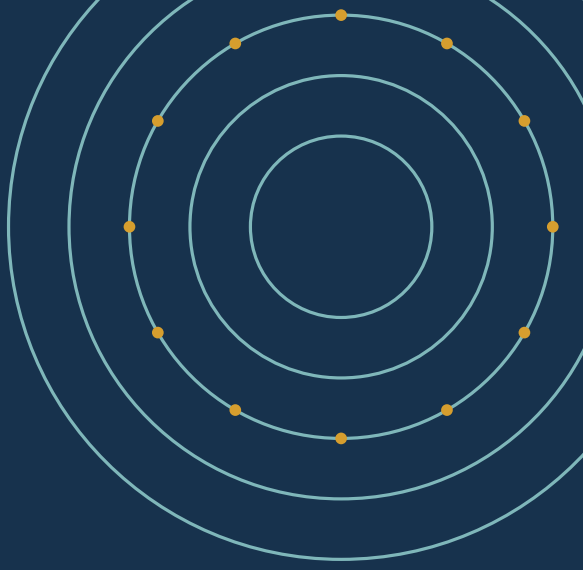
AN INTRODUCTION TO NUMBER THEORY



Patterns, Proofs, and Primes

A visual, inquiry-driven course with exercises

One-semester course packet

Euclid → continued fractions → congruences → groups → prime counting

How to use this course

Big idea

Number theory is an unusually good place to learn what proof feels like. The objects are familiar, experiments are cheap, and elementary questions quickly demand new ways of thinking. This course repeatedly follows the same rhythm:

$$\text{compute} \longrightarrow \text{notice} \longrightarrow \text{conjecture} \longrightarrow \text{prove} \longrightarrow \text{reuse}.$$

Audience and prerequisites

This packet is designed for a first proof-oriented course: strong secondary students, first- or second-year undergraduates, or mathematically mature non-majors. Students should be comfortable with algebra, functions, and mathematical induction. No calculus or abstract algebra is required. A short "proof studio" in the first week can supply any missing experience with direct proof, contradiction, and induction.

Learning goals

By the end, students should be able to:

1. use the Euclidean algorithm both computationally and as a source of proofs;
2. prove unique factorization from Euclid's lemma and use prime exponents to prove $\sqrt{2}$ irrational;
3. generate primes with the sieve of Eratosthenes and justify its stopping rule;
4. encode continued-fraction convergents by unimodular 2×2 matrix products, connect them to extended-Euclidean coefficients, and use them as alternating bounds;
5. explain rational base-b periodicity and prove the Lagrange–Galois characterization of periodic continued fractions using fixed points, conjugation, and reduced surds;
6. prove the symmetry of square-root periods and identify exactly which convergents solve the positive and negative Pell equations;
7. use extended gcd to compute modular inverses and solve linear congruences;
8. construct CRT selectors and recognize Lagrange interpolation as polynomial CRT;

9. prove Fibonacci periodicity modulo an integer using reversible pairs of residues;
10. prove Fermat and Euler, and use multiplicative functions and Möbius inversion;
11. prove basic identities and divisibility statements for binomial coefficients;
12. understand why central binomial coefficients control the density of primes;
13. write short proofs that clearly separate evidence, claim, and justification.

A 14-week route

The design assumes two 75-minute meetings per week. "Studio" means that at least half the meeting is devoted to student work at boards or in small groups.

Week	Core mathematics	Suggested studio or assessment
1	Divisibility; gcd; Euclidean algorithm	Tiling model; proof diagnostic
2	Bézout identity; linear Diophantine equations	Extended Euclid relay
3	Primes; unique factorization; $\sqrt{2} \notin \mathbb{Q}$; the sieve	Sieve studio; repair a flawed proof; Problem Set 1
4	Finite continued fractions; matrix products; extended Euclid; calendar design	Match remainders with convergent coefficients
5	Alternating convergents; Lagrange and Galois; Pell equations	Fixed points and reduced-state cycles; Problem Set 2
6	Congruences; extended gcd; inverses; linear congruences	Bézout certificates and multiplication checks
7	CRT; Lagrange interpolation; Fibonacci pairs	Reconstruct remainders and values; prove periodicity
8	Necklaces; Fermat; prime powers and Euler	Count periods; assemble with CRT; Problem Set 3
9	Multiplicative functions; Möbius inversion; groups	Recover exact counts; inspect Cayley tables
10	Lagrange's theorem; Fermat and Euler via groups	Compare the proofs; Problem Set 4
11	Binomial coefficients and Pascal's triangle	Identity stations
12	p-adic valuations; Kummer's carry idea	Color divisibility patterns; Problem Set 5
13	Central binomial coefficient; Chebyshev functions	Prime-counting proof workshop
14	Chebyshev estimate; synthesis	Capstone presentations or oral exam

Chapter 3 proves Lagrange and Galois through fixed points, conjugation, bounded states, and unique predecessors; use Exercises 20–26 for practice. Section 3.10 and Exercises 27–31 cover square-root symmetry and Pell solutions. The cattle application (Section 3.11, Exercises 32–34, Notebook 1), metric theory, and astronomical asides are optional.

In Chapter 4, make extended-gcd certificates the default method for division. Use them for linear congruences and CRT, then recognize the same selectors in Lagrange interpolation. Prove Fibonacci periodicity using reversible pairs; systematic order calculations can wait.

Chapter 5 extends necklaces to Euler, then develops multiplicative functions and Möbius inversion. Bonus sections connect posets, inclusion–exclusion, interpolation, and Fourier inversion. Fermat's sum formula becomes a polynomial identity in Chapter 7. Optional Appendix D follows Chapters 4–7:

develop rings, characteristic, bases, and prime-power size; build quadratic extensions; prove cyclicity and the existence and uniqueness of Galois fields; then study matrix counts, Frobenius, subfields, and Fibonacci periods. A final bonus proves Wedderburn's theorem on finite division rings. Its class equation and cyclotomic-polynomial lemmas are developed there.

Assessment design

One workable balance is 45% problem sets, 20% two short proof interviews, 15% computational or visual labs, and 20% capstone. Problem sets should mix three modes:

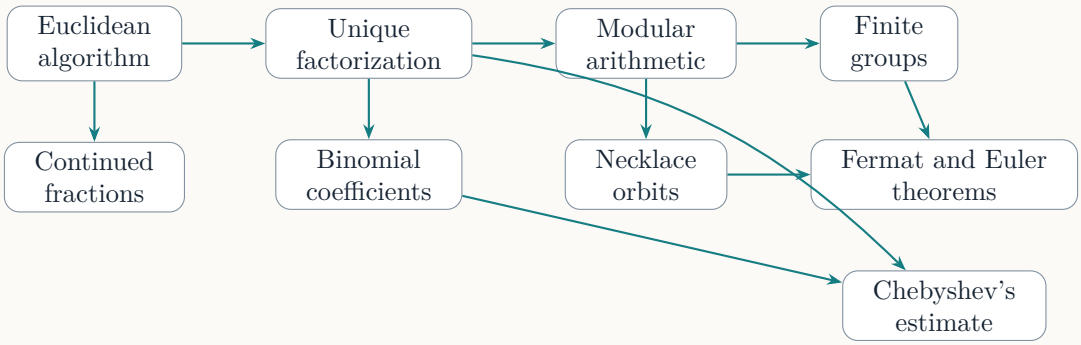
- **Fluency**: focused calculations that make later proofs readable.
- **Reasoning**: multi-step proofs, often with a hint or intermediate claim.
- **Exploration**: generate data, state a conjecture, then test its limits.

The exercises in each chapter carry these labels. A typical weekly assignment is two fluency problems, two reasoning problems, and one exploration. The appendices supply a short matrix toolkit, continued-fraction arithmetic, an astronomical Venus case study, finite fields and division rings with Fibonacci applications, selected hints and solutions, and reference sheets. An instructor can remove the solution pages when distributing assignments.

Teaching principles

- **Let the picture precede the notation.** Tiling, winding around a clock, rotating a necklace, and counting lattice paths give students something to reason about before symbols compress the idea.
- **Prove a theorem twice.** Fermat and Euler connect the course: necklace rotations and CRT give concrete proofs, while finite groups reveal the reusable structure.
- **Make proof choices visible.** Ask not only "Is this correct?" but "Why did this definition unlock that step?"
- **Use computation as a microscope, not an oracle.** The optional Sage notebook helps students find examples and conjectures; written work must still explain why a pattern continues.

Dependency map



Checkpoint

Opening prompt. What is the largest square tile that covers a 391×299 rectangle without cutting? Ask for a justification before naming the gcd.

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Chapter 1

The algorithm hidden in a rectangle

Divisibility, gcd, and Bézout's identity

Guiding question 1.1. What is the largest square tile that exactly covers a 391×299 rectangle?

If a square has integer side length d, then it fits precisely only when $d \mid 391$ and $d \mid 299$. We are looking for the *greatest common divisor*, written $\gcd(391, 299)$.

1.1 Divisibility as structure

For integers a, b, we say $a \mid b$ when $b = ak$ for some $k \in \mathbb{Z}$. Divisibility is not an operation like division; it is a yes-or-no relation.

Proposition 1.2 (Division algorithm)**.** *If $a \in \mathbb{Z}$ and $b > 0$, there are unique integers q, r such that*

$$a = bq + r, \qquad 0 \le r < b.$$

The quotient q says how many full copies of b fit inside a; the remainder r is what is left. The crucial observation is

$$\gcd(a, b) = \gcd(b, r) \quad \text{whenever } a = bq + r.$$

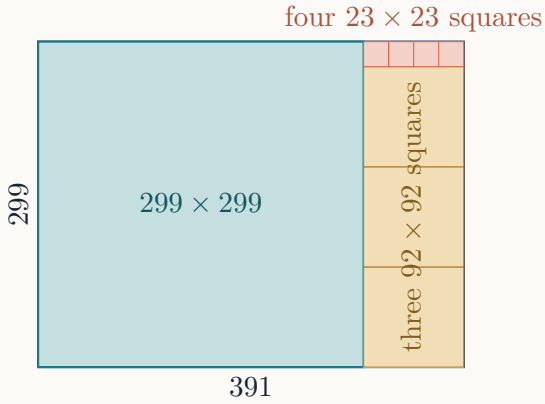
Indeed, an integer divides both a and b exactly when it divides both b and $a - bq = r$.

1.2 Euclid's algorithm

Repeat the division algorithm until the remainder is zero:

$$\begin{aligned} 391 &= 1 \cdot 299 + 92, \\ 299 &= 3 \cdot 92 + 23, \\ 92 &= 4 \cdot 23 + 0. \end{aligned}$$

The last nonzero remainder is 23, so $\gcd(391, 299) = 23$.



The same remainders appear when successively removing the largest possible squares.

Theorem 1.3 (Euclidean algorithm)**.** *For positive integers $a \geq b$, repeated division terminates, and the last nonzero remainder is* $\gcd(a, b)$.

Proof. Each remainder is a nonnegative integer smaller than the previous positive remainder, so the process must stop. Replacing (a, b) by (b, r) does not change the common divisors. At the last line $r_{k-1} = qr_k$, hence the common divisors are precisely the divisors of r_k. □

1.3 Running the movie backward

The remainders do more than find the gcd. Reverse the equations:

$$\begin{aligned} 23 &= 299 - 3 \cdot 92 \\ &= 299 - 3(391 - 299) \\ &= 4 \cdot 299 - 3 \cdot 391. \end{aligned}$$

Thus 23 is an integer linear combination of 391 and 299.

Theorem 1.4 (Bézout's identity)**.** *For integers a, b, not both zero, there are integers x, y such that*

$$ax + by = \gcd(a, b).$$

Moreover, $\gcd(a, b)$ *is the smallest positive integer of the form $ax + by$.*

Corollary 1.5. *The equation $ax + by = c$ has integer solutions exactly when* $\gcd(a, b) \mid c$.

Proof. If $ax + by = c$, every common divisor of a, b divides c. Conversely, multiply a Bézout identity for the gcd by $c/\gcd(a, b)$. □

Common trap

The gcd is positive, but Bézout coefficients need not be. Restricting x, y to positive integers makes the statement false.

Explore

Choose two 3-digit integers. Make a two-column record: Euclid's divisions on the left, backward substitutions on the right. How large can the coefficients become? The optional Sage notebook automates the arithmetic but also asks you to certify its output by substitution.

Exercises

1. Compute $\gcd(2024, 748)$ by Euclid's algorithm. [Fluency]

2. Express $\gcd(527, 341)$ in the form $527x + 341y$. [Fluency]

3. Prove: if $d \mid a$ and $d \mid b$, then $d \mid (ma + nb)$ for all $m, n \in \mathbb{Z}$. [Reasoning]

4. Find all integer solutions to $35x + 22y = 1$. Describe how one solution generates all the others. [Reasoning]

5. Show that $\gcd(a, b) = \gcd(a + kb, b)$ for every $k \in \mathbb{Z}$. Explain why this is the engine of Euclid's algorithm. [Reasoning]

6. A rectangle has integer sides $a > b$. Repeatedly remove the largest square available. Prove that the final square has side $\gcd(a, b)$. [Exploration]

7. Consecutive Fibonacci numbers make Euclid's algorithm take many steps. Test this claim, then prove $\gcd(F_{n+1}, F_n) = 1$. [Exploration]

Chapter 2

Every integer has an address

Primes and the fundamental theorem of arithmetic

Positive integers factor. But why should the factorization into primes be unique? The arithmetic step that makes uniqueness possible is modest enough to miss.

Definition 2.1. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p. An integer $n > 1$ that is not prime is *composite*.

2.1 Euclid's lemma

Lemma 2.2 (Euclid)**.** *If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Suppose $p \nmid a$. Since the only positive divisors of p are $1, p$, we have $\gcd(p, a) = 1$. Bézout gives $px + ay = 1$. Multiply by b:

$$pbx + aby = b.$$

Both terms on the left are divisible by p, so $p \mid b$. □

Common trap

The word "prime" is essential. We have $6 \mid 2 \cdot 3$, but 6 divides neither factor.

Applying the lemma repeatedly shows: if a prime divides $a_1 a_2 \cdots a_k$, it divides at least one factor.

2.2 Existence and uniqueness

Theorem 2.3 (Fundamental theorem of arithmetic)**.** *Every integer $n > 1$ can be written as a product of primes. Apart from rearranging the factors, this product is unique.*

Existence. Use strong induction on n. If n is prime, we are done. If it is composite, write $n = ab$ with $1 < a, b < n$. By induction, both a and b are products of primes, so n is too. □

Uniqueness. Suppose

$$p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$$

with all p_i, q_j prime. Since p_1 divides the right side, Euclid's lemma says $p_1 = q_j$ for some j. Cancel this common factor. Repeat. No factor can remain on only one side, so the lists agree up to order. □

2.3 An exponent-parity proof of irrationality

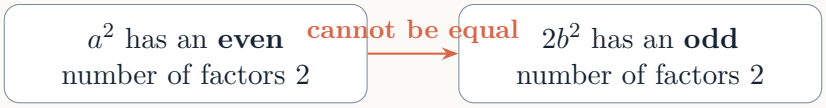
Unique factorization can rule out rational numbers without trying to compute their decimal expansions.

Theorem 2.4. *The number $\sqrt{2}$ is irrational.*

Proof. Suppose instead that $\sqrt{2} = a/b$ for integers a, b with $b \neq 0$. Squaring and clearing the denominator gives

$$a^2 = 2b^2.$$

By the fundamental theorem of arithmetic, the exponent of 2 in the prime factorization of a^2 is even: every exponent occurring in a has been doubled. The exponent of 2 in $2b^2$ is odd: it is one plus twice the exponent occurring in b. Uniqueness says that equal positive integers must have the same exponent of every prime, so this is impossible. □



The contradiction is not merely that one side is even; it is that the two sides contain incompatible numbers of factors 2.

The same proof works with any prime in place of 2. More generally, a rational number can be a perfect square only when every prime exponent in its reduced numerator and denominator is even.

Aside: existence without a witness

Here is a famous one-line use of the law of excluded middle. Set

$$x = (\sqrt{2})^{\sqrt{2}}.$$

Either x is rational or it is irrational. In the first case, the two irrational numbers $a = b = \sqrt{2}$ satisfy $a^b = x \in \mathbb{Q}$. In the second case, the two irrational numbers $a = x$ and $b = \sqrt{2}$ satisfy

$$a^b = \left((\sqrt{2})^{\sqrt{2}}\right)^{\sqrt{2}} = 2.$$

Thus, classically, there exist irrational a and b for which a^b is rational.

The proof is valid in classical mathematics, but it is not a constructive proof: it uses the disjunction "x is rational or x is not rational" without deciding which alternative holds, so it does not identify which of the two proposed pairs is a witness. Constructive and intuitionistic mathematics do not in general accept this unrestricted use of excluded middle.

Unique factorization gives an elementary explicit witness. Take

$$a = \sqrt{2}, \qquad b = \log_{\sqrt{2}} 3.$$

Then $a^b = 3$. Moreover b is irrational: if $b = p/q$ with $q > 0$, then raising $(\sqrt{2})^{p/q} = 3$ to the power $2q$ would give

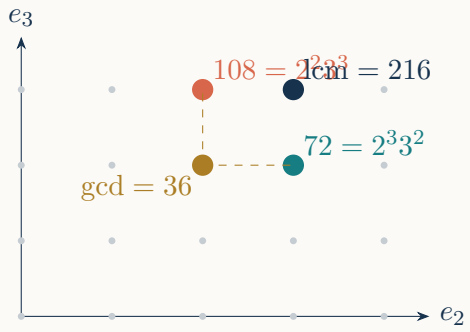
$$2^p = 3^{2q},$$

contradicting unique prime factorization. This time the proof supplies the pair that works. Much more deeply, the Gelfond–Schneider theorem implies that $(\sqrt{2})^{\sqrt{2}}$ is transcendental, so it retrospectively selects the second branch of the classical argument.

Collecting repeated primes gives a coordinate system

$$n = 2^{e_2} 3^{e_3} 5^{e_5} \cdots,$$

where only finitely many exponents are nonzero. Multiplication adds exponent vectors; gcd takes coordinatewise minima; lcm takes coordinatewise maxima.



Prime exponents turn gcd and lcm into geometry.

Proposition 2.5. *If* $n = \prod p^{e_p}$ *and* $m = \prod p^{f_p}$, *then*

$$\gcd(n, m) = \prod p^{\min(e_p, f_p)}, \qquad \operatorname{lcm}(n, m) = \prod p^{\max(e_p, f_p)}.$$

In particular, $\gcd(n, m) \operatorname{lcm}(n, m) = nm$.

2.4 The sieve of Eratosthenes

Unique factorization also explains a practical way to find every prime up to a chosen bound N. Write down the integers $2, 3, \ldots, N$. Circle the first uncrossed number, then cross out its larger multiples. Repeat. The circled numbers are precisely the primes.

1	**2**	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50

Sieve to 50: the boxed survivors are prime; every composite has been crossed out.

Proposition 2.6 (Why the sieve stops early)**.** *If $n > 1$ is composite, then it has a prime divisor at most $\sqrt{n}$. Consequently, to find all primes up to N, it is enough to cross out multiples of primes $p \le \sqrt{N}$.*

Proof. Write $n = ab$ with $1 < a \le b$. Then $a \le \sqrt{n}$. A prime factor p of a therefore satisfies $p \le a \le \sqrt{n}$. Thus if an integer $n \le N$ survives all sieving primes through $\sqrt{N}$, it cannot be composite. □

Checkpoint

For the display above, only the multiples of $2, 3, 5$, and 7 need to be crossed out. The next prime is $11 > \sqrt{50}$. Notice also that one may begin crossing out at p^2: every smaller multiple of p has already been handled by an earlier prime.

2.5 There is no last prime

Theorem 2.7 (Euclid)**.** *There are infinitely many primes.*

Proof. If $p_1, \dots, p_k$ were all primes, consider $N = p_1 \cdots p_k + 1$. Some prime divides N, but none of the listed primes does: division by any p_i leaves remainder 1. Contradiction. □

This proof does not say that N is prime. It says that N has a prime factor missing from the proposed complete list.

Exercises

1. Factor 7560 and use the exponent vectors to find the number of its positive divisors. [Fluency]
2. Compute $\gcd(2^7 3^2 11, 2^4 3^5 5)$ and the lcm without multiplying first. [Fluency]
3. Prove that if $\gcd(a, b) = 1$ and $a \mid c$, $b \mid c$, then $ab \mid c$. [Reasoning]
4. Reproduce the exponent-parity proof that $\sqrt{2}$ is irrational, then prove that $\sqrt{p}$ is irrational for every prime p. State exactly where unique factorization is used. [Reasoning]
5. Prove that every composite $n > 1$ has a prime divisor at most $\sqrt{n}$. Turn this into a primality test. [Reasoning]
6. Use the sieve of Eratosthenes to list every prime at most 100. Explain why, after sieving by $2, 3, 5$, and 7, no further crossing-out step is needed. [Fluency]
7. Is $n! + 1$ always prime? Find the first counterexample, then explain why Euclid's proof is unaffected. [Exploration]
8. Prove that infinitely many primes are congruent to 3 modulo 4 by adapting Euclid's idea to $4p_1 \cdots p_k - 1$. You may use the fact that a product of numbers congruent to 1 modulo 4 is again 1 modulo 4. [Challenge]
9. Verify directly that $a = \sqrt{2}$ and $b = \log_{\sqrt{2}} 3$ are irrational and satisfy $a^b = 3$. What does this explicit witness add to the excluded-middle proof? [Reasoning]

Chapter 3

Fractions inside fractions

Continued fractions and best approximations

The quotients from Euclid's algorithm are not bookkeeping debris. Read them forward and they produce a new numerical language.

3.1 Euclid, folded up

From

$$415 = 4 \cdot 93 + 43, \quad 93 = 2 \cdot 43 + 7, \quad 43 = 6 \cdot 7 + 1, \quad 7 = 7 \cdot 1,$$

we obtain

$$\frac{415}{93} = 4 + \cfrac{1}{2 + \cfrac{1}{6 + \cfrac{1}{7}}} = [4; 2, 6, 7].$$

Definition 3.1. A finite simple continued fraction is

$$[a_0; a_1, \ldots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}},$$

where $a_0 \in \mathbb{Z}$ and $a_i \in \mathbb{N}$ for $i \geq 1$.

The truncations $[a_0; \ldots, a_k] = p_k/q_k$ are called *convergents*. They can be computed without rebuilding the nested fraction:

$$p_k = a_k p_{k-1} + p_{k-2}, \qquad q_k = a_k q_{k-1} + q_{k-2},$$

with $p_{-2} = 0, p_{-1} = 1, q_{-2} = 1, q_{-1} = 0$.

3.2 Linear fractional transformations

A *linear fractional transformation*, also called a Möbius transformation, is a function of the form

$$T_M(t) = \frac{at+b}{ct+d}, \qquad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad ad - bc \neq 0.$$

We will use integer or rational coefficients. The formula is defined when $ct + d \neq 0$. The determinant condition excludes a constant function and ensures that the transformation can be undone.

Why does a 2×2 matrix act on a single number? Represent t by the column $(t, 1)^T$, multiply, and take the ratio of the two coordinates:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} t \\ 1 \end{pmatrix} = \begin{pmatrix} at+b \\ ct+d \end{pmatrix} \quad \longmapsto \quad \frac{at+b}{ct+d}.$$

Thus multiplying M by any nonzero scalar does not change T_M. In this chapter, $T_M(t)$ means this ratio, not ordinary multiplication of a matrix by a scalar. One may also include a point ∞: when $c \neq 0$, $T_M(\infty) = a/c$ and $T_M(-d/c) = \infty$; when $c = 0$, ∞ stays fixed.

Proposition 3.2 (Composition is matrix multiplication)**.** *For invertible* 2×2 *matrices* M, N*,*

$$T_M \circ T_N = T_{MN}, \qquad (T_M)^{-1} = T_{M^{-1}}.$$

Proof. If $N = \left(\begin{smallmatrix} e & f \\ g & h \end{smallmatrix}\right)$, substitution gives

$$T_M(T_N(t)) = \frac{a(et+f) + b(gt+h)}{c(et+f) + d(gt+h)} = \frac{(ae+bg)t + (af+bh)}{(ce+dg)t + (cf+dh)}.$$

These four coefficients are the entries of MN. Taking $N = M^{-1}$ proves the inverse rule. Explicitly, solving $y = (at+b)/(ct+d)$ gives $t = (dy-b)/(a-cy)$. □

As with functions, the rightmost factor acts first. In particular,

$$D(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix} \quad \text{acts by} \quad T_{D(a)}(t) = a + \frac{1}{t}.$$

Therefore

$$[a_0; a_1, \ldots, a_k, t] = T_{D(a_0)D(a_1)\cdots D(a_k)}(t).$$

Here the final entry t is an entire remaining tail, not necessarily an integer. The nesting of a continued fraction has become a product of matrices.

3.3 The matrix engine

Package one continued-fraction digit into the matrix

$$D(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}.$$

Repeated matrix multiplication stores every convergent at once:

$$D(a_0)D(a_1)\cdots D(a_k) = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix}. \tag{3.1}$$

Indeed, multiplying the right-hand side by $D(a_{k+1})$ produces exactly the two convergent recurrences. The first column advances to the next convergent, while the old first column becomes the second.

Each digit matrix has determinant -1, not $+1$. Thus it lies in $\mathrm{GL}_2(\mathbb{Z})$, and a product with an even number of factors lies in $\mathrm{SL}_2(\mathbb{Z})$. More important than the sign is *unimodularity*: every product has determinant ± 1 and therefore has an inverse with integer entries. Appendix A develops the small amount of matrix algebra behind this observation. Appendix B then uses the same Möbius machinery to perform exact arithmetic directly on continued-fraction streams.

For example,

$$D(4)D(2)D(6)D(7) = \begin{pmatrix} 415 & 58 \\ 93 & 13 \end{pmatrix}, \qquad \begin{pmatrix} 415 & 58 \\ 93 & 13 \end{pmatrix}^{-1} = \begin{pmatrix} 13 & -58 \\ -93 & 415 \end{pmatrix}.$$

The entries of the inverse already display the certificate $1 = 13 \cdot 415 - 58 \cdot 93$. This is the first payoff of invertibility; the next section explains why it is exactly the extended Euclidean algorithm.

3.4 Extended Euclid in the convergents

The convergents do not merely reuse Euclid's quotients: their numerators and denominators are exactly the coefficients produced by the extended Euclidean algorithm.

Theorem 3.3 (Remainders from convergents)**.** *Let $A > B > 0$, and write the Euclidean divisions as*

$$r_{-2} = A, \qquad r_{-1} = B, \qquad r_{k-2} = a_k r_{k-1} + r_k \quad (k = 0, 1, \ldots, n),$$

where $r_{n-1} = \gcd(A, B)$ and $r_n = 0$. If

$$\frac{A}{B} = [a_0; a_1, \ldots, a_n] \quad \textit{and} \quad \frac{p_k}{q_k} = [a_0; a_1, \ldots, a_k],$$

then

$$\boxed{r_k = (-1)^k (q_k A - p_k B)} \qquad (0 \le k \le n).$$

In particular, the penultimate convergent gives a Bézout identity for the gcd.

Proof. For $k = 0$, the formula reads $r_0 = A - a_0 B$, the first Euclidean division. For $k = 1$,

$$r_1 = B - a_1 r_0 = -a_1 A + (a_0 a_1 + 1)B = -q_1 A + p_1 B.$$

If the formula holds for the preceding two remainders, then

$$\begin{aligned} r_k &= r_{k-2} - a_k r_{k-1} \\ &= (-1)^k \big((a_k q_{k-1} + q_{k-2})A - (a_k p_{k-1} + p_{k-2})B\big) \\ &= (-1)^k (q_k A - p_k B), \end{aligned}$$

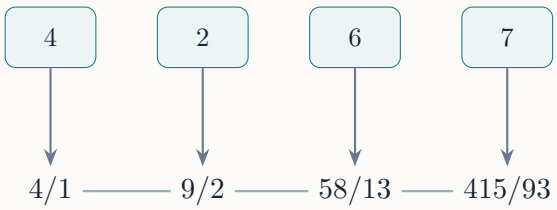
using the convergent recurrences. This proves the formula by induction. □

For $A/B = 415/93$, the two algorithms can be read in the same table:

k	a_k	convergent p_k/q_k	signed linear combination r_k
0	4	$4/1$	$415 - 4(93) = 43$
1	2	$9/2$	$-(2(415) - 9(93)) = 7$
2	6	$58/13$	$13(415) - 58(93) = 1$
3	7	$415/93$	$-(93(415) - 415(93)) = 0$

Thus the convergent just before $415/93$ contains the certificate

$$1 = 13 \cdot 415 - 58 \cdot 93.$$



Each convergent remembers all earlier partial quotients.

Theorem 3.4 (Determinant identity)**.** *Consecutive convergents satisfy*

$$p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}.$$

Consequently, each p_k/q_k is in lowest terms and

$$\left| \frac{p_k}{q_k} - \frac{p_{k-1}}{q_{k-1}} \right| = \frac{1}{q_k q_{k-1}}.$$

Proof. Take determinants in (3.1). The determinant on the left is $(-1)^{k+1}$, while the determinant on the right is $p_k q_{k-1} - p_{k-1} q_k$. Since $(-1)^{k+1} = (-1)^{k-1}$, the identity follows. □

3.5 Infinite continued fractions

For an irrational real number x, repeatedly take integer parts:

$$a_0 = \lfloor x \rfloor, \qquad x_1 = \frac{1}{x - a_0}, \qquad a_1 = \lfloor x_1 \rfloor, \quad \dots$$

This produces $x = [a_0; a_1, a_2, \dots]$.

Example 3.5 (The silver staircase)**.** Let $x = \sqrt{2}$. Since

$$\frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1 = 2 + (\sqrt{2} - 1),$$

the same remainder repeats, so

$$\sqrt{2} = [1; \overline{2}] = [1; 2, 2, 2, \dots].$$

Its first convergents are

$$1, \quad \frac{3}{2}, \quad \frac{7}{5}, \quad \frac{17}{12}, \quad \frac{41}{29}, \dots$$

They alternately fall below and above $\sqrt{2}$.

Theorem 3.6 (The alternating squeeze)**.** *Let $c_k = p_k/q_k$ be the zero-indexed convergents of an irrational number $x = [a_0; a_1, a_2, \dots]$. Then*

$$c_0 < c_2 < c_4 < \cdots < x < \cdots < c_5 < c_3 < c_1.$$

Thus, when the convergents are counted starting with the first, the odd-numbered ones increase and stay below x, while the even-numbered ones decrease and stay above x.

The Möbius derivative trick. Fix k and leave the tail variable. The continued fraction with prefix through a_k is the fractional linear, or Möbius, map

$$M_k(t) = [a_0; a_1, \dots, a_k, t] = \frac{p_k t + p_{k-1}}{q_k t + q_{k-1}}.$$

The determinant identity gives

$$M_k'(t) = \frac{p_k q_{k-1} - p_{k-1} q_k}{(q_k t + q_{k-1})^2} = \frac{(-1)^{k+1}}{(q_k t + q_{k-1})^2}.$$

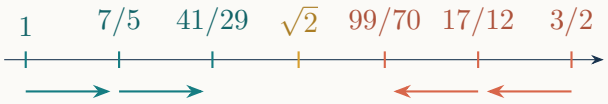
The complete tail $\alpha_{k+1} = [a_{k+1}; a_{k+2}, \dots]$ is finite and positive, and

$$x = M_k(\alpha_{k+1}), \qquad c_k = M_k(\infty).$$

When k is even, M_k is decreasing, so $c_k < x$; when k is odd, M_k is increasing, so $x < c_k$. The same comparison with $c_{k+2} = M_k([a_{k+1}; a_{k+2}])$ shows that $c_k < c_{k+2}$ for even k and $c_{k+2} < c_k$ for odd k. These are exactly the two monotone chains displayed above. □

Big idea

Every new convergent switches sides, but every second convergent moves inward. Continued fractions therefore trap x between a rising sequence of lower bounds and a falling sequence of upper bounds.



Schematic, not to scale: both parity subsequences move inward toward $\sqrt{2}$.

Theorem 3.7 (Approximation by convergents)**.** *If p_k/q_k is a convergent of an irrational x, then*

$$\left| x - \frac{p_k}{q_k} \right| < \frac{1}{q_k q_{k+1}} < \frac{1}{q_k^2}.$$

Moreover, if $0 < q < q_{k+1}$, then no fraction p/q is closer to x than both neighboring convergents in the appropriate interval. In this precise sense, convergents are best approximations with small denominators.

The determinant identity is the engine: distinct reduced fractions $a/b, c/d$ are at least $1/(bd)$ apart. There is simply not enough room for another small-denominator fraction between a convergent and x.

Theorem 3.8 (Legendre's approximation criterion)**.** *Let x be irrational and let p/q be in lowest terms with $q > 0$. If*

$$\left| x - \frac{p}{q} \right| < \frac{1}{2q^2},$$

then p/q is a convergent of x.

The theorem is a partial converse to the approximation theorem: an *exceptionally* good rational approximation cannot have arisen by accident. A standard proof chooses the two consecutive convergents whose denominators surround q and writes the integer vector (p, q) in their unimodular basis. If (p, q) is not one of the basis vectors, the determinant ± 1 forces it to be at least $1/(2q^2)$ away from x. This is another place where the integral inverse of the convergent matrix does real work.

Aside: continued fractions keep the calendar

A calendar must use whole days. Write the tropical year as $365+\alpha$ days, where $\alpha \approx 0.2421988$. A q-year cycle with p leap days therefore asks for $p/q \approx \alpha$. Now

$$\alpha = [0; 4, 7, 1, 3, 5, \dots], \qquad \text{with convergents} \qquad \frac{1}{4}, \frac{7}{29}, \frac{8}{33}, \frac{31}{128}, \frac{163}{673}, \dots$$

The first convergent is the Julian rule: one leap day every four years. The convergent $31/128 = 0.2421875$ is remarkably accurate, but a 128-year civil cycle is awkward.
The Gregorian rule adds a practical constraint: organize corrections by centuries. Writing $q = 100Q$ changes the approximation problem:

$$q = 100Q: \qquad \frac{p}{Q} \approx 100\alpha \approx 24.21988, \qquad [24; 4] = \frac{97}{4}.$$

Thus the first nonintegral convergent of 100α gives 97 leap days in 400 years and the mean length

$$365 + \frac{97}{400} = 365.2425.$$

Operationally, the rule counts 100 multiples of 4, removes the four century years, and restores the one multiple of 400: $100 - 4 + 1 = 97$.
Notice that $97/400$ is not a convergent of α itself; it emerges after the century constraint. This is a mathematical reconstruction, not a claim about the reformers' historical method. It shows that "best" depends on which denominators an application permits.

Aside: continued fractions are roughly "base 11"

Khinchin's theorem says that for almost every real number,

$$(a_1 a_2 \cdots a_n)^{1/n} \longrightarrow K_0 \approx 2.685452,$$

the universal *Khinchin constant*. For efficiency, Lévy's companion theorem is more direct:

$$q_n^{1/n} \longrightarrow L = e^{\pi^2/(12 \log 2)} \approx 3.275823.$$

Since convergent error is on the scale q_n^{-2}, it follows that

$$|x - p_n/q_n|^{-1/n} \longrightarrow L^2 = e^{\pi^2/(6\log 2)} \approx 10.731.$$

That is the "base 11" joke: one generic continued-fraction term buys about $\log_{10}(10.731) \approx 1.0306$ decimal digits. Lochs' theorem makes the comparison exact: N known decimal digits determine about $(6\log 2 \log 10/\pi^2)N \approx 0.97027N$ continued-fraction terms.
"Almost every" matters: a partial quotient is not literally a fixed-base digit, and quadratic surds such as $\sqrt{2}$ have exceptional periodic growth.

Experiment: π looks typical; e does not

Write

$$G_n(x) = (a_1 a_2 \cdots a_n)^{1/n}, \qquad B_n(x) = q_n^{2/n}.$$

Khinchin and Lévy predict, for almost every x, that $G_n(x) \to 2.685452\ldots$ and $B_n(x) \to 10.731016\ldots$. For

$$\pi = [3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, \ldots],$$

a computation gives the following rounded values:

n	$G_n(\pi)$	$B_n(\pi)$	$\log_{10} B_n(\pi)$
100	2.68315	10.68768	1.02888
1,000	2.66270	10.54434	1.02302
10,000	2.66320	10.57847	1.02442
100,000	2.67956	10.67835	1.02850
almost-everywhere target	2.68545	10.73102	1.03064

The wobble is real: rare large partial quotients, such as the early 292, produce visible jumps. A published computation using 17,001,303 terms found $G_n(\pi) = 2.686393$. This is compelling evidence, but no theorem is known that π satisfies either limiting law.
The number e is a useful contrast, not a second generic-looking example:

$$e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \ldots].$$

For $n = 3m$, its first n noninitial partial quotients have product $2^m m!$. Stirling's formula therefore gives

$$G_{3m}(e) = (2^m m!)^{1/(3m)} \sim (2m/e)^{1/3} \longrightarrow \infty.$$

Also $q_{3m} \geq a_1 \cdots a_{3m} = 2^m m!$, so $B_{3m}(e) \to \infty$ as well. Thus e is provably exceptional for both laws, while π is numerically persuasive but theoretically mysterious.

Common trap

Legendre, Lagrange, and Galois all appear in this subject. Legendre's theorem recognizes very good rational approximations. Lagrange's theorem characterizes eventually periodic continued fractions of quadratic irrationals; Galois's refinement says exactly when the period begins immediately.

3.6 A parallel periodicity: digits in base b

Continued fractions are not the first place students meet repeating expansions. Let $0 < m < q$ and write $x = m/q$ in base $b \geq 2$. Long division produces digits $d_1, d_2, \ldots$ and remainders $r_0, r_1, \ldots$ by

$$r_0 = m, \qquad br_n = qd_{n+1} + r_{n+1}, \qquad 0 \leq r_{n+1} < q.$$

There are only q possible remainders. If a remainder becomes 0, the expansion terminates; otherwise some remainder repeats, and from that point the deterministic long-division process repeats. Thus every rational number has an eventually periodic base-b expansion.

The converse uses only a geometric series. If a block of ℓ digits with integer value N repeats immediately, then

$$0.\overline{d_1 \cdots d_{\ell b}} = \frac{N}{b^\ell} + \frac{N}{b^{2\ell}} + \cdots = \frac{N}{b^\ell - 1}.$$

An initial nonrepeating block only adds a rational number with denominator a power of b.

Theorem 3.9 (When the repeating block begins immediately)**.** *If the reduced fraction m/q has a purely periodic base-b expansion, then* $\gcd(q, b) = 1$*. Conversely, if* $\gcd(q, b) = 1$*, its expansion is purely periodic.*

The direction available now. A purely periodic expansion has the form $N/(b^\ell - 1)$. Its reduced denominator divides $b^\ell - 1$, which is relatively prime to b. □

The reverse implication is deliberately postponed until Chapter 6. Euler's theorem will give

$$b^{\varphi(q)} \equiv 1 \pmod{q},$$

so $q \mid b^{\varphi(q)} - 1$; rewriting m/q with that denominator creates a repeating block (possibly with leading zeros). In decimal, for example,

$$\frac{1}{7} = 0.\overline{142857}, \qquad \frac{1}{8} = 0.125000\ldots.$$

The first expansion is purely periodic because $\gcd(7, 10) = 1$; the second is only eventually periodic.

3.7 Periodic tails are fixed points

The matrix language answers a question that the nested notation conceals: what kind of number can a repeating continued fraction represent? Throughout this discussion the continued fractions are *simple*: $a_0 \in \mathbb{Z}$ and $a_n \geq 1$ for $n \geq 1$.

Definition 3.10. An infinite continued fraction is *eventually periodic* if its digits repeat after a finite prefix. It is *purely periodic* if repetition starts at the first digit:

$$[\overline{b_0, \ldots, b_{\ell-1}}] = [b_0; b_1, \ldots, b_{\ell-1}, b_0, b_1, \ldots].$$

In a purely periodic simple continued fraction, every b_i is positive, including b_0. Rational numbers have finite simple continued fractions; we do not call those eventually periodic here.

Definition 3.11. A *quadratic irrational* (or quadratic surd) is an irrational real root of a quadratic polynomial with integer coefficients. Its other root is its *conjugate*, written x'. Equivalently,

$$x = u + v\sqrt{D}, \qquad x' = u - v\sqrt{D} \quad (u, v \in \mathbb{Q},\ v \neq 0),$$

where D is a positive nonsquare integer. Replacing $\sqrt{D}$ by $-\sqrt{D}$ respects addition, multiplication, and division by nonzero numbers. In particular, $(T_M(x))' = T_M(x')$ for rational M.

Lemma 3.12 (A nontrivial fixed point is quadratic)**.** *If an invertible rational matrix $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is not a scalar multiple of the identity and fixes an irrational number t under its linear fractional action, then t is a quadratic irrational.*

Proof. Multiplying $t = (at + b)/(ct + d)$ by its denominator gives

$$ct^2 + (d - a)t - b = 0. \tag{3.2}$$

The coefficients are not all zero: that would say $c = b = 0$ and $d = a$, making M scalar. Moreover, c cannot be zero, since a nonzero linear polynomial with rational coefficients cannot vanish at an irrational number. Thus this is a genuine quadratic equation. Clear denominators to obtain integer coefficients. □

Common trap

The qualification *nontrivial* matters. The identity transformation fixes every real number, so $t = T_I(t)$ by itself says nothing about t. For a period matrix there is no such problem: its lower-left entry is strictly positive.

Theorem 3.13 (Lagrange)**.** *A real number has an eventually periodic simple continued fraction if and only if it is a quadratic irrational.*

Proof that periodicity implies a quadratic equation. First let $x = [\overline{b_0, \ldots, b_{\ell-1}}]$. Removing one complete period leaves the same infinite tail, so

$$x = T_N(x), \qquad N = D(b_0) \cdots D(b_{\ell-1}) = \begin{pmatrix} A & B \\ C & D_0 \end{pmatrix}.$$

Because every $b_i \geq 1$, we have $B, C > 0$. Also $\det N = (-1)^\ell$. The fixed-point equation is

$$Cx^2 + (D_0 - A)x - B = 0.$$

The alternating-convergent argument also applies to an infinite positive digit sequence: it gives a limit and an error at most $1/(q_k q_{k+1})$. That limit cannot be rational. Indeed, the convergents are in lowest terms and $q_k \to \infty$, so they are eventually distinct from any fixed rational p/q. Their distance from p/q is then at least $1/(qq_k)$, contradicting the error bound once $q_{k+1} > q$. Thus x is quadratic.

Now allow a prefix:

$$y = [a_0; \ldots, a_{r-1}, \overline{b_0, \ldots, b_{\ell-1}}] = T_M(x), \qquad M = D(a_0) \cdots D(a_{r-1}).$$

If the prefix is empty, take $M = I$. Since T_M is invertible, substituting $x = T_{M^{-1}}(y)$ into $x = T_N(x)$ gives

$$\boxed{y = T_{MNM^{-1}}(y).} \tag{3.3}$$

Both M and N have determinant ± 1, so $K = MNM^{-1}$ is an integer matrix. It is not scalar: if $K = \lambda I$, then $N = M^{-1}KM = \lambda I$, contrary to $C > 0$.

Finally, write $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The number y is irrational. Otherwise the inverse formula $x = T_{M^{-1}}(y)$ would make x rational. Its denominator cannot vanish at $y = T_M(x)$: direct substitution gives $a - cy = (ad - bc)/(cx + d) \neq 0$. The fixed-point lemma applied to K therefore gives a genuine quadratic equation for y as well. □

Big idea

A repeating tail is a fixed point. Attaching a finite prefix *conjugates the transformation that fixes it.* This is the missing algebraic step between a purely periodic tail and an eventually periodic number; it requires only matrix multiplication and inversion.

Example 3.14 (A prefix changes the equation, not its degree)**.** Take

$$x = [\overline{1, 2}], \qquad N = D(1)D(2) = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}.$$

Then $2x^2 - 2x - 1 = 0$, so $x = (1 + \sqrt{3})/2$. Prefix the digit 1:

$$y = [1; \overline{1, 2}] = 1 + \frac{1}{x}, \qquad M = D(1) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The conjugated matrix is

$$MNM^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}.$$

Thus $y = (2y + 3)/(y + 2)$, giving $y^2 = 3$. Since $y > 0$, $y = \sqrt{3}$. The same matrix will reappear as a generator of Pell solutions.

3.8 Why every quadratic irrational repeats

We now prove the other direction of Lagrange's theorem. There are two tasks: find a bounded set of integer states, and show that the continued fraction eventually enters it. The conjugate supplies exactly the inequalities we need.

Definition 3.15. A quadratic irrational x is *reduced* if

$$x > 1 \qquad \text{and} \qquad -1 < x' < 0.$$

Step 1: every quadratic irrational has a reduced tail

Write $x_0 = x$, $a_n = \lfloor x_n \rfloor$, and $x_{n+1} = 1/(x_n - a_n)$. For $n \geq 1$, $x_n > 1$ and $a_n \geq 1$. Conjugating the recurrence gives

$$x'_{n+1} = \frac{1}{x'_n - a_n}.$$

Consequently, once $x'_n < 0$ with $n \geq 1$, the next conjugate belongs to $(-1, 0)$. Every later complete quotient is reduced too.

Why must a negative conjugate appear? Invert the matrix for the first n digits and then conjugate. For $n \geq 2$ this gives

$$x'_n = -\frac{q_{n-2}x' - p_{n-2}}{q_{n-1}x' - p_{n-1}}.$$

The convergents p_k/q_k tend to x, and $x' \neq x$. Hence, for all sufficiently large k, the numbers $x' - p_k/q_k$ have the same sign as $x' - x$. The numerator and denominator in the displayed ratio therefore have the same sign for large n, since the q_k are positive. It follows that $x'_n < 0$ eventually, as required. This argument also applies when the original x is negative or less than 1.

Step 2: integer states evolve by an exact recurrence

Choose an equation $Ax^2 + Bx + C = 0$ with integer coefficients and $A \neq 0$. After changing all three signs if necessary, the quadratic formula reads

$$x = \frac{-B + \sqrt{D}}{2A}, \qquad D = B^2 - 4AC > 0.$$

Here D is a nonsquare integer; it need not be squarefree. Put $P_0 = -B$ and $Q_0 = 2A$. Then $Q_0 \neq 0$ and $Q_0 \mid D - P_0^2$, because $D - P_0^2 = -4AC$. The sign change above is important: Q_0 may initially be negative.

Suppose $x_n = (P_n + \sqrt{D})/Q_n$ with integers P_n, Q_n satisfying $Q_n \mid D - P_n^2$. Rationalizing $1/(x_n - a_n)$ yields

$$\boxed{P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = \frac{D - P_{n+1}^2}{Q_n}.} \tag{3.4}$$

Indeed,

$$\frac{1}{x_n - a_n} = \frac{Q_n}{\sqrt{D} - P_{n+1}} = \frac{P_{n+1} + \sqrt{D}}{(D - P_{n+1}^2)/Q_n}.$$

Since $P_{n+1} \equiv -P_n \pmod{Q_n}$, the new Q_{n+1} is an integer. It is nonzero because D is not a square. The equality $D - P_{n+1}^2 = Q_n Q_{n+1}$ supplies the divisibility condition for the next step. Thus the representation persists forever with the same D.

Step 3: reduced states lie in a finite box

For a reduced complete quotient,

$$x_n = \frac{P_n + \sqrt{D}}{Q_n} > 1, \qquad -1 < \frac{P_n - \sqrt{D}}{Q_n} < 0.$$

Their difference is $2\sqrt{D}/Q_n > 1$, so $0 < Q_n < 2\sqrt{D}$. The conjugate inequalities then give $\sqrt{D} - Q_n < P_n < \sqrt{D}$. In particular,

$$\boxed{|P_n| < \sqrt{D}, \qquad 0 < Q_n < 2\sqrt{D}.} \tag{3.5}$$

Only finitely many integer pairs satisfy these bounds. By Step 1 all sufficiently late states lie in this box, and by Step 2 each state determines its next state, including the digit $a_n = \left\lfloor (P_n + \sqrt{D})/Q_n \right\rfloor$. Two states must repeat. Every later state, and hence every later digit, then repeats with the same period. This completes the proof of Theorem 3.13.

Example 3.16 (The states for $\sqrt{3}$)**.** Starting with $(P_0, Q_0) = (0, 1)$ and $D = 3$ gives

n	P_n	Q_n	x_n	a_n
0	0	1	$\sqrt{3}$	1
1	1	2	$(1+\sqrt{3})/2$	1
2	1	1	$1+\sqrt{3}$	2
3	1	2	$(1+\sqrt{3})/2$	1

The initial state is not reduced. The next two are reduced and form a cycle, giving $\sqrt{3} = [1; \overline{1, 2}]$.

3.9 Galois: when repetition starts immediately

The fixed-point equation tells us that periodic continued fractions are quadratic. It does *not* tell us which quadratic irrationals have no initial nonrepeating segment. That distinction is Galois's theorem.

Theorem 3.17 (Galois's pure-periodicity criterion)**.** *The simple continued fraction of a quadratic irrational x is purely periodic if and only if x is reduced:*

$$x > 1 \qquad \textit{and} \qquad -1 < x' < 0.$$

Proof. Suppose first that x is purely periodic with period length ℓ. The fixed-point argument shows that it is quadratic, and Step 1 above shows that all sufficiently late complete quotients are reduced. But $x_{k\ell} = x_0 = x$ for every k. Taking k large proves that x itself is reduced.

Conversely, suppose x is reduced. All its complete quotients remain reduced, and the finite-state proof gives $x_i = x_j$ for some $i < j$. To conclude that the period starts at 0, we need one more fact: *a reduced complete quotient has at most one reduced predecessor.*

Indeed, suppose the continued-fraction step sends a reduced u to z, and let $a = \lfloor u \rfloor$. Then $z = 1/(u - a)$, so

$$-\frac{1}{z'} = a - u'.$$

Since $-1 < u' < 0$, we have $a < -1/z' < a + 1$. Therefore

$$\boxed{a = \left\lfloor -1/z' \right\rfloor, \qquad u = a + \frac{1}{z}.} \tag{3.6}$$

Both the previous digit and the previous reduced number are determined by z. If $i > 0$, equality $x_i = x_j$ therefore forces $x_{i-1} = x_{j-1}$. Repeating the argument gives $x_0 = x_{j-i}$. The deterministic continued-fraction algorithm now repeats from its first step, proving pure periodicity. □

Big idea

Finitely many states give *eventual* periodicity. Recovering the previous reduced state removes the possible initial segment and gives *pure* periodicity. These are different steps. For instance, the finite map $0 \mapsto 1$, $1 \mapsto 1$ has an initial segment when started at 0; finiteness alone does not exclude one.

Example 3.18 (Two inequalities, both necessary)**.** The number $\sqrt{2} = [1; \overline{2}]$ is not reduced, because $-\sqrt{2} < -1$. It is eventually but not purely periodic. In contrast,

$$1 + \sqrt{2} = [\overline{2}], \qquad \frac{1+\sqrt{5}}{2} = [\overline{1}]$$

are reduced and purely periodic. The number $\sqrt{2}/2$ has conjugate in $(-1, 0)$ but is less than 1, so it is not purely periodic either.

A displayed prefix does not necessarily represent a genuine preperiod:

$$[2; \overline{1, 2}] = 1 + \sqrt{3} = [\overline{2, 1}].$$

Here the conjugate $1 - \sqrt{3}$ lies in $(-1, 0)$, so Galois correctly detects pure periodicity despite the first way of writing the digits.

Corollary 3.19 (The conjugate reverses the period)**.** *If* $x = [\overline{b_0, \ldots, b_{\ell-1}}]$*, then*

$$-\frac{1}{x'} = [\overline{b_{\ell-1}, \ldots, b_0}].$$

Proof. Each $D(b_i)$ is symmetric, so reversing the digits transposes their product $N = \left(\begin{smallmatrix} A & B \\ C & D_0 \end{smallmatrix}\right)$. Its fixed-point equation is $Cx^2 + (D_0 - A)x - B = 0$. Substituting $x' = -1/z$ gives

$$Bz^2 + (D_0 - A)z - C = 0,$$

which is the fixed-point equation for N^T. Since $B, C > 0$, this quadratic has exactly one positive root. Both $-1/x'$ and the reversed periodic continued fraction are that root. □

Three names, three different conclusions

Legendre: an exceptionally close rational approximation must be a convergent. **Lagrange**: the numbers with eventually periodic simple continued fractions are exactly the quadratic irrationals. **Galois**: among them, pure periodicity means exactly $x > 1$ and $-1 < x' < 0$. The first direction of Lagrange is the fixed-point calculation; Galois's additional content is the precise conjugate criterion.

3.10 Periodicity meets Pell's equation

Fix a nonsquare integer $D \geq 2$. *Pell's equation* asks for positive integers p, q such that

$$p^2 - Dq^2 = 1.$$

We will also study the equation with right-hand side -1. There are two questions to settle: why must a solution appear among the convergents of $\sqrt{D}$, and which convergents actually work? An approximation estimate answers the first; an exact identity answers the second.

Every Pell solution is a convergent

Suppose $p^2 - Dq^2 = \pm 1$, with $p, q > 0$. Any common divisor of p, q would divide 1, so p/q is reduced. Factoring a difference of squares gives

$$\left|\sqrt{D} - \frac{p}{q}\right| = \frac{1}{q^2(\sqrt{D} + p/q)} < \frac{1}{2q^2}. \tag{3.7}$$

For the strict inequality, note that $(p/q)^2 \geq D - 1 \geq 1$, while $\sqrt{D} > 1$. Legendre's criterion now makes p/q a convergent of $\sqrt{D}$. This does not yet prove that a solution exists: we still need a convergent whose difference of squares is exactly 1 or -1.

The palindrome in a square-root period

Write $a_0 = \lfloor \sqrt{D} \rfloor$. The first complete quotient is

$$x_1 = \frac{1}{\sqrt{D} - a_0} > 1, \qquad x_1' = -\frac{1}{\sqrt{D} + a_0} \in (-1, 0).$$

Galois's theorem therefore makes x_1 purely periodic. Let L be its *least* period length, so that $\sqrt{D} = [a_0; \overline{a_1, \ldots, a_L}]$.

Theorem 3.20 (The square-root palindrome)**.** *The last digit of the least period is $a_L = 2a_0$, and*

$$a_j = a_{L-j} \qquad (1 \leq j \leq L - 1).$$

Thus

$$\boxed{\sqrt{D} = [a_0; \overline{a_1, \ldots, a_{L-1}, 2a_0}], \quad (a_1, \ldots, a_{L-1}) \text{ is a palindrome.}}$$

For $L = 1$ the palindrome is the empty block.

Proof. Apply Corollary 3.19 to x_1:

$$-\frac{1}{x_1'} = \sqrt{D} + a_0 = [\overline{a_L, a_{L-1}, \ldots, a_1}].$$

Its integer part is $2a_0$, so $a_L = 2a_0$. On the other hand,

$$\sqrt{D} + a_0 = 2a_0 + \frac{1}{x_1} = [2a_0; \overline{a_1, \ldots, a_L}].$$

These are two expansions of the same irrational number. The continued-fraction algorithm determines its digits uniquely. Comparing the next $L - 1$ digits gives $(a_1, \ldots, a_{L-1}) = (a_{L-1}, \ldots, a_1)$. □

For example,

$$\sqrt{23} = [4; \overline{1, 3, 1, 8}], \qquad \sqrt{14} = [3; \overline{1, 2, 1, 6}], \qquad \sqrt{2} = [1; \overline{2}].$$

The final $2a_0$ marks the end of the symmetric block; it is not itself part of the palindrome asserted by the theorem.

There is a useful consequence for the integer states $x_n = (P_n + \sqrt{D})/Q_n$ of (3.4), starting with $(P_0, Q_0) = (0, 1)$. Since $x_{L+1} = x_1$,

$$x_L = a_L + \frac{1}{x_{L+1}} = 2a_0 + (\sqrt{D} - a_0) = \sqrt{D} + a_0.$$

Consequently $(P_L, Q_L) = (a_0, 1)$, and the same state occurs at every positive multiple of L. Conversely, if $n \geq 1$ and $Q_n = 1$, reducedness gives $-1 < P_n - \sqrt{D} < 0$, forcing $P_n = a_0$. Thus $x_n = x_L$. The L states in a least cycle are distinct, so

$$\boxed{Q_n = 1 \iff L \mid n \qquad (n \geq 1).} \tag{3.8}$$

An exact identity for every convergent

Lemma 3.21 (The complete quotient measures the Pell remainder)**.** *For every $n \geq 1$,*

$$\boxed{p_{n-1}^2 - Dq_{n-1}^2 = (-1)^n Q_n.} \tag{3.9}$$

Proof. Keep the entire tail in the matrix product:

$$\sqrt{D} = \frac{p_{n-1}x_n + p_{n-2}}{q_{n-1}x_n + q_{n-2}}, \qquad x_n = \frac{P_n + \sqrt{D}}{Q_n}.$$

Substitute the second expression into the first and multiply through. Because $\sqrt{D}$ is irrational, the rational coefficients and the coefficients of $\sqrt{D}$ must agree separately. They give

$$\begin{aligned} p_{n-1} &= P_n q_{n-1} + Q_n q_{n-2}, \\ Dq_{n-1} &= P_n p_{n-1} + Q_n p_{n-2}. \end{aligned}$$

Multiply the first equality by p_{n-1} and the second by q_{n-1}, then subtract:

$$\begin{aligned} p_{n-1}^2 - Dq_{n-1}^2 &= Q_n(p_{n-1}q_{n-2} - p_{n-2}q_{n-1}) \\ &= (-1)^n Q_n. \end{aligned}$$

The last step is the determinant of the product of the n digit matrices $D(a_0), \ldots, D(a_{n-1})$. The initial conventions $p_{-1} = 1, q_{-1} = 0$ also cover $n = 1$. □

Theorem 3.22 (Exactly which convergents solve Pell)**.** *For each $m \geq 1$,*

$$p_{mL-1}^2 - Dq_{mL-1}^2 = (-1)^{mL}. \tag{3.10}$$

These are all positive integer solutions to the two equations $p^2 - Dq^2 = \pm 1$. In particular:

- *If L is even, $+1$ occurs at every index $mL - 1$, and the equation with right-hand side -1 has no integer solution.*
- *If L is odd, -1 occurs for odd m and $+1$ for even m.*

The least positive solution of the $+1$ equation is (p_{L-1}, q_{L-1}) for even L and (p_{2L-1}, q_{2L-1}) for odd L.

Proof. All Q_n with $n \geq 1$ are positive. The lemma says that a convergent solves one of the two equations exactly when $Q_n = 1$. By (3.8), this happens exactly when $n = mL$; (3.10) and its sign follow. Legendre's criterion, already applied above, ensures that no positive solution lies outside the convergents. Finally, the recurrence makes $p_0, p_1, \ldots$ strictly increasing, so the first admissible index gives the least p, and hence the least q as well. If a solution of the -1 equation had either coordinate zero, it would require $p^2 = -1$ or $Dq^2 = 1$, both impossible; changing signs reduces every other integer solution to a positive one. □

Stop before the final digit

The first candidate is

$$\frac{p_{L-1}}{q_{L-1}} = [a_0; a_1, \ldots, a_{L-1}],$$

which stops *before* the final $2a_0$. There are L digits here, counting a_0. For $\sqrt{3} = [1; \overline{1, 2}]$, the candidate is $[1; 1] = 2$, and $2^2 - 3 \cdot 1^2 = 1$. Including the next digit gives $[1; 1, 2] = 5/3$, whose remainder is $25 - 27 = -2$.

Example 3.23 (An odd period: $D = 13$)**.** The state calculation gives $\sqrt{13} = [3; \overline{1, 1, 1, 1, 6}]$, with $L = 5$. The lemma explains every entry of the last column below.

k	a_k	p_k/q_k	Q_{k+1}	$p_k^2 - 13q_k^2$
0	3	3/1	4	−4
1	1	4/1	3	3
2	1	7/2	3	−3
3	1	11/3	4	4
4	1	18/5	1	−1
5	6	119/33	4	4
6	1	137/38	3	−3
7	1	256/71	3	3
8	1	393/109	4	−4
9	1	649/180	1	1

Thus the first negative solution is $(18, 5)$, while the first positive one is $(649, 180)$. Squaring the former gives the latter:

$$(18 + 5\sqrt{13})^2 = 649 + 180\sqrt{13}.$$

The period matrix becomes a Pell matrix

Set $p = p_{L-1}$, $q = q_{L-1}$ and $M = D(a_0) \cdots D(a_{L-1})$. At the returning state $P_L = a_0, Q_L = 1$, the two coefficient identities in the lemma give

$$M \begin{pmatrix} 1 & a_0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & a_0 p + p_{L-2} \\ q & a_0 q + q_{L-2} \end{pmatrix} = \begin{pmatrix} p & Dq \\ q & p \end{pmatrix}.$$

The shift by a_0 accounts for the fact that M sends the tail $\sqrt{D} + a_0$ to $\sqrt{D}$. The resulting matrix fixes $\sqrt{D}$ and has determinant $p^2 - Dq^2 = (-1)^L$. For odd L, square it to obtain determinant $+1$.

Why one fundamental solution generates all the others

For $z = r + s\sqrt{D}$, define its *norm* by $N(z) = zz' = r^2 - Ds^2$. Conjugation respects multiplication, so $N(zw) = N(z)N(w)$. If (u, v) is the least positive solution of the $+1$ equation, put $\varepsilon = u + v\sqrt{D} > 1$. Its inverse is $u - v\sqrt{D}$, again with integer coefficients.

Proposition 3.24 (All positive Pell solutions are powers)**.** *Every positive solution of* $p^2 - Dq^2 = 1$ *has the form*

$$p + q\sqrt{D} = \varepsilon^j \qquad (j = 1, 2, \ldots).$$

Proof. Powers have integer coefficients, norm 1, and positive coefficients for $j \geq 1$. Conversely, let $\alpha = p + q\sqrt{D} > 1$ have norm 1. Choose $j \geq 0$ with $\varepsilon^j \leq \alpha < \varepsilon^{j+1}$ and put $\beta = \alpha\varepsilon^{-j} = r + s\sqrt{D}$. Then r, s are integers, $N(\beta) = 1$, and $1 \leq \beta < \varepsilon$. Since $\beta' = 1/\beta$,

$$r = \frac{\beta + \beta^{-1}}{2} > 0, \qquad s = \frac{\beta - \beta^{-1}}{2\sqrt{D}} \geq 0.$$

If $\beta > 1$, the function $t - t^{-1}$ is strictly increasing for $t > 0$, so $0 < s < v$. This would be a smaller positive Pell solution, a contradiction. Therefore $\beta = 1$ and $\alpha = \varepsilon^j$. Since $q > 0$, we have $j \geq 1$. □

Equivalently, multiplication by ε is the matrix recurrence

$$\begin{pmatrix} p^{(j+1)} \\ q^{(j+1)} \end{pmatrix} = \begin{pmatrix} u & Dv \\ v & u \end{pmatrix} \begin{pmatrix} p^{(j)} \\ q^{(j)} \end{pmatrix}, \qquad (p^{(0)}, q^{(0)}) = (1, 0).$$

The superscript counts powers; the subscript on p_k/q_k counts continued-fraction convergents. For $D = 2$, $\varepsilon = 3 + 2\sqrt{2}$ gives $(3, 2), (17, 12), (99, 70)$.

Big idea

Periodicity gives the exact return $Q_{mL} = 1$. The determinant identity then turns that return into $p_{mL-1}^2 - Dq_{mL-1}^2 = (-1)^{mL}$. Legendre proves completeness, and powers of the least positive solution give a second way to list every positive solution. Each step has a different job.

Checkpoint

Compute $99/70$ as a decimal. Its accuracy follows from $99^2 - 2 \cdot 70^2 = 1$, which forces

$$\left|\sqrt{2} - 99/70\right| = \frac{1}{70^2(\sqrt{2} + 99/70)}.$$

For a second check, use the period of $\sqrt{23}$ above to find the fundamental solution $(24, 5)$ without searching through denominators.

3.11 The Cattle of the Sun: Archimedes' challenge

The cattle problem attributed to Archimedes asks for a herd belonging to the Sun, with bulls and cows in four colors: white, black, yellow, and dappled. Its proportions can be written as linear equations. Two further conditions ask for a square and a triangle. Together they lead to a Pell equation with a remarkably large first solution.

This optional application begins with the ancient problem's conditions. For a detailed modern exposition, see Ilan Vardi, "Archimedes' Cattle Problem," *American Mathematical Monthly* **105** (1998), 305–319; an accessible preprint is also available. The calculations below can be reproduced with exact integer arithmetic in the first notebook.

Eight unknowns, seven linear conditions

Let W, B, Y, S count white, black, yellow, and spotted (dappled) bulls; let w, b, y, s count cows of the corresponding colors. The conditions are

$$W = \left(\frac{1}{2}+\frac{1}{3}\right)B + Y, \qquad B = \left(\frac{1}{4}+\frac{1}{5}\right)S + Y, \qquad S = \left(\frac{1}{6}+\frac{1}{7}\right)W + Y,$$
$$w = \left(\frac{1}{3}+\frac{1}{4}\right)(B+b), \qquad b = \left(\frac{1}{4}+\frac{1}{5}\right)(S+s),$$
$$y = \left(\frac{1}{6}+\frac{1}{7}\right)(W+w), \qquad s = \left(\frac{1}{5}+\frac{1}{6}\right)(Y+y).$$

The cow equations use an entire herd of the specified color, including its bulls. The two additional conditions concern bulls only:

$$W + B = \text{a square}, \qquad Y + S = \frac{h(h+1)}{2} \quad \text{for some integer } h \geq 1. \tag{3.11}$$

Eliminate B, Y, S from the first three equations to obtain

$$B = \frac{267}{371}W, \qquad Y = \frac{297}{742}W, \qquad S = \frac{790}{1113}W.$$

The four cow equations form a cycle. Substituting around that cycle expresses w as a known multiple of W plus $\frac{7}{12}\frac{9}{20}\frac{11}{30}\frac{13}{42}w$. Since this coefficient is less than 1, w, and then every cow count, is uniquely determined by W. Clearing denominators gives

Color	Bulls	Cows
White	$10\,366\,482t$	$7\,206\,360t$
Black	$7\,460\,514t$	$4\,893\,246t$
Yellow	$4\,149\,387t$	$5\,439\,213t$
Dappled	$7\,358\,060t$	$3\,515\,820t$

Substitution verifies all seven equations, and the eight coefficients have gcd 1. Therefore all eight counts are integers exactly when t is an integer: a Bézout combination of those counts equals t. Every positive solution of the linear part has $t \geq 1$, and the total number of cattle is

$$T = 50\,389\,082t. \tag{3.12}$$

The square and the triangle produce Pell

The square condition is

$$W + B = 17\,826\,996t = 4 \cdot 4\,456\,749t, \qquad 4\,456\,749 = 3 \cdot 11 \cdot 29 \cdot 4657.$$

The four factors in the last product are distinct primes (trial division checks 4657). Unique factorization therefore says that this is a square exactly when

$$t = 4\,456\,749r^2 \qquad (r \geq 1).$$

Indeed, every prime in the displayed squarefree product must have odd exponent in t, and every other prime must have even exponent. Now the triangular condition becomes

$$11\,507\,447t = \frac{h(h+1)}{2}, \qquad 51\,285\,802\,909\,803r^2 = \frac{h(h+1)}{2}.$$

Multiplying by 8 and completing the square gives

$$\boxed{X^2 - 410\,286\,423\,278\,424r^2 = 1, \qquad X = 2h+1.} \tag{3.13}$$

Conversely, a positive solution of this equation has odd X, since the coefficient of r^2 is divisible by 8. Thus $h = (X-1)/2$ is an integer and reconstructs the triangle. Together with the table, this proves an equivalence, not just a necessary condition.

The coefficient is large, but contains a large square factor:

$$410\,286\,423\,278\,424 = 4\,729\,494 \cdot 9314^2.$$

Put $Z = 9314r$. The equivalent smaller-coefficient problem is

$$\boxed{X^2 - 4\,729\,494Z^2 = 1, \qquad 9314 \mid Z.} \tag{3.14}$$

Keeping the divisibility condition is essential: an arbitrary solution of the smaller Pell equation need not give an integer r.

A short period, followed by a divisibility test

The integer-state algorithm gives $a_0 = 2174$ and least period length $L = 92$ for $\sqrt{4\,729\,494}$. The final digit is 4348; the preceding 91 digits form a palindrome. Since 92 is even, the convergent p_{91}/q_{91} gives the fundamental solution

$$u = 1099319867328297349798662328214335439010880 49,$$
$$v = 5054948523431503307447781973554040898634 0.$$

These values can be checked by the exact equality $u^2 - 4\,729\,494v^2 = 1$. Write $\varepsilon = u + v\sqrt{4\,729\,494}$ and

$$\varepsilon^j = X_j + Z_j\sqrt{4\,729\,494}.$$

Proposition 3.24 proves that this lists every positive solution of the smaller Pell equation. We need the first j for which $9314 \mid Z_j$.

We can find it without ever forming the enormous powers. Reduce the multiplication recurrence modulo 9314. Here

$$u \equiv 9063, \quad v \equiv 7708, \quad 4\,729\,494 \equiv 7296 \pmod{9314}.$$

The following finite computation checks every exponent up to the first success, using only small integers:

```
modulus = 9314
u, v, d = 9063, 7708, 7296
a, b = 1, 0
for j in range(1, 2330):
    a, b = ((u*a + d*v*b) % modulus,
            (v*a + u*b) % modulus)
    if b == 0:
        print(j, a, b)
        break
```

Its output is `2329 9313 0`; no earlier exponent passes the test. Thus ε^{2329} produces the smallest admissible r and hence the smallest herd. This is a reproducible finite calculation, distinct from the general Pell proofs above. Vardi explains a shorter divisibility test using a quadratic finite field.

An enormous answer with a compact description

Let $X + Z\sqrt{4\,729\,494} = \varepsilon^{2329}$. Then the least total is

$$T = 50\,389\,082 \cdot 4\,456\,749 \left(\frac{Z}{9314}\right)^2. \tag{3.15}$$

This is an exact specification using integer arithmetic. The first notebook computes X, Z by repeated squaring, verifies the seven proportions and both shape conditions, and checks

$$10^{206544} \le T < 10^{206545}.$$

The decimal expansion of the smallest herd's size therefore has 206 545 digits:

$$T \approx 7.760271406 \times 10^{206544}.$$

Its size is a mathematical consequence of the conditions, rather than a plausible census of animals.

Conjugation also gives a compact real-number expression, of the kind developed in Vardi's exposition. Since

$$Z = \frac{\varepsilon^{2329} - \varepsilon^{-2329}}{2\sqrt{4\,729\,494}},$$

substitution into (3.15) yields

$$T = \frac{25\,194\,541}{184\,119\,152}\left(\varepsilon^{4658} + \varepsilon^{-4658} - 2\right).$$

An expression can be short even when its decimal expansion is long.

Big idea

The linear conditions leave one scale factor. Prime-exponent parity makes that factor a fixed number times a square. The triangular condition becomes Pell, and a divisibility test selects the allowed powers of its fundamental solution. The same elementary tools that solve $p^2 - 2q^2 = 1$ organize the cattle problem.

Exercises

1. Find the continued fraction of $355/113$. List all convergents. [Fluency]

2. Compute the first six convergents of $[1; \overline{1}]$. What familiar sequence appears? [Fluency]

3. Prove the determinant identity by induction. Use it to prove $\gcd(p_k, q_k) = 1$. [Reasoning]

4. Show that $[a_0; \ldots, a_n] = [a_0; \ldots, a_n - 1, 1]$ when $a_n > 1$. Explain the only ambiguity in finite expansions. [Reasoning]

5. Derive $\sqrt{3} = [1; \overline{1, 2}]$ and compute four convergents. [Reasoning]

6. Write the tropical year as $365 + \alpha$ with $\alpha \approx 0.2421988$. Compute the first four nonzero

convergents of α and identify the Julian rule. Then impose a century-based cycle by approximating 100α; explain how $97/4$ becomes the Gregorian rule $97/400$. Compare it with $31/128$. [Exploration]

7. Investigate which convergents to $\sqrt{2}$ solve $p^2 - 2q^2 = 1$ and which solve -1. State and prove a pattern using the recurrence. [Challenge]

8. Let $A/B = [a_0; \ldots, a_n]$ and use the convergent recurrences to prove $r_k = (-1)^k(q_kA - p_kB)$. Apply the penultimate convergent of $527/341$ to recover a Bézout identity for $\gcd(527, 341)$. [Reasoning]

9. For $x = \sqrt{3}$, compute enough convergents to display three lower bounds and three upper bounds. Verify that the lower bounds increase and the upper bounds decrease, then explain why the Möbius derivative has the sign needed for each chain. [Reasoning]

10. Prove the matrix product formula (3.1) by induction. Explain why its inverse always has integer entries. [Reasoning]

11. Long-divide $1/7$ and $5/12$ in base 10, recording the remainder at every step. Use repeated remainders to predict the preperiod and period. [Fluency]

12. Prove directly from a geometric series that every eventually periodic base-b expansion is rational. [Reasoning]

13. After studying Euler's theorem, return to prove that $\gcd(q, b) = 1$ implies the base-b expansion of m/q is purely periodic. Give an explicit upper bound for a period length. [Reasoning]

14. Verify that $355/113$ satisfies Legendre's inequality for π using a calculator value of π. What conclusion does the theorem certify? [Exploration]

15. Let $t = \overline{[1, 2]}$. Use the matrix $D(1)D(2)$ to find a quadratic equation for t, and identify the positive root. [Reasoning]

16. Check Galois's inequalities for $1 + \sqrt{2}$ and $(1 + \sqrt{5})/2$. Explain why their continued fractions are purely periodic. [Reasoning]

17. Let $p^2 - Dq^2 = \pm 1$ with $p, q > 0$. Derive (3.7), verify its strict Legendre bound, and conclude that p/q is a convergent of $\sqrt{D}$. [Reasoning]

18. Compute the period of $\sqrt{3}$. Use its length to predict whether one period solves the $+1$ or -1 Pell equation. Then use powers of $\left(\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right)$ to generate three solutions of $p^2 - 3q^2 = 1$. [Exploration]

19. Define $G_n(x) = (a_1 \cdots a_n)^{1/n}$ and $B_n(x) = q_n^{2/n}$. Use the Sage notebook to reproduce several rows of the π table above. Explain why agreement with the Khinchin–Lévy constants is evidence but not proof. Then derive the displayed formula for $G_{3m}(e)$ directly from the pattern in the continued fraction of e. [Exploration]

20. Compute $D(1)D(2)$ and $D(2)D(1)$, and write the linear fractional transformation associated to each. Explain the order of composition and find each transformation's positive fixed point. [Fluency]

21. Let $x = [\overline{1,2}]$ and $N = D(1)D(2)$. For $j = 1, 2$, set $M = D(j)$ and compute MNM^{-1}. Use its fixed-point equation to identify $y = [j; \overline{1,2}]$. Which of these numbers is purely periodic? Write its period without an initial prefix. [Reasoning]

22. Prove that an invertible rational linear fractional transformation sends every irrational real number to an irrational real number. Explain why a pole cannot occur at an irrational input. Then identify the missing hypothesis in the assertion that $t = T_M(t)$ always makes t quadratic. Give a counterexample. [Reasoning]

23. Use Galois's criterion to decide which of

$$\sqrt{2}, \quad 1+\sqrt{2}, \quad \frac{1+\sqrt{5}}{2}, \quad \frac{1+\sqrt{3}}{2}, \quad \frac{\sqrt{2}}{2}, \quad -\sqrt{2}$$

have purely periodic continued fractions. Explain why all six are eventually periodic. [Fluency]

24. Starting from $D = 13$ and $(P_0, Q_0) = (0, 1)$, use (3.4) to compute the continued fraction of $\sqrt{13}$. Record (P_n, Q_n, a_n) until a state first repeats, and verify the reduced-state bounds after the first step. [Exploration]

25. Recover the unique reduced predecessor of $z = (1+\sqrt{3})/2$ using (3.6). For $D = 3$, enumerate the integer pairs satisfying (3.5) that actually represent reduced surds and satisfy $Q \mid 3 - P^2$. Show how the continued-fraction step permutes them. Why would finiteness without reversibility not suffice to prove pure periodicity? [Reasoning]

26. Let $x = [\overline{1,2,3}]$. Compute its period matrix and its positive fixed point. Transpose the matrix to obtain the equation for $[\overline{3,2,1}]$, and verify directly that this number is $-1/x'$. [Reasoning]

27. Compute the integer states and least period for $\sqrt{23}$. Identify the palindrome, find the fundamental Pell solution, and calculate what happens if you include the final digit $2a_0$ in the first candidate. [Fluency]

28. For $D = 6$, calculate four convergents and the corresponding Q_{k+1}. Verify (3.9) in each row. Use (3.8) to classify every convergent that solves either sign of Pell. [Reasoning]

29. Decide whether $p^2 - Dq^2 = -1$ is solvable for $D = 2, 3, 5, 7, 13, 14$. Give the least positive solution when it exists. For $D = 3$, also give a one-line impossibility proof modulo 3. [Reasoning]

30. Let $p^2 - 3q^2 = 1$ with $p, q > 0$. Show that multiplying $p + q\sqrt{3}$ by $2 - \sqrt{3}$ gives $(1, 0)$ from $(2, 1)$, and otherwise a positive solution with smaller second coordinate. Use this descent to prove directly that all positive solutions are powers of $2 + \sqrt{3}$. [Reasoning]

31. Use the integer-state algorithm to find the least period of $\sqrt{61}$. Find the first negative and positive Pell solutions, and explain why the theorem proves that each is the first of its sign. Use exact integers, not a decimal approximation to $\sqrt{61}$. [Exploration]

32. Verify the cattle table at $t = 1$, including its gcd and total. Explain why the seven linear equations have only one free scale factor, and why the gcd certifies that this scale is an integer. Does the table at $t = 1$ solve the full problem? [Reasoning]

33. For positive integers c, r, prove that cr^2 is triangular exactly when $X^2 - 8cr^2 = 1$ has a positive odd solution X. Apply this to the cattle problem, and prove both directions of the passage from (3.13) to (3.14). [Reasoning]

34. Reproduce the modular search for the cattle problem in the first notebook. Explain why its first hit proves minimality. Using $\varepsilon^{2329} \equiv -1 \pmod{9314}$ and the absence of an earlier hit, prove that the admissible exponents are exactly the positive multiples of 2329. Why does 4658 appear in the total's compact formula? [Exploration]

Chapter 4

Arithmetic on a clock

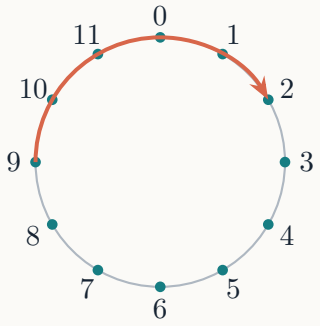
Extended Euclid, Chinese remainders, and interpolation

Big idea

The extended gcd algorithm is the computational engine of modular arithmetic. It tells us when division is possible and supplies the inverse when it is. We will use those inverses to reconstruct an integer from its remainders, then recognize the same construction in polynomial interpolation. Finally, a reversible recurrence will explain why Fibonacci residues repeat.

Definition 4.1. For $n \geq 2$, we write $a \equiv b \pmod{n}$ if $n \mid (a-b)$. The equivalence class of a is written $[a]$ or $\bar{a}$.

Congruence remembers the remainder and forgets the number of complete turns around the clock.



$9 + 5 \equiv 2 \pmod{12}$
one complete turn disappears

4.1 Legal arithmetic

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d \pmod{n}, \qquad ac \equiv bd \pmod{n}.$$

Powers are therefore legal too. Division is different.

These rules do not depend on the chosen representatives. For example, if $a = b + kn$ and $c = d + \ell n$, then $ac - bd = n(b\ell + dk + k\ell n)$. Thus addition and multiplication are well-defined operations on

the residue classes $\mathbb{Z}/n\mathbb{Z}$. Reduce after each step or at the end: the resulting residue class is the same.

4.2 Extended gcd: division comes with a certificate

Definition 4.2. A residue a is a *unit modulo* n if there exists b with $ab \equiv 1 \pmod{n}$. Such a b is an inverse of a.

Theorem 4.3 (Invertibility criterion)**.** *a has an inverse modulo n exactly when* $\gcd(a, n) = 1$.

Proof. The congruence $ab \equiv 1 \pmod{n}$ means $ab + nk = 1$ for some k, so any common divisor of a, n divides 1. Conversely, when $\gcd(a, n) = 1$, extended Euclid produces u, v such that $au + nv = 1$. Reducing this *integer identity* modulo n gives $au \equiv 1$: the coefficient u, reduced modulo n, is the required inverse. □

The theorem is an algorithm, not a direction to try every residue until one works. Here is a forward version of the backward substitutions from Chapter 1. First reduce a so that $0 \le a < n$. Keep three entries in each row:

$$r_i = ns_i + at_i.$$

Start with $(r_0, s_0, t_0) = (n, 1, 0)$ and $(r_1, s_1, t_1) = (a, 0, 1)$. Whenever $r_i \neq 0$, divide $r_{i-1} = q_i r_i + r_{i+1}$ and update

$$(r_{i+1}, s_{i+1}, t_{i+1}) = (r_{i-1}, s_{i-1}, t_{i-1}) - q_i(r_i, s_i, t_i).$$

The same subtraction in all three columns preserves the displayed identity. The last nonzero row gives the gcd and its Bézout coefficients.

Example 4.4 (An inverse without guessing)**.** Find $17^{-1} \pmod{43}$. The divisions are

$$43 = 2 \cdot 17 + 9, \quad 17 = 9 + 8, \quad 9 = 8 + 1, \quad 8 = 8 \cdot 1.$$

The coefficient table records $r = 43s + 17t$ throughout:

r	s	t	Certificate
43	1	0	$43 = 1 \cdot 43 + 0 \cdot 17$
17	0	1	$17 = 0 \cdot 43 + 1 \cdot 17$
9	1	-2	$9 = 43 - 2 \cdot 17$
8	-1	3	$8 = -43 + 3 \cdot 17$
1	2	-5	$1 = 2 \cdot 43 - 5 \cdot 17$
0	-17	43	$0 = -17 \cdot 43 + 43 \cdot 17$

Hence $17^{-1} \equiv -5 \equiv 38 \pmod{43}$. Check the result directly: $17 \cdot 38 = 646 = 15 \cdot 43 + 1$.

Checkpoint

In a certificate $au + nv = 1$, which coefficient is the inverse of a modulo n? Read the coefficient of a, even if the table began with n in its first row. The final multiplication check catches a swapped coefficient or a sign error.

Common trap

Cancellation can fail. Modulo 6, $2 \cdot 1 \equiv 2 \cdot 4$, but $1 \not\equiv 4$. You may cancel a modulo n only when a is a unit, or after carefully reducing the modulus.

For a prime p, every nonzero residue is a unit. We write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ and call it a *field*: addition and multiplication obey the usual commutative arithmetic laws, and every nonzero element permits division. The rational and real numbers are other examples of fields. For composite n, $\mathbb{Z}/n\mathbb{Z}$ need not permit division by a nonzero residue.

4.3 Solving a linear congruence with the same algorithm

Proposition 4.5 (Linear congruence)**.** *The congruence $ax \equiv b \pmod{n}$ has a solution exactly when $d = \gcd(a, n)$ divides b. When it does, there are exactly d incongruent solutions modulo n.*

Construction and completeness. Run extended Euclid once to obtain $au + nv = d$. If $ax \equiv b \pmod{n}$, then $b = ax - nk$ for an integer k, so $d \mid b$ is necessary. If $d \mid b$, multiply the certificate by b/d to obtain a solution

$$x_0 = u\frac{b}{d}.$$

If $d = n$, then both a and b are multiples of n, so all n residues are solutions. Otherwise $n/d \geq 2$, and any two solutions satisfy $(a/d)(x - x_0) \equiv 0 \pmod{n/d}$. Since a/d is a unit modulo n/d, this is equivalent to $x \equiv x_0 \pmod{n/d}$. Therefore the complete list modulo n is

$$x_0,\ x_0 + \frac{n}{d},\ \ldots,\ x_0 + (d-1)\frac{n}{d}.$$

These d residues are distinct. □

Example 4.6 (A nonunit can give several solutions)**.** For $14x \equiv 21 \pmod{35}$, extended Euclid gives $7 = 35 - 2 \cdot 14$. Multiplying by 3 gives $21 = 3 \cdot 35 - 6 \cdot 14$, so $x \equiv -6 \equiv 4 \pmod{5}$. The seven solutions modulo 35 are $4, 9, 14, 19, 24, 29, 34$. The same calculation rules out $14x \equiv 20 \pmod{35}$, because $7 \nmid 20$.

4.4 Reassembling remainders

Suppose a number leaves remainder 2 modulo 3 and remainder 3 modulo 5. Listing possibilities gives 8 modulo 15. The Chinese remainder theorem says this always works for coprime clocks.

Theorem 4.7 (Chinese remainder theorem)**.** *If $n_1, \ldots, n_r$ are pairwise coprime, then for any $a_1, \ldots, a_r$, the system*

$$x \equiv a_i \pmod{n_i} \qquad (i = 1, \ldots, r)$$

has a solution, unique modulo $N = n_1 \cdots n_r$.

Construction. Set $N_i = N/n_i$. Since $\gcd(N_i, n_i) = 1$, *use extended Euclid* to find M_i, v_i with $N_i M_i + n_i v_i = 1$. Then

$$x = \sum_{i=1}^{r} a_i N_i M_i$$

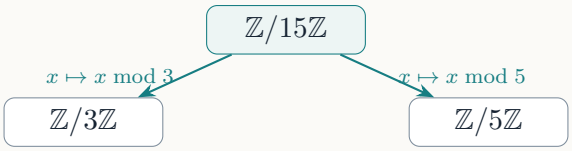
has every prescribed remainder: modulo n_i, its ith term is a_i and every other term is 0. If x, y both work, each n_i divides $x - y$; coprimality makes $N \mid x - y$. □

The quantities $e_i = N_i M_i$ are *selectors*: e_i is 1 at its own modulus and 0 at every other modulus. The term $a_i e_i$ installs one prescribed remainder without disturbing the others.

Example 4.8 (Construct, reduce, and check)**.** Solve $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$. Here $N = 105$, and the construction is

n_i	a_i	N_i	M_i	$e_i = N_i M_i$
3	2	35	2	70
5	3	21	1	21
7	2	15	1	15

For instance, $2 \cdot 35 - 23 \cdot 3 = 1$ certifies the first inverse. Thus $x = 2 \cdot 70 + 3 \cdot 21 + 2 \cdot 15 = 233 \equiv 23 \pmod{105}$. Finally, 23 has remainders $2, 3, 2$, as required. Every solution is $23 + 105k$ for an integer k.



One clock of size 15 is equivalent to two independent clocks.

The map from one clock to its component clocks preserves addition and multiplication, since reduction does. CRT says it is also a bijection. For example, the selectors modulo 15 are 10 and 6, giving $x \equiv 10a + 6b \pmod{15}$ for remainders a modulo 3 and b modulo 5.

Proposition 4.9 (When two clocks share a factor)**.** *The conditions $x \equiv a \pmod m$, $x \equiv b \pmod n$ are compatible exactly when $a \equiv b \pmod{\gcd(m, n)}$. If compatible, the solution is unique modulo $\operatorname{lcm}(m, n)$.*

Proof. Write $x = a + mt$. The remaining condition is $mt \equiv b - a \pmod n$, so the linear-congruence construction supplies a solution exactly when $\gcd(m, n) \mid b - a$. Differences between two solutions are precisely the common multiples of m and n. □

4.5 Chinese remainders become Lagrange interpolation

CRT builds an integer with prescribed remainders. Interpolation builds a polynomial with prescribed values. Polynomial division makes these the same kind of problem.

Let K be a field, such as $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{F}_p$. Write $K[X]$ for polynomials in an indeterminate X with coefficients in K. Division by $X - c$ gives

$$f(X) = (X - c)q(X) + r,$$

where r is constant. Substituting $X = c$ shows $r = f(c)$. In other words,

$$f(X) \equiv b \pmod{X - c} \quad \Longleftrightarrow \quad f(c) = b.$$

Here polynomial congruence means that the difference is divisible by the polynomial modulus.

The same construction, with polynomial moduli

The division algorithm works in $K[X]$: cancel the leading term using division in K, subtract, and continue while the degree decreases. Extended Euclid therefore works too. For coprime polynomials A, B, it produces U, V with $UA + VB = 1$.

Consequently, for pairwise coprime monic moduli $m_1, \ldots, m_r$, set $P = \prod_i m_i$ and $P_i = P/m_i$, and use extended Euclid to find $U_i P_i + V_i m_i = 1$. The polynomial

$$f = \sum_i b_i P_i U_i$$

has prescribed polynomial remainders b_i modulo m_i. The proof is the integer CRT proof: reduce at each modulus separately. Uniqueness is modulo P, since the pairwise coprime factors m_i all divide the difference of two solutions. Dividing by the monic polynomial P selects one representative of degree less than $\deg P$.

For interpolation, take $m_i = X - c_i$ with the c_i distinct. Two such linear polynomials are coprime: subtracting them gives the nonzero constant $c_j - c_i$, which is invertible in K. In this special case, the inverse required by CRT is particularly simple. For

$$P_i(X) = \prod_{j \neq i} (X - c_j),$$

reduction modulo $X - c_i$ gives the nonzero number $P_i(c_i) = \prod_{j \neq i}(c_i - c_j)$. Thus define

$$\ell_i(X) = \frac{P_i(X)}{P_i(c_i)} = \prod_{j \neq i} \frac{X - c_j}{c_i - c_j}.$$

These are polynomial selectors: $\ell_i(c_i) = 1$ and $\ell_i(c_j) = 0$ for $j \neq i$.

Theorem 4.10 (Lagrange interpolation)**.** *Let $c_1, \ldots, c_r$ be distinct points of a field K. Given values $b_i \in K$, the unique polynomial of degree less than r with $f(c_i) = b_i$ for every i is*

$$f(X) = \sum_{i=1}^{r} b_i \ell_i(X).$$

Every polynomial with these values has the form $f(X) + h(X) \prod_i (X - c_i)$, with $h \in K[X]$.

Proof. The selector values prove existence. If f, g have the required values, each $X - c_i$ divides $f - g$, and hence so does their product. A nonzero multiple of that degree-r product cannot have degree less than r. This proves uniqueness and the description of all solutions. The same argument proves the useful *root bound*: a nonzero polynomial of degree d over a field has at most d distinct roots. □

	Integer CRT	Lagrange interpolation
Local data	$x \equiv a_i \pmod{n_i}$	$f(c_i) = b_i$
Moduli	pairwise coprime n_i	distinct linear factors $X - c_i$
Selector	$N_i(N_i^{-1} \bmod n_i)$	$P_i(X)/P_i(c_i)$
Formula	$\sum_i a_i e_i$	$\sum_i b_i \ell_i(X)$
Ambiguity	multiples of N	multiples of $\prod_i (X - c_i)$
Choice	$0 \leq x < N$	$\deg f < r$

Example 4.11 (Three values determine a quadratic)**.** Over $\mathbb{Q}$, prescribe $f(0) = 1$, $f(1) = 2$, $f(2) = 5$. The selectors are

$$\ell_0 = \frac{(X-1)(X-2)}{2}, \qquad \ell_1 = -X(X-2), \qquad \ell_2 = \frac{X(X-1)}{2}.$$

Combining them gives $f = \ell_0 + 2\ell_1 + 5\ell_2 = X^2 + 1$. All other solutions are $X^2 + 1 + X(X-1)(X-2)h(X)$.

The same computation works in $\mathbb{F}_5$, where the prescribed values are $1, 2, 0$ and division by 2 means multiplication by 3. The certificate $2 \cdot 3 - 5 = 1$ makes that division legitimate. The result is again $X^2 + 1$, now with coefficients modulo 5.

Common trap

Distinct interpolation points must have invertible differences. This is automatic in a field. It can fail for composite moduli: over $\mathbb{Z}/8\mathbb{Z}$, the distinct points 0 and 2 have difference 2, which has no inverse. Indeed, $f(2) - f(0)$ is even for every polynomial, so the data $f(0) = 0$, $f(2) = 1$ cannot be interpolated.

Checkpoint

Both reconstruction formulas first make an object that vanishes at every unwanted coordinate, then multiply by an inverse to make its remaining coordinate equal to 1. Identify those two steps before memorizing either formula.

4.6 Fast powers and patterns

Modular reduction prevents numbers from exploding. To compute $7^{222} \pmod{13}$, square and reduce:

$$7^2 \equiv 10, \quad 7^4 \equiv 9, \quad 7^8 \equiv 3, \quad 7^{16} \equiv 9 \pmod{13},$$

then use the binary expansion $222 = 128 + 64 + 16 + 8 + 4 + 2$. Later, Fermat's theorem will make this even faster.

Explore

Make a multiplication table modulo n and highlight the rows that are permutations of all residues. Conjecture which row labels have inverses. Then compare with the gcd criterion.

4.7 Fibonacci residues: why the pattern must repeat

Set $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Modulo an integer $m \geq 2$, the next term depends on the previous *two* terms. The appropriate state is therefore

$$S_n = (F_n, F_{n+1}) \pmod{m}, \qquad T(a, b) = (b, a + b) \pmod{m}.$$

There are only m^2 pairs, so a state must eventually repeat. But finiteness alone only gives eventual periodicity. Here we can do better:

$$T^{-1}(u, v) = (v - u, u) \pmod{m}.$$

Every step has a unique predecessor, using only subtraction.

Theorem 4.12 (Fibonacci periodicity from the starting pair)**.** *For every $m \geq 2$, the Fibonacci sequence modulo m is periodic from the start. Its least period $\pi(m)$ is the smallest positive t for which*

$$(F_t, F_{t+1}) \equiv (0, 1) \pmod{m}.$$

In particular, $\pi(m) \leq m^2 - 1$.

Proof. If $S_i = S_j$ with $i < j$, apply the inverse step i times to get $S_0 = S_{j-i}$. Repeating the forward step now gives a period beginning at S_0, not just after an initial segment. Conversely, a period of the sequence must return both F_0 and F_1, so it returns the starting pair. Before the first return all states are distinct. The pair $(0, 0)$ is fixed by T and has no other predecessor, so the orbit of $(0, 1)$ never visits it. There are at most $m^2 - 1$ available states. □

Example 4.13 (A zero is not yet a restart)**.** Modulo 3, the successive states are

n	0	1	2	3	4	5	6	7	8
F_n	0	1	1	2	0	2	2	1	0
F_{n+1}	1	1	2	0	2	2	1	0	1

The pair first returns at $n = 8$, so $\pi(3) = 8$. Although $F_4 \equiv 0$, its successor is 2, not 1. Looking only for a repeated Fibonacci value would give the wrong answer.

Proposition 4.14 (CRT combines periods)**.** *If $\gcd(m, n) = 1$, then*

$$\pi(mn) = \operatorname{lcm}(\pi(m), \pi(n)).$$

Proof. CRT says that a pair returns modulo mn exactly when it returns modulo both m and n. The return times modulo m are precisely the multiples of $\pi(m)$: divide any return time by $\pi(m)$ and use the minimality of the remainder. Likewise for n. The first common return time is their least common multiple. □

For example, the orbit modulo 2 has length 3, so $\pi(6) = \operatorname{lcm}(3, 8) = 24$. The connection is operational: CRT reassembles the two coordinates of every state as well as the arithmetic on the clocks.

Explore

For a small modulus, start at $(0, 1)$, repeatedly replace (a, b) by $(b, a + b \bmod m)$, and stop on the first return to $(0, 1)$. The theorem proves this procedure terminates and that the answer is exact. Record pairs, not just zeros. These small experiments need no extension fields.

What we can prove now, and what comes next

The reversible-pair argument proves periodicity for *every* modulus, prime or composite. The estimate $\pi(m) \leq m^2 - 1$ is an upper bound; it does not assert that $\pi(m)$ divides $m^2 - 1$. For example, direct iteration gives $\pi(5) = 20$, and $20 \nmid 24$.

A structural computation uses the roots of $X^2 - X - 1$. Their $\sqrt{5}$ exists in $\mathbb{F}_p$ for $p = 11$, but not for $p = 7$. Appendix D first builds arithmetic in an extension field, then uses Frobenius and

multiplicative orders to compute exact periods. Those tools belong to a later unit; pair iteration and CRT require only the arithmetic developed here.

Exercises

1. Find the inverse of 37 modulo 101 using extended Euclid. [Fluency]

2. Solve $18x \equiv 30 \pmod{42}$ and list all solutions modulo 42. [Fluency]

3. Prove that congruence modulo n is an equivalence relation. [Reasoning]

4. Show that an integer is divisible by 9 exactly when its decimal digit sum is divisible by 9. [Reasoning]

5. Solve $x \equiv 2 \pmod{7}$, $x \equiv 3 \pmod{8}$, $x \equiv 4 \pmod{9}$ by the constructive CRT proof. [Reasoning]

6. Determine every possible last two digits of a square. Explain how CRT can split the work into moduli 4 and 25. [Exploration]

7. Prove that $x^2 \equiv 1 \pmod{p}$ has only the solutions $x \equiv \pm 1$ when p is prime. Find a composite modulus where there are more than two solutions. [Challenge]

8. Run the forward extended-gcd table for 83 and 29, recording $r = 83s + 29t$ in every row. Find $29^{-1} \pmod{83}$ and check it by multiplication. [Fluency]

9. Solve $18x \equiv 24 \pmod{42}$ using a Bézout certificate and list all residues modulo 42. Explain why $18x \equiv 25 \pmod{42}$ has no solution. [Fluency]

10. For the CRT selectors, prove $e_i^2 \equiv e_i$, $e_i e_j \equiv 0$ when $i \neq j$, and $\sum_i e_i \equiv 1 \pmod{N}$. Explain how these identities recover componentwise arithmetic. [Reasoning]

11. Solve $x \equiv 2 \pmod{6}$, $x \equiv 5 \pmod{9}$. Give the modulus of uniqueness. Why is $x \equiv 1 \pmod{6}$, $x \equiv 2 \pmod{9}$ incompatible? [Reasoning]

12. Construct the polynomial of degree at most 2 over $\mathbb{Q}$ with $f(-1) = 2$, $f(0) = 1$, $f(2) = 3$ using Lagrange selectors. State all polynomials with these values. [Fluency]

13. Work in $\mathbb{F}_7$. Use Lagrange interpolation to find the polynomial of degree at most 2 with $f(0) = 1$, $f(1) = 4$, $f(3) = 2$. Certify each denominator inverse by an integer Bézout identity. [Reasoning]

14. Prove that no polynomial over $\mathbb{Z}/8\mathbb{Z}$ can satisfy $f(0) = 0$, $f(2) = 1$. Which hypothesis of the interpolation theorem fails? [Reasoning]

15. List the Fibonacci pair orbits modulo 2 and 3. Why is the first positive zero of $F_n \pmod{3}$ not the period? [Fluency]

16. Use CRT and $\pi(2) = 3$, $\pi(3) = 8$, $\pi(5) = 20$ to find $\pi(6)$ and $\pi(15)$. Explain why the least common multiple, rather than the product, is the appropriate operation. [Reasoning]

17. Iterate $(a, b) \mapsto (b, 2a + b)$ modulo 2, starting at $(1, 0)$. Show that the orbit has an initial segment before it repeats. Locate the failure of the Fibonacci reversibility proof. [Reasoning]

18. List the squares modulo 7 and show that $X^2 - X - 1$ has no root in $\mathbb{F}_7$. Explain why the real Binet formula cannot simply be reduced modulo 7 without first enlarging the arithmetic system. [Bridge]

Chapter 5

A theorem you can rotate

Necklaces, Fermat, Euler, and Möbius inversion

Why should a prime p divide $a^p - a$ for every integer a? Instead of manipulating the expression, we count objects that naturally arrive in packets of p.

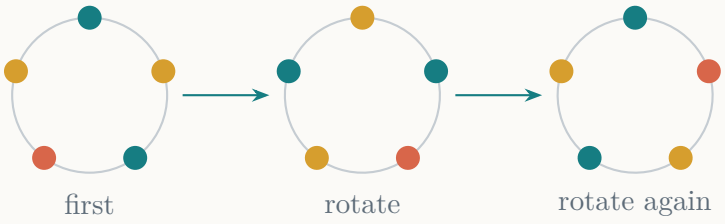
5.1 Words on a circular loom

Fix a prime p and a colors. First imagine p labeled positions around a circle. A coloring is a word

$$(c_0, c_1, \ldots, c_{p-1}),$$

and there are a^p such words. Rotation by one step sends it to

$$(c_{p-1}, c_0, \ldots, c_{p-2}).$$



Group words together when one is a rotation of another. Such a group is called a rotational *orbit*. How large can an orbit be?

Lemma 5.1 (Prime orbit dichotomy)**.** *For p prime, a length-p word has either 1 or p distinct rotations. It has one exactly when it is constant.*

Proof. If rotation by k positions, with $1 \le k < p$, fixes the word, then entries i and $i + k$ have the same color (indices modulo p). Since $\gcd(k, p) = 1$, repeatedly adding k visits every position. Thus all entries have one color. A nonconstant word has no repeated rotation before the pth, so its orbit has size p. □

There are a constant words, one for each color. The remaining $a^p - a$ words split into orbits of exactly p words.

Theorem 5.2 (Fermat's little theorem, necklace form)**.** *If p is prime, then for every integer a,*

$$a^p \equiv a \pmod{p}.$$

If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. For $a \geq 0$, the orbit count above shows $p \mid a^p - a$. This congruence depends only on the residue of a modulo p, so it holds for every integer a. If $p \nmid a$, then a has an inverse modulo p, and we may cancel it from $a^p \equiv a$. □

Big idea

Divisibility by p appears because symmetry packages nonexceptional objects into groups of p. The exceptions are exactly the colorings fixed by rotation. This pattern – count, isolate fixed points, divide the rest into orbits – reaches far beyond number theory.

5.2 A subtle choice: labeled words or necklaces?

A physical necklace usually ignores where position 0 is. Here we first count *labeled* circular words, then use rotation to partition them. Calling each orbit a necklace is convenient, but the proof depends on keeping those two levels distinct.

Common trap

When the length is composite, an orbit can have an intermediate size. The word $RBRB$ has two distinct rotations, not 1 or 4. To extend the counting proof, we must account for these shorter periods.

5.3 From prime lengths to prime powers

Length 9 gives us a useful next experiment. The word $ABCABCABC$ returns after three rotations, while $AAAAAAAAA$ returns after one. To obtain packets of nine words, we must remove all words with a shorter period. The following elementary fact tells us exactly which periods can occur.

Lemma 5.3 (A rotation period divides the length)**.** *For a circular word of length N, let d be the least positive number of one-step rotations that returns the word to itself. Then $d \mid N$, and the word has exactly d distinct rotations.*

Proof. Write $N = qd + r$ with $0 \leq r < d$. Rotating N times does nothing, and rotating qd times also returns the word. Hence rotating r times returns it, so minimality forces $r = 0$. The first d rotations are distinct: equality at steps $i < j < d$ would give a return after $j - i$ steps. All later rotations repeat these. □

Example 5.4 (Nine positions, one subtraction)**.** Use $a \geq 1$ colors on nine labeled positions. The possible least periods are $1, 3, 9$. A word with period 1 or 3 is precisely three copies of a block of

length 3:

$$\underbrace{c_0c_1c_2}_{\text{choose this block}} \quad \underbrace{c_0c_1c_2}_{\text{repeat}} \, \underbrace{c_0c_1c_2}_{\text{repeat}}.$$

There are a^3 such words, including the constant ones. Thus the remaining $a^9 - a^3$ words have full period 9 and split into rotation orbits of size 9. Consequently

$$9 \mid a^9 - a^3 = a^3(a^6 - 1).$$

If $3 \nmid a$, the factor a^3 is a unit modulo 9, so we may cancel it and conclude $a^6 \equiv 1 \pmod 9$.

For two colors, $2^9 - 2^3 = 512 - 8 = 504 = 9 \cdot 56$: there are 56 full-period necklaces. Cancellation gives $2^6 \equiv 1 \pmod 9$. If desired, certify the cancellation by extended Euclid: $8 \cdot 8 - 7 \cdot 9 = 1$.

Proposition 5.5 (The prime-power necklace congruence)**.** *For every prime p, integer $k \geq 1$, and integer a,*

$$a^{p^k} \equiv a^{p^{k-1}} \pmod{p^k}.$$

Proof. First take $a \geq 1$ colors. Every proper divisor of p^k divides p^{k-1}. Therefore the words of length p^k with a shorter period are exactly p repetitions of a block of length p^{k-1}. The first block determines the whole labeled word uniquely, even when that block itself repeats. There are exactly $a^{p^{k-1}}$ such words.

All remaining $a^{p^k} - a^{p^{k-1}}$ words have full period p^k. Their rotation orbits have size p^k, proving the divisibility. For an arbitrary integer a, choose a positive representative modulo p^k; the congruence depends only on that residue. □

If $p \nmid a$, factor and cancel the invertible factor:

$$a^{p^{k-1}}\left(a^{p^{k-1}(p-1)} - 1\right) \equiv 0 \pmod{p^k} \quad \Longrightarrow \quad a^{p^{k-1}(p-1)} \equiv 1 \pmod{p^k}.$$

For $k = 1$, this is the original necklace proof of Fermat. For larger k, one subtraction still removes every shorter period; no inclusion–exclusion formula is needed.

5.4 Counting units with CRT

Definition 5.6. Euler's *totient function* counts integers coprime to the modulus:

$$\varphi(n) = \#\{r : 1 \leq r \leq n,\ \gcd(r, n) = 1\}.$$

For $n \geq 2$, this is the number of units in $\mathbb{Z}/n\mathbb{Z}$. The definition also gives $\varphi(1) = 1$.

For a prime power p^k, the only excluded integers are the multiples of p. There are p^{k-1} of them between 1 and p^k, so

$$\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1).$$

This is exactly the exponent produced by the necklace argument.

Proposition 5.7 (Multiplicativity from CRT)**.** *If* $\gcd(m, n) = 1$*, then* $\varphi(mn) = \varphi(m)\varphi(n)$.

Proof. For $m, n \geq 2$, CRT gives a bijection

$$\mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \qquad x \longmapsto (x \bmod m, x \bmod n).$$

An integer is coprime to mn exactly when it is coprime to both m and n. Thus this bijection restricts to a bijection between units modulo mn and pairs of units modulo m, n. Counting the pairs proves the formula. If either modulus is 1, use $\varphi(1) = 1$. □

For example, $72 = 8 \cdot 9$, so $\varphi(72) = \varphi(8)\varphi(9) = 4 \cdot 6 = 24$. The coprimality hypothesis matters: $\varphi(9) = 6$, whereas $\varphi(3)\varphi(3) = 4$.

If $n = p_1^{k_1} \cdots p_r^{k_r}$ with distinct primes p_i, repeated use of CRT yields

$$\varphi(n) = \prod_{i=1}^{r} p_i^{k_i - 1}(p_i - 1) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right).$$

The familiar product formula follows from counting units at each prime power. Inclusion–exclusion gives another derivation, but it is not a prerequisite here.

5.5 Euler's theorem: put the prime powers together

Theorem 5.8 (Euler, from necklaces and CRT)**.** *If $n \geq 2$ and $\gcd(a, n) = 1$, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. Factor $n = \prod_i p_i^{k_i}$. For each prime-power factor, the necklace argument and cancellation give

$$a^{\varphi(p_i^{k_i})} \equiv 1 \pmod{p_i^{k_i}}.$$

The totient product formula shows that each $\varphi(p_i^{k_i})$ divides $\varphi(n)$. Raise the corresponding congruence to the integer power $\varphi(n)/\varphi(p_i^{k_i})$. Thus $a^{\varphi(n)}$ is 1 modulo every prime-power factor of n. CRT makes it 1 modulo their product n. □

Example 5.9 (A composite modulus)**.** For $n = 45 = 9 \cdot 5$, the component exponents are $\varphi(9) = 6$ and $\varphi(5) = 4$, and $\varphi(45) = 24$. The congruences $2^6 \equiv 1 \pmod 9$ and $2^4 \equiv 1 \pmod 5$ therefore give $2^{24} \equiv 1 \pmod{45}$. Since $2026 = 24 \cdot 84 + 10$,

$$2^{2026} \equiv 2^{10} = 1024 \equiv 34 \pmod{45}.$$

Check the remainder on the component clocks: $34 \equiv 7 \pmod 9$ and $34 \equiv 4 \pmod 5$.

Common trap

The necklace congruence holds for every a, but Euler's conclusion requires a to be a unit. For example, $3^9 - 3^3$ is divisible by 9, yet $3^6 \equiv 0 \pmod 9$, not 1; the factor 3^3 cannot be cancelled. Also, $\varphi(n)$ is a guaranteed exponent, not necessarily the least one for a particular unit. Modulo 8, $3^2 \equiv 1$ although $\varphi(8) = 4$.

Checkpoint

The proof uses three familiar moves: count full rotation orbits at a prime power, cancel using a modular inverse, and reassemble congruences using CRT. The totient formula records the number of units in the same prime-power decomposition. Chapter 6 will give another proof using finite groups.

5.6 Using Fermat for a prime modulus

To compute $3^{1000} \pmod 7$, note that $3^6 \equiv 1$ and $1000 = 6 \cdot 166 + 4$:

$$3^{1000} \equiv (3^6)^{166} 3^4 \equiv 81 \equiv 4 \pmod 7.$$

5.7 The sum formula over the prime field

There is an immediate consequence of the theorem we have just proved. For $a, b \in \mathbb{F}_p$, Fermat gives

$$(a+b)^p = a + b = a^p + b^p.$$

No binomial expansion is needed for this conclusion: the map $a \mapsto a^p$ fixes *every* element of $\mathbb{F}_p$. For integers a, b, the same statement is the congruence $(a+b)^p \equiv a^p + b^p \pmod p$.

There is also a stronger statement, whose proof will use the prime row of Pascal's triangle: $(X+Y)^p = X^p + Y^p$ as a *polynomial identity* over $\mathbb{F}_p$. Unlike an assertion just about the p residues, that identity can be substituted into any extension field of $\mathbb{F}_p$. The distinction matters because a new element u of an extension need not satisfy $u^p = u$.

Big idea

On $\mathbb{F}_p$, the pth-power map is the identity. Its name, *Frobenius*, becomes informative when there are additional field elements for it to move. We will first justify the polynomial identity in Chapter 7, then construct a finite extension and compute this map in Appendix D.

5.8 Multiplicative functions: work one prime at a time

The identity $\varphi(mn) = \varphi(m)\varphi(n)$ for coprime m, n suggests a useful class of functions. Many counts become products when their prime factors can be chosen independently.

Definition 5.10. An *arithmetic function* assigns a real number $f(n)$ to each positive integer n. It is *multiplicative* if $f(1) = 1$ and

$$f(mn) = f(m)f(n) \qquad \text{whenever } \gcd(m, n) = 1.$$

It is *completely multiplicative* if this equality holds for all positive m, n, without the coprimality condition.

Unique factorization now gives

$$f\left(\prod_i p_i^{k_i}\right) = \prod_i f(p_i^{k_i}).$$

Thus a multiplicative function is determined by its values at prime *powers*. A completely multiplicative function needs only its values at primes, since then $f(p^k) = f(p)^k$. The constant function $\mathbf{1}(n) = 1$ and the identity function $\mathrm{id}(n) = n$ are completely multiplicative. Euler's totient is multiplicative by CRT, but $\varphi(4) = 2 \neq 1 = \varphi(2)^2$.

Two more examples: count divisors and add them

Write $\sum_{d|n}$ for a sum over the positive divisors of n, and set

$$\tau(n) = \sum_{d|n} 1, \qquad \sigma(n) = \sum_{d|n} d.$$

If $n = \prod_i p_i^{k_i}$, each divisor chooses an exponent between 0 and k_i independently at each prime. Counting choices, or summing the resulting products, gives

$$\tau(n) = \prod_i (k_i + 1), \qquad \sigma(n) = \prod_i (1 + p_i + \cdots + p_i^{k_i}).$$

In particular, $\tau(p^k) = k + 1$ and $\sigma(p^k) = (p^{k+1} - 1)/(p - 1)$. Both functions are multiplicative: coprime factors have disjoint prime choices. For $72 = 2^3 \cdot 3^2$,

$$\tau(72) = 4 \cdot 3 = 12, \qquad \sigma(72) = (1 + 2 + 4 + 8)(1 + 3 + 9) = 195.$$

Divisor sums and Dirichlet convolution

Definition 5.11. The *Dirichlet convolution* of arithmetic functions f, g is

$$(f * g)(n) = \sum_{d|n} f(d) g(n/d).$$

Its identity is the function ε given by $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$.

For example, $f * \mathbf{1}$ is the divisor-sum function $n \mapsto \sum_{d|n} f(d)$, so $\tau = \mathbf{1} * \mathbf{1}$ and $\sigma = \mathrm{id} * \mathbf{1}$. Swapping d and n/d proves commutativity. Both ways to form a triple convolution sum $f(a)g(b)h(c)$ over $abc = n$, proving associativity. Only the term $d = n$ survives in $f * \varepsilon$, giving $f(n)$. These are finite sums; no convergence questions enter.

Proposition 5.12. *The convolution of two multiplicative functions is multiplicative. In particular, divisor sums preserve multiplicativity.*

Proof. For coprime m, n, each divisor of mn has a unique form ab with $a \mid m$ and $b \mid n$. The pairs a, b and $m/a, n/b$ are coprime. Therefore

$$\begin{aligned}(f * g)(mn) &= \sum_{a|m} \sum_{b|n} f(a) f(b) g(m/a) g(n/b) \\ &= \left(\sum_{a|m} f(a) g(m/a)\right) \left(\sum_{b|n} f(b) g(n/b)\right).\end{aligned}$$

Also $(f * g)(1) = f(1)g(1) = 1$. □

Common trap

Convolution is not pointwise multiplication. For example, $(\mathbf{1}*\mathbf{1})(6) = 4$, although $\mathbf{1}(6)\mathbf{1}(6) = 1$. And multiplicativity does not let us replace $f(p^k)$ by $f(p)^k$: the coprimality hypothesis is doing real work.

The totient identity to be inverted

Proposition 5.13. *For every positive integer n,*

$$\sum_{d\mid n} \varphi(d) = n, \qquad \text{or equivalently} \qquad \varphi * \mathbf{1} = \mathrm{id}\,.$$

Proof. Partition the integers $1 \le r \le n$ according to $g = \gcd(r, n)$. For each divisor $g \mid n$, put $d = n/g$ and write $r = gs$. Then $1 \le s \le d$ and $\gcd(s, d) = 1$, so this class has $\varphi(d)$ elements. Every r belongs to exactly one class. Adding their sizes counts all n integers, including $r = n$ in the class corresponding to $d = 1$. □

For $n = 12$, this reads $\varphi(1) + \varphi(2) + \varphi(3) + \varphi(4) + \varphi(6) + \varphi(12) = 1 + 1 + 2 + 2 + 2 + 4 = 12$. Can we recover the individual values from their divisor sums?

5.9 Möbius inversion: undo a divisor sum

If $F(n) = \sum_{d\mid n} f(d)$, we can solve successively: $f(1) = F(1)$ and $f(n) = F(n) - \sum_{d\mid n,\, d<n} f(d)$. Möbius inversion packages those repeated subtractions into one formula. The Möbius function introduced here is a different object from the linear fractional transformations in Chapter 3.

Definition 5.14. An integer is *squarefree* if no prime square divides it. The *Möbius function* is

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes}, \\ 0, & \text{some } p^2 \text{ divides } n. \end{cases}$$

For example, $\mu(6) = 1$, $\mu(12) = 0$, and $\mu(30) = -1$. For coprime inputs, square factors remain square factors; if neither input has one, their disjoint lists of prime factors concatenate. This proves that μ is multiplicative. It is not completely multiplicative: $\mu(p^2) = 0$ but $\mu(p)^2 = 1$.

Lemma 5.15 (Cancellation over the divisors)**.** *For every positive integer n,*

$$\sum_{d\mid n} \mu(d) = \varepsilon(n).$$

Proof. For $n = 1$, the sum is $\mu(1) = 1$. If $n > 1$, choose a prime $p \mid n$. Only squarefree divisors contribute. Pair each squarefree divisor not containing p with the divisor obtained by adjoining p. The two terms have opposite signs, and every nonzero term belongs to exactly one pair. The sum is zero. □

Theorem 5.16 (Möbius inversion on divisors)**.** *For arbitrary arithmetic functions f, F,*

$$F(n) = \sum_{d|n} f(d) \quad \text{for every } n \quad \Longleftrightarrow \quad f(n) = \sum_{d|n} \mu(d) F(n/d) \quad \text{for every } n.$$

Equivalently, $f(n) = \sum_{d|n} \mu(n/d) F(d)$. Neither function needs to be multiplicative.

Proof. Substitute the first formula into the proposed inverse and collect the coefficient of each $f(e)$:

$$\begin{aligned} \sum_{d|n} \mu(d) F(n/d) &= \sum_{d|n} \mu(d) \sum_{e|n/d} f(e) \\ &= \sum_{e|n} f(e) \sum_{d|n/e} \mu(d) = f(n). \end{aligned}$$

The inner sum vanishes unless $e = n$. In convolution notation the cancellation lemma says $\mu * \mathbf{1} = \varepsilon$, so the reverse implication follows as well: $f = \mu * F$ gives $\mathbf{1} * f = (\mathbf{1} * \mu) * F = F$. □

Example 5.17 (Recover the totient)**.** Invert $\sum_{d|n} \varphi(d) = n$ to obtain

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d} = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

For the last equality, expand the finite product: choosing a subset of the distinct prime factors produces exactly one squarefree divisor, with its Möbius sign. For $n = 12$, only $d = 1, 2, 3, 6$ contribute:

$$\varphi(12) = 12 - 6 - 4 + 2 = 4.$$

CRT proved this product formula earlier; inversion explains the alternating subtractions behind it.

Return to necklaces of any length

Fix $a \geq 1$ colors. Let $P_a(n)$ count labeled words of length n whose *least* rotational period is n; call them primitive words. Every word of length n has a unique least period $d \mid n$ and is obtained by repeating a unique primitive block of length d. Consequently

$$a^n = \sum_{d|n} P_a(d), \qquad P_a(n) = \sum_{d|n} \mu(d) a^{n/d}.$$

The primitive words have rotation orbits of size n. Thus

$$\frac{1}{n} \sum_{d|n} \mu(d) a^{n/d}$$

is the integer number of primitive necklaces. At $n = p^k$, only $d = 1, p$ survive, recovering $a^{p^k} - a^{p^{k-1}}$ and the Euler proof. For binary words of length 6,

$$P_2(6) = 2^6 - 2^3 - 2^2 + 2 = 54,$$

so there are nine primitive necklaces.

Common trap

This counts necklaces of *full* period. To count all length-n necklaces, include the shorter periods too:

$$N_a(n) = \sum_{d \mid n} \frac{P_a(d)}{d}.$$

Indeed, a primitive block of length d has d distinct rotations even after it is repeated to length n. For $a = 2, n = 6$, the contributions are $2, 1, 2, 9$ for $d = 1, 2, 3, 6$, giving $N_2(6) = 14$.

Checkpoint

Divisor sums record a total after grouping objects by a divisor. Möbius inversion recovers the exact groups. In the totient example the grouping uses a gcd; in the necklace example it uses a least period. The next section explains how the same method works for other ways of ordering the groups. It is bonus material and can be read independently of the later group-theory chapter.

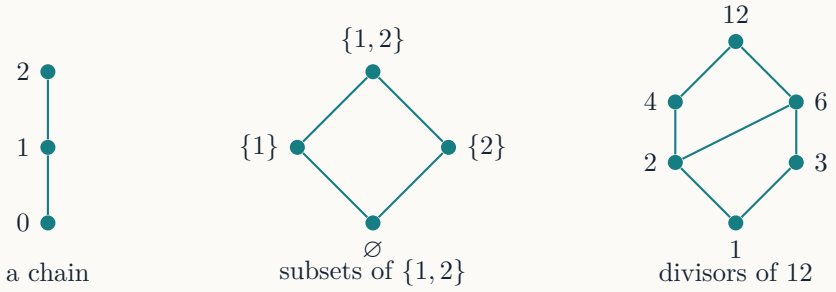
5.10 Bonus: Möbius inversion on partially ordered sets

For ordinary running totals $G(j) = F(0) + \cdots + F(j)$, subtraction recovers $F(j) = G(j) - G(j-1)$ for $j \geq 1$, with $F(0) = G(0)$. Divisor sums have a more complicated inverse because divisibility branches: 2 and 3 both precede 6, but neither precedes the other. We can describe the common operation as adding all contributions below a given object in an order.

5.10.1 Orders, intervals, and diagrams

A *partial order* $\leq$ on a set P is reflexive ($x \leq x$), antisymmetric ($x \leq y$ and $y \leq x$ imply $x = y$), and transitive. The pair is a *partially ordered set*, or *poset*. Two elements need not be comparable. Examples are integers in their usual order, positive divisors of an integer ordered by divisibility, and subsets of a set ordered by inclusion. The symbol $\leq$ below always means the chosen order.

The *interval* $[x, y]$ is $\{z : x \leq z \leq y\}$. In a *Hasse diagram*, larger elements appear higher; join x to y when $x < y$ and there is no element strictly between them. Transitive comparisons are represented by upward paths.



For now assume P is finite. Given exact contributions $F(x)$, define the cumulative values $G(y) = \sum_{x \leq y} F(x)$. We can always recover $F(y) = G(y) - \sum_{x < y} F(x)$, starting with minimal elements. The

inverse therefore exists and is unique. Its coefficients depend only on the order, not on the values being added.

5.10.2 Define the inverse by cancellation

Definition 5.18. For comparable elements of a finite poset, define its *Möbius function* recursively by

$$\mu_P(x,x) = 1, \qquad \mu_P(x,y) = -\sum_{x \le z < y} \mu_P(x,z) \quad (x < y).$$

Equivalently, $\sum_{x \le z \le y} \mu_P(x,z) = \delta_{xy}$, where δ_{xy} is 1 if $x = y$ and 0 otherwise.

Theorem 5.19 (Möbius inversion on a finite poset)**.** *For functions $F, G : P \to \mathbb{R}$,*

$$G(y) = \sum_{x \le y} F(x) \quad \textit{for all } y \quad \Longleftrightarrow \quad F(y) = \sum_{x \le y} \mu_P(x,y) G(x) \quad \textit{for all } y.$$

Proof. Let $H(y)$ denote the proposed inverse on the right. Its cumulative sum is

$$\begin{aligned} \sum_{y \le t} H(y) &= \sum_{y \le t} \sum_{x \le y} \mu_P(x,y) G(x) \\ &= \sum_{x \le t} G(x) \sum_{x \le y \le t} \mu_P(x,y) = G(t). \end{aligned}$$

Thus H has cumulative values G. Successive subtraction gives a unique such function, so $H = F$. This calculation also proves the reverse implication. □

There is an equally useful form for totals *above* an element:

$$G(x) = \sum_{y \ge x} F(y) \quad \Longleftrightarrow \quad F(x) = \sum_{y \ge x} \mu_P(x,y) G(y).$$

To verify the inverse, substitute the upper sum for $G(y)$ and collect the coefficient of $F(z)$; it is $\sum_{x \le y \le z} \mu_P(x,y) = \delta_{xz}$. Successive subtraction from maximal elements proves uniqueness.

5.10.3 Three familiar inversions

A chain gives finite differences. On $0 < 1 < \cdots < N$, the recurrence gives

$$\mu_P(i,j) = \begin{cases} 1, & j = i, \\ -1, & j = i+1, \\ 0, & j \ge i+2. \end{cases}$$

The first two terms cancel in every longer interval, so all later terms are zero. The theorem recovers ordinary subtraction of running totals.

Divisibility gives arithmetic Möbius inversion. If $r \mid s$, the elements of $[r,s]$ correspond to divisors of s/r by $t \mapsto t/r$. The arithmetic cancellation lemma gives

$$\sum_{r \mid t \mid s} \mu(t/r) = \varepsilon(s/r) = \delta_{rs}.$$

Uniqueness in the recursive definition therefore gives $\mu_P(r,s) = \mu(s/r)$. The poset theorem becomes precisely the divisor-sum formula from Section 5.9.

Subset inclusion gives inclusion–exclusion. For subsets $S \subseteq T$ of a fixed finite set,

$$\mu_P(S,T) = (-1)^{|T\setminus S|}.$$

To check the recurrence, sum $(-1)^{|U\setminus S|}$ over $S \subseteq U \subseteq T$. If $S \neq T$, choose an element of $T \setminus S$ and pair subsets by inserting or removing it. The signs cancel; for $S = T$, the sum is 1. Thus

$$G(T) = \sum_{S\subseteq T} F(S) \quad \Longleftrightarrow \quad F(T) = \sum_{S\subseteq T} (-1)^{|T\setminus S|} G(S).$$

To see the usual counting formula, let $A_1, \ldots, A_r$ be subsets of a finite universe U. Let $F(T)$ count elements whose set of memberships is *exactly* T, and put $G(S) = |\bigcap_{i\in S} A_i|$, with $G(\varnothing) = |U|$. Then $G(S) = \sum_{T\supseteq S} F(T)$: an intersection allows additional memberships. Apply the *upper* version of inversion at the empty set:

$$\left| U \setminus \bigcup_{i=1}^{r} A_i \right| = \sum_{S\subseteq\{1,\ldots,r\}} (-1)^{|S|} \left| \bigcap_{i\in S} A_i \right|.$$

Subtract from $|U|$ to obtain the usual alternating formula for the union.

5.10.4 A coefficient larger than one: partitions

A partition divides a set into nonempty disjoint blocks. Order partitions by *refinement*: $\pi \leq \sigma$ when every block of π lies inside a block of σ. For $\{1,2,3\}$ there are five partitions:

$$\widehat{0} = 1|2|3, \qquad 12|3,\ 13|2,\ 23|1, \qquad \widehat{1} = 123.$$

The three middle partitions are incomparable, each above $\widehat{0}$ and below $\widehat{1}$. The recursive coefficients from $\widehat{0}$ are 1 at the bottom, -1 at each middle element, and

$$\mu_P(\widehat{0},\widehat{1}) = -(1-1-1-1) = 2.$$

So poset Möbius coefficients need not belong to $\{-1,0,1\}$.

Color the three labeled positions with a colors. Let $F(\sigma)$ count colorings whose equality classes are exactly the blocks of σ. Let $G(\pi)$ count colorings constant on every block of π, allowing different blocks to receive the same color. Then $G(\pi) = a^{b(\pi)}$, where $b(\pi)$ is its number of blocks, and $G(\pi) = \sum_{\sigma\geq\pi} F(\sigma)$. Upper inversion at $\widehat{0}$ gives the number with all colors distinct:

$$F(\widehat{0}) = a^3 - 3a^2 + 2a = a(a-1)(a-2).$$

The coefficient 2 corrects the three overlapping ways of merging a pair. It is determined by the partition order.

5.10.5 Independent orders multiply the coefficients

The product poset $P \times Q$ has $(x,u) \leq (y,v)$ exactly when $x \leq y$ and $u \leq v$. Its Möbius function factors:

$$\mu_{P\times Q}((x,u),(y,v)) = \mu_P(x,y)\mu_Q(u,v).$$

Indeed, summing the proposed coefficients over the product interval factors as

$$\left(\sum_{x\le z\le y}\mu_P(x,z)\right)\left(\sum_{u\le w\le v}\mu_Q(u,w)\right)=\delta_{xy}\delta_{uv}.$$

This is the defining cancellation rule, so uniqueness proves the claim.

The divisors of $n=\prod_i p_i^{k_i}$ form the product of exponent chains $0<1<\cdots<k_i$. A jump of two or more steps in any chain has coefficient zero. This explains structurally why $\mu(n)$ vanishes when a prime square divides n, and otherwise equals one factor of -1 for each prime. Likewise, the subsets of an r-element set form a product of r two-element chains, explaining the signs in inclusion–exclusion.

5.10.6 The algebra behind the cancellation

For functions A, B on comparable pairs of a finite poset, define

$$(A*B)(x,y)=\sum_{x\le z\le y}A(x,z)B(z,y).$$

These functions, with pointwise addition and this convolution, form the *incidence algebra*. The identity is $\delta(x,y)=\delta_{xy}$; associativity follows by summing over $x\le z\le w\le y$. The *zeta function* is $\zeta(x,y)=1$ for $x\le y$. The recurrence and inversion theorem say

$$\mu_P*\zeta=\zeta*\mu_P=\delta.$$

If the elements are listed so that smaller ones precede larger ones, these are upper triangular matrices: zeta performs cumulative summation and its inverse supplies the Möbius coefficients.

On positive integers ordered by divisibility, the pair function $A_f(r,s)=f(s/r)$ satisfies

$$(A_f*A_g)(r,s)=\sum_{r\mid t\mid s}f(t/r)g(s/t)=(f*g)(s/r).$$

Dirichlet convolution is therefore this incidence convolution for functions depending only on the quotient of the endpoints.

Common trap

An infinite poset is *locally finite* when each interval $[x,y]$ is finite. That suffices to define the incidence algebra and its Möbius function. For sums over *all* $x\le y$, also require finitely many contributing terms, for example finite lower sets or finitely supported data. Local finiteness alone does not ensure this: the usual order on $\mathbb{Z}$ has finite intervals but infinitely many elements below each integer. All examples computed here use finite sums.

Big idea

First identify what "exactly" means and what the cumulative count includes. Then choose the order connecting those two counts. Divisibility leads to arithmetic Möbius inversion, inclusion to inclusion–exclusion, and refinement to partition coefficients. In each case the inverse cancels all unwanted intermediate contributions.

5.11 Bonus: interpolation and Fourier inversion

Möbius inversion recovers exact contributions from cumulative sums. Interpolation recovers a polynomial from its values. Both invert a linear transformation: the observations are known linear combinations of the data we want. Fourier inversion belongs to the same family. At roots of unity, its connection with interpolation is exact.

5.11.1 Evaluate a polynomial on a circle

For this section, work over the complex numbers. Write $i^2 = -1$ and choose a *primitive nth root of unity* ω: its powers $1, \omega, \dots, \omega^{n-1}$ are distinct and $\omega^n = 1$. One choice is $\omega = \cos(2\pi/n) + i\sin(2\pi/n)$. For $n = 4$, we can use $\omega = i$, giving $1, i, -1, -i$.

Let

$$f(X) = \sum_{k=0}^{n-1} c_k X^k, \qquad y_j = f(\omega^j) = \sum_{k=0}^{n-1} c_k \omega^{jk} \quad (0 \le j < n).$$

With this choice of sign, the map $(c_k) \mapsto (y_j)$ is the *discrete Fourier transform*. Its matrix is the Vandermonde matrix $(\omega^{jk})_{j,k}$: each row evaluates the monomials $1, X, \dots, X^{n-1}$ at one root of unity. Some conventions put a minus sign in the forward transform; then the inverse uses the opposite sign as well.

Lemma 5.20 (Orthogonality of roots of unity)**.** *For an integer m,*

$$\sum_{j=0}^{n-1} \omega^{jm} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

Proof. If $n \mid m$, every term is 1. Otherwise $q = \omega^m \neq 1$ and $q^n = 1$, so the geometric sum is $(q^n - 1)/(q - 1) = 0$. □

The functions $j \mapsto \omega^{jk}$ are the *characters* of the cyclic index set: addition of indices becomes multiplication of their values. Orthogonality says that different characters cancel when we sum over a complete cycle.

5.11.2 Lagrange selectors are the inverse Fourier transform

Recall the interpolation selectors from Chapter 4: the polynomial ℓ_j of degree less than n is 1 at the jth sample point and 0 at the others. At roots of unity it has the particularly simple form

$$\ell_j(X) = \frac{1}{n} \sum_{k=0}^{n-1} \omega^{-jk} X^k.$$

Indeed, $\ell_j(\omega^r) = n^{-1} \sum_k \omega^{(r-j)k} = \delta_{jr}$ by orthogonality. These are exactly the Lagrange basis polynomials: the degree bound and their values characterize them uniquely. Interpolation therefore gives

$$f(X) = \sum_{j=0}^{n-1} y_j \ell_j(X) = \sum_{k=0}^{n-1} \left(\frac{1}{n} \sum_{j=0}^{n-1} y_j \omega^{-jk} \right) X^k.$$

Comparing coefficients proves the inversion formula

$$\boxed{c_k = \frac{1}{n}\sum_{j=0}^{n-1} y_j \omega^{-jk}.}$$

Thus interpolation at the roots of unity *is* discrete Fourier inversion. The factor $1/n$ comes from the surviving sum of n ones. The degree restriction matters: values at these points determine a general polynomial only modulo $X^n - 1$.

Example 5.21 (Four samples recover four coefficients)**.** For $f(X) = 1 + 2X + 3X^2 + 4X^3$ and $\omega = i$, the samples are

$$(y_0, y_1, y_2, y_3) = (10, -2 - 2i, -2, -2 + 2i).$$

The inverse formula becomes

$$\begin{aligned} c_0 &= (y_0 + y_1 + y_2 + y_3)/4 = 1, \\ c_1 &= (y_0 - iy_1 - y_2 + iy_3)/4 = 2, \\ c_2 &= (y_0 - y_1 + y_2 - y_3)/4 = 3, \\ c_3 &= (y_0 + iy_1 - y_2 - iy_3)/4 = 4. \end{aligned}$$

Changing the sign of the root in the inverse, as well as dividing by 4, is essential to recover the coefficients in the stated order.

5.11.3 CRT explains multiplication in Fourier coordinates

The factorization $X^n - 1 = \prod_{j=0}^{n-1}(X - \omega^j)$ has distinct linear factors. Polynomial CRT identifies a remainder modulo $X^n - 1$ with its n values at these roots. The selectors ℓ_j reconstruct that remainder, exactly as integer CRT selectors reconstruct residues.

If $f(X) = \sum c_k X^k$ and $g(X) = \sum d_k X^k$, their product modulo $X^n - 1$ has coefficients

$$(c \star d)_r = \sum_{k=0}^{n-1} c_k d_{r-k \bmod n}.$$

This is *cyclic convolution*: exponents wrap around because $X^n = 1$. Evaluation takes a product to the product of its values, so

$$\widehat{c \star d}_j = \widehat{c}_j \, \widehat{d}_j.$$

Fourier coordinates turn cyclic convolution into pointwise multiplication. Dirichlet convolution instead combines factor pairs $ab = n$, while incidence convolution sums over intermediate elements of an interval. Each convolution reflects the structure of its indexing set.

5.11.4 What the three inversions share

In each setting, compose the forward and inverse kernels and all unwanted contributions cancel:

$$\begin{aligned} &\text{Möbius:} && \sum_{x \le z \le y} \mu_P(x, z) = \delta_{xy}, \\ &\text{Interpolation:} && \ell_j(x_r) = \delta_{jr}, \\ &\text{Fourier:} && \frac{1}{n}\sum_{j=0}^{n-1} \omega^{j(r-k)} = \delta_{rk} \quad (0 \le r, k < n). \end{aligned}$$

Möbius coefficients undo accumulation along an order; Lagrange polynomials undo evaluation at distinct points; Fourier coefficients are the explicit interpolation inverse at roots of unity. This is the common principle behind the three constructions: choose coefficients that isolate one contribution and cancel the rest.

Explore

No analysis or limiting process was used in the Fourier argument. The same formulas work over a field containing a primitive nth root of unity, provided n is invertible there. For example, in $\mathbb{F}_5$, 2 has order 4, so the four-point transform uses $1, 2, 4, 3$ and $1/4 = 4$. Notebook 6 checks both this example and the complex four-point example with exact arithmetic. If the required roots are missing, an extension field may supply them; that is a later use of the arithmetic developed in Appendix D.

Exercises

1. List the rotational orbits of binary words of length 3. Identify the constant words. [Fluency]

2. Compute $11^{2026} \pmod{17}$ using Fermat's theorem. [Fluency]

3. In the orbit lemma, prove carefully that $0, k, 2k, \ldots, (p-1)k$ are all distinct modulo p. [Reasoning]

4. Give an orbit-counting proof that $p \mid a^p - a$ using linear strings and cyclic shift, without drawing a circle. [Reasoning]

5. Find every possible orbit size for binary words of length 6, and give an example of each. [Exploration]

6. Use Fermat's theorem directly, without expanding a power, to prove $(x_1 + \cdots + x_m)^p \equiv x_1^p + \cdots + x_m^p \pmod{p}$ for integers x_i. Explain why this argument alone does not apply to new elements of a field extension. [Reasoning]

7. If n is composite, must there be an a with $a^n \not\equiv a \pmod{n}$? Test $n = 341$ and 561. This opens the door to pseudoprimes and Carmichael numbers. [Challenge]

8. Count the binary words of length 9 with least rotational period 1, 3, and 9. How many necklaces have full period 9? [Fluency]

9. With three colors and eight labeled positions, count the words and necklaces of full period 8. Use the count and a modular inverse to deduce $3^4 \equiv 1 \pmod{8}$. [Fluency]

10. Why does subtracting only the constant words fail for length 4? Compare $2^4 - 2$ with $2^4 - 2^2$, and explain which words must be removed. [Reasoning]

11. Compute $\varphi(72)$, $\varphi(100)$, and $\varphi(81)$ using prime powers and CRT. Explain why multiplicativity cannot be applied directly to $9 = 3 \cdot 3$. [Fluency]

12. Use $72 = 8 \cdot 9$ to prove $5^{24} \equiv 1 \pmod{72}$ from the two prime-power congruences. Then compute $5^{2026} \pmod{72}$. [Reasoning]

13. Locate the invalid cancellation in an attempted proof of $3^6 \equiv 1 \pmod{9}$. Separately, show that every odd integer has square 1 modulo 8, even though $\varphi(8) = 4$. [Reasoning]

14. For length 6, show that the words of shorter period are the union of words whose periods divide 2 and those whose periods divide 3. Their intersection consists of constant words. Deduce that $a^6 - a^3 - a^2 + a$ is divisible by 6. Explain why the prime-power case needs only one subtraction. [Exploration]

15. Classify $n \mapsto n^2$, $\mathbf{1}$, φ, τ, and μ as multiplicative or completely multiplicative. For each failure of complete multiplicativity, give a counterexample. [Fluency]

16. Compute $\tau(360)$ and $\sigma(360)$ by prime powers. Explain the independent choices behind each product. [Fluency]

17. Partition $1, \dots, 18$ by gcd with 18 and verify $\sum_{d \mid 18} \varphi(d) = 18$. Which class accounts for $\varphi(1)$? [Reasoning]

18. Tabulate $\mu(1), \dots, \mu(12)$. List the nonzero terms in $\sum_{d \mid 60} \mu(d)$ and in $\varphi(60) = \sum_{d \mid 60} \mu(d)(60/d)$. [Fluency]

19. Find f when $\sum_{d \mid n} f(d)$ equals $\tau(n)$, $\sigma(n)$, or $n + 1$. Why is inversion legitimate in the third case even though $n \mapsto n + 1$ is not multiplicative? [Reasoning]

20. Compute $(\mu * \mu)(4)$ and explain why the inverse of μ under convolution is $\mathbf{1}$ rather than μ. [Reasoning]

21. Count primitive binary words and primitive binary necklaces of length 12. Identify which divisors contribute to the inversion. [Fluency]

22. Starting with $N_a(n) = \sum_{d \mid n} P_a(d)/d$, derive

$$N_a(n) = \frac{1}{n} \sum_{d \mid n} \varphi(d) a^{n/d}.$$

Use the inverted totient identity to collect coefficients. Compute the number of all binary necklaces of length 12. [Challenge]

23. Bonus. On subsets of $\{1, 2\}$, the cumulative values at $\varnothing, \{1\}, \{2\}, \{1, 2\}$ are $2, 5, 7, 16$. Recover the four exact values and check by summing again. [Fluency]

24. Bonus. On the chain $0 < 1 < 2 < 3$, recover exact values from cumulative values $2, 7, 8, 20$. Compute $\mu_P(0, 3)$; why is "alternate signs along the chain" the wrong rule? [Reasoning]

25. Bonus. In a universe of 40 elements, three sets have sizes $20, 15, 12$, pairwise intersections of sizes $8, 5, 4$, and triple intersection of size 2. How many elements lie in their union, and how many in none? Specify the upper cumulative count being inverted. [Fluency]

26. Bonus. Compute the Möbius coefficients from the bottom of the partition poset of $\{1, 2, 3\}$. Use upper inversion to count its injective colorings with four colors, and check by direct choice. [Reasoning]

27. Bonus. View the divisors of 12 as a product of exponent chains. Compute $\mu_P(1, d)$ for every $d \mid 12$. Explain the difference between $\mu_P(1, 6)$ and $\mu_P(1, 12)$. [Reasoning]

28. Bonus. Explain why the usual order on $\mathbb{Z}$ is locally finite but the sum $\sum_{x \le y} 1$ is not finite. State a condition on F that makes every lower sum well defined. [Reasoning]

29. Bonus. Use $\omega = i$ to evaluate $2 - X + X^3$ at the fourth roots of unity, then recover its coefficients by Fourier inversion. Explain why the samples cannot distinguish this polynomial from $2 - X + X^3 + (X^4 - 1)$. [Fluency]

30. Bonus. Prove directly that $\ell_j(X) = n^{-1} \sum_{k=0}^{n-1} \omega^{-jk} X^k$ equals the usual Lagrange product $\prod_{r \neq j} (X - \omega^r)/(\omega^j - \omega^r)$. Use their degree bounds and values. [Reasoning]

31. Bonus. In $\mathbb{F}_5$, verify that 2 has order 4. Evaluate $1 + 2X + 3X^2 + 4X^3$ at $1, 2, 4, 3$, and use the inverse transform to recover its coefficients. Identify the two inverses used in the formula. [Reasoning]

Chapter 6

The reusable machine

A quick introduction to finite groups

Necklace rotations and CRT have proved both Fermat's and Euler's theorems. Group theory isolates the rules shared by those rotations, invertible residues, permutations, and many other systems. It will give a second proof that applies directly to the units modulo any integer.

6.1 One operation, four rules

Definition 6.1. A *group* is a set G with an operation $*$ such that:

1. $a * b \in G$ for all $a, b \in G$ (closure);

2. $(a * b) * c = a * (b * c)$ (associativity);

3. there is an identity e with $e * a = a * e = a$;

4. each a has an inverse a^{-1} with $a * a^{-1} = a^{-1} * a = e$.

If also $a * b = b * a$, the group is *abelian*.

Group	Operation	Identity / inverse
$\mathbb{Z}/n\mathbb{Z}$	addition	0 / $-a$
$(\mathbb{Z}/n\mathbb{Z})^\times$	multiplication of units	1 / modular inverse
rotations of a regular n-gon	composition	no rotation / reverse rotation
permutations of a set	composition	do nothing / undo

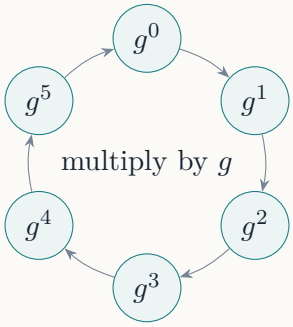
Not every familiar set works: all residues modulo n under multiplication fail because nonunits have no inverse.

6.2 Powers and cyclic subgroups

For $g \in G$, the list

$$e, g, g^2, g^3, \dots$$

eventually repeats when G is finite. The first positive m with $g^m = e$ is the *order* of g, written $\operatorname{ord}(g)$. The powers form the cyclic subgroup $\langle g \rangle$.



A cyclic group of order 6: repeated multiplication traces a loop.

6.3 Cosets and Lagrange's theorem

If H is a subgroup of G and $g \in G$, the left coset

$$gH = \{gh : h \in H\}$$

is a translated copy of H. Two cosets are either equal or disjoint, and every coset has $|H|$ elements because $h \mapsto gh$ is a bijection.

Theorem 6.2 (Lagrange)**.** *If G is finite and $H \leq G$, then $|H|$ divides $|G|$.*

Proof. The distinct cosets of H partition G into equally sized blocks. If there are r blocks, then $|G| = r\,|H|$. □

Corollary 6.3. *For every g in a finite group G, $\operatorname{ord}(g) \mid |G|$ and hence $g^{|G|} = e$.*

6.4 Fermat and Euler through Lagrange

For prime p, every nonzero residue has an inverse, so

$$(\mathbb{Z}/p\mathbb{Z})^\times = \{1, 2, \ldots, p-1\}$$

is a group of order $p - 1$. Lagrange immediately yields the standard group-theoretic proof:

$$a^{p-1} \equiv 1 \pmod{p} \qquad (p \nmid a).$$

The same argument now recovers Euler's theorem from Section 5.5, using the group of units directly.

Theorem 6.4 (Euler)**.** *If $n \geq 2$ and $\gcd(a, n) = 1$, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n},$$

where $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.

Proof. Apply the corollary of Lagrange to the group $(\mathbb{Z}/n\mathbb{Z})^\times$. □

Big idea

The earlier Euler proof counted full rotation orbits at each prime power, then used CRT. Lagrange partitions the units modulo n into cosets of the subgroup generated by a. Both prove divisibility by partitioning a finite set into equal blocks.

6.5 A second standard proof without naming Lagrange

Let $u_1, \dots, u_{\varphi(n)}$ represent the units modulo n. If $\gcd(a, n) = 1$, multiplication by a permutes them: multiplying by a^{-1} reverses the operation. Comparing products gives

$$\prod_{i=1}^{\varphi(n)} u_i \equiv \prod_{i=1}^{\varphi(n)} (au_i) = a^{\varphi(n)} \prod_{i=1}^{\varphi(n)} u_i \pmod{n}.$$

Every factor u_i has an inverse, so their product can be cancelled. This proves Euler's theorem again. For a prime modulus, the product is $(p-1)!$, recovering the familiar permutation proof of Fermat. Commutativity permits us to collect the factors in this product proof; Lagrange's theorem also works in noncommutative groups.

Exercises

1. List $(\mathbb{Z}/12\mathbb{Z})^\times$ and make its multiplication table. Find the order of every element. [Fluency]
2. Find $\varphi(20)$ and use Euler's theorem to reduce 3^{2026} modulo 20. [Fluency]
3. Prove that the identity of a group is unique and that each element has a unique inverse. [Reasoning]
4. Prove that two left cosets of H are equal or disjoint. [Reasoning]
5. Show that a group of prime order is cyclic. [Reasoning]
6. Draw the subgroup lattice of $(\mathbb{Z}/12\mathbb{Z}, +)$. Check Lagrange's theorem at every node. [Exploration]
7. Explain how the CRT bijection on units preserves multiplication and inverses. Recover the formula $\varphi(mn) = \varphi(m)\varphi(n)$ for $\gcd(m, n) = 1$. [Reasoning]
8. Compare the necklace, permutation, and Lagrange proofs of Fermat's theorem. For each, identify the set being partitioned or permuted and the exact place primality enters. [Synthesis]

Chapter 7

Arithmetic in Pascal's triangle

Binomial coefficients and divisibility

Definition 7.1. For $0 \le k \le n$,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

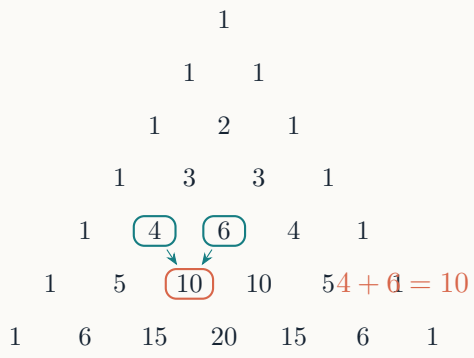
It counts the k-element subsets of an n-element set.

The counting meaning proves integrality without asking the fraction to look integral. It also makes identities visible.

Proposition 7.2 (Pascal and symmetry)**.**

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

For Pascal's identity, distinguish k-subsets that contain a chosen element from those that do not.



7.1 Three ways to see the binomial theorem

Theorem 7.3 (Binomial theorem)**.**

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

In expanding $(x+y)^n$, choose which k of the n factors contribute y. This proof also explains why the coefficients are integers and why their sum is 2^n.

Proposition 7.4 (Vandermonde)**.** *For nonnegative integers r, s, n,*

$$\sum_k \binom{r}{k}\binom{s}{n-k} = \binom{r+s}{n}.$$

Count n-person committees from two groups: either all at once, or according to how many come from the first group.

7.2 Prime rows

Theorem 7.5. *If p is prime and $1 \le k \le p-1$, then*

$$p \mid \binom{p}{k}.$$

Proof. The identity $k\binom{p}{k} = p\binom{p-1}{k-1}$ shows p divides $k\binom{p}{k}$. Since $p \nmid k$, Euclid's lemma gives $p \mid \binom{p}{k}$. □

Thus the pth row of Pascal's triangle becomes $1, 0, \dots, 0, 1$ modulo p, which again yields

$$(x+y)^p \equiv x^p + y^p \pmod{p}.$$

A polynomial identity, not only a statement about values

More precisely, reducing the *coefficients* in the binomial theorem gives the identity

$$(X+Y)^p = X^p + Y^p \qquad \text{in } \mathbb{F}_p[X, Y].$$

If K is a field containing $\mathbb{F}_p$, then adding 1 to itself p times still gives 0 in K; we say K has *characteristic* p. Every intermediate binomial coefficient is therefore zero in K, and substitution gives

$$(a+b)^p = a^p + b^p \qquad \text{for all } a, b \in K.$$

This proof does not assume $a^p = a$ or $b^p = b$. It is the justification we will need for Frobenius on finite extensions.

Common trap

Agreeing at every element of a finite field does not make two polynomials identical. Fermat says that $X^p - X$ has value zero at every element of $\mathbb{F}_p$, but it is a nonzero polynomial of degree p. Thus the prime-field argument in Section 5.7 and the coefficient argument here have different scopes.

Once a finite extension has been constructed, the map $a \mapsto a^p$ preserves addition by this identity and preserves multiplication because $(ab)^p = a^p b^p$. Appendix D proves that it is a bijection and shows it exchanging two roots of the Fibonacci characteristic polynomial. That is where the term *Frobenius automorphism* acquires a concrete role.

7.3 Measuring divisibility one prime at a time

Definition 7.6. For nonzero integer m, $v_p(m)$ is the exponent of p in its prime factorization. Thus $v_2(40) = 3$.

Valuations turn multiplication into addition:

$$v_p(ab) = v_p(a) + v_p(b), \qquad v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b)$$

when the quotient is an integer.

Theorem 7.7 (Legendre)**.**

$$v_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots .$$

Proof. Among $1, \ldots, n$, exactly $\lfloor n/p \rfloor$ contribute at least one p; $\lfloor n/p^2 \rfloor$ contribute one additional p; and so on. □

Consequently,

$$v_p\binom{n}{k} = \sum_{j \geq 1} \left(\left\lfloor \frac{n}{p^j} \right\rfloor - \left\lfloor \frac{k}{p^j} \right\rfloor - \left\lfloor \frac{n-k}{p^j} \right\rfloor \right).$$

Each parenthesis is 0 or 1: it records whether adding k and $n-k$ creates a carry at the p^j place.

Theorem 7.8 (Kummer's carry rule)**.** *$v_p\binom{n}{k}$ equals the number of carries when k and $n-k$ are added in base p.*

The formula above is already the proof once carries are read from right to left. It gives a remarkable visual test: $\binom{n}{k}$ is not divisible by p precisely when subtracting k from n in base p needs no borrowing.

Example 7.9. To find $v_2\binom{10}{3}$, add $3 = 0011_2$ and $7 = 0111_2$:

$$\begin{array}{r} 0011 \\ +0111 \\ \hline 1010 \end{array}$$

There are three carries, so $v_2\binom{10}{3} = 3$. Indeed, $\binom{10}{3} = 120 = 2^3 \cdot 15$.

Explore

Color entry $\binom{n}{k}$ when it is nonzero modulo 2, then modulo 3. Zoom out. The self-similar patterns are the visual shadow of base-p digits and lead to Lucas's theorem.

Exercises

1. Prove $\sum_k \binom{n}{k} = 2^n$ and $\sum_k (-1)^k \binom{n}{k} = 0$ for $n > 0$. [Fluency]

2. Compute $v_2(100!)$, $v_5(100!)$, and the number of trailing zeros of $100!$. [Fluency]

3. Give a counting proof of $k\binom{n}{k} = n\binom{n-1}{k-1}$. [Reasoning]

4. Use Vandermonde's identity to prove $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$. [Reasoning]
5. Determine all n for which every interior entry of row n is even. Use Kummer's rule. [Reasoning]
6. For a fixed prime p, explore the first p^2 rows of Pascal's triangle modulo p. State a digit-based conjecture for when $\binom{n}{k} \not\equiv 0 \pmod{p}$. [Exploration]
7. Prove that $\operatorname{lcm}(1, 2, \ldots, 2n)$ is divisible by $\binom{2n}{n}$ by comparing p-adic valuations. [Bridge]
8. Compare the polynomials X^p and X over $\mathbb{F}_p$. Explain why they give the same function on $\mathbb{F}_p$ but are different polynomials. Why does the coefficient proof, rather than equality of these functions, justify substituting extension-field elements into $(X+Y)^p = X^p + Y^p$? [Reasoning]

Chapter 8

How crowded are the primes?

Chebyshev's estimate from the middle of Pascal's triangle

Euclid proved that the primes never end. A sharper question asks how many lie below x:

$$\pi(x) = \#\{p \le x : p \text{ prime}\}.$$

The prime number theorem says $\pi(x) \sim x/\log x$, but its proof lies beyond this course. Chebyshev's elementary argument already captures the correct scale.

8.1 The central binomial squeeze

The largest coefficient in $(1+1)^{2n}$ is central. Since there are $2n+1$ nonnegative coefficients with sum 4^n,

$$\frac{4^n}{2n+1} \le \binom{2n}{n} < 4^n. \tag{8.1}$$

This elementary inequality contains surprising information about primes.

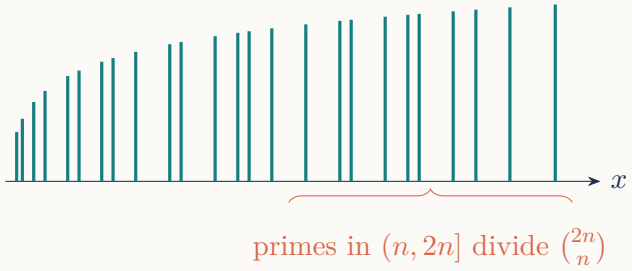
Define two weighted prime counts:

$$\vartheta(x) = \sum_{p \le x} \log p, \qquad \psi(x) = \sum_{p^k \le x} \log p.$$

The first logs the product of primes at most x; the second logs the least common multiple:

$$\psi(x) = \log \operatorname{lcm}(1, 2, \dots, \lfloor x \rfloor).$$

A logarithmic weight turns products into sums.



8.2 An upper bound for the prime product

Every prime $n < p \le 2n$ occurs in the numerator $(n+1)(n+2)\cdots(2n)$ and not in $n!$. Therefore it divides $\binom{2n}{n}$, and

$$\vartheta(2n) - \vartheta(n) \le \log \binom{2n}{n} < 2n \log 2.$$

Adding these inequalities over dyadic intervals gives the following deliberately non-optimized bound.

Lemma 8.1. *For $x \ge 2$,*

$$\vartheta(x) \le 2(\log 2)x.$$

Proof. Split $(1, x]$ into $(x/2, x], (x/4, x/2], \ldots$ and apply the preceding estimate, with integer endpoints rounded outward. The resulting geometric series is bounded by

$$x \log 2 + \frac{x}{2} \log 2 + \frac{x}{4} \log 2 + \cdots = 2x \log 2.$$

Endpoint rounding can only add primes already included in the neighboring interval. □

8.3 A lower bound from an lcm

The bridge exercise in the last chapter showed

$$\binom{2n}{n} \mid \operatorname{lcm}(1, 2, \ldots, 2n).$$

Taking logarithms and using (8.1),

$$\psi(2n) \ge \log \binom{2n}{n} \ge 2n \log 2 - \log(2n+1). \tag{8.2}$$

Prime powers make ψ larger than ϑ, but not by too much:

$$\psi(x) = \vartheta(x) + \vartheta(x^{1/2}) + \vartheta(x^{1/3}) + \cdots .$$

By Lemma 8.1, all terms after the first contribute at most

$$2\sqrt{x} \log x$$

for $x \ge 4$: there are at most $\log_2 x$ terms, and every argument is at most $\sqrt{x}$. Combining this with (8.2) shows that, once x is sufficiently large,

$$\vartheta(x) \ge \frac{\log 2}{2} x. \tag{8.3}$$

The constant $1/2$ is chosen for convenience; better bookkeeping improves it.

8.4 From weighted primes to ordinary counting

The lower bound is immediate because every term in $\vartheta(x)$ is at most $\log x$:

$$\vartheta(x) \le \pi(x)\log x \quad \Longrightarrow \quad \pi(x) \ge \frac{\log 2}{2}\frac{x}{\log x}.$$

For the upper bound, primes between $\sqrt{x}$ and x each contribute at least $\frac{1}{2}\log x$:

$$\vartheta(x) \ge (\pi(x) - \pi(\sqrt{x}))\frac{\log x}{2}.$$

Using $\pi(\sqrt{x}) \le \sqrt{x}$ and Lemma 8.1 gives

$$\pi(x) \le 4\log 2\,\frac{x}{\log x} + \sqrt{x} \le 4\frac{x}{\log x}$$

for all sufficiently large x.

Theorem 8.2 (Chebyshev estimate)**.** *For all sufficiently large x,*

$$\boxed{\frac{\log 2}{2}\frac{x}{\log x} \le \pi(x) \le 4\frac{x}{\log x}}$$

In particular, $\pi(x) \asymp x/\log x$: their ratio stays between two positive constants.

Common trap

The symbol $\asymp$ does not mean $\sim$. Chebyshev's estimate gets the correct order of magnitude. The prime number theorem makes the much stronger claim

$$\lim_{x\to\infty} \frac{\pi(x)}{x/\log x} = 1.$$

Big idea

Why did binomial coefficients see primes? Factorials contain every integer up to a boundary. Dividing $(2n)!$ by two copies of $n!$ cancels most small-scale information while exposing primes that cross the boundary from n to $2n$. The central coefficient is both arithmetically rich and analytically easy to estimate.

Exercises

1. Compute $\pi(10^k)$ and $10^k/\log(10^k)$ for $k = 1, 2, 3, 4$. Track the ratio. [Fluency]
2. Prove both sides of the central binomial squeeze (8.1). [Reasoning]
3. Prove that every prime p with $n < p \le 2n$ divides $\binom{2n}{n}$ exactly once. [Reasoning]
4. Fill in the valuation proof that $\binom{2n}{n}$ divides $\operatorname{lcm}(1, \dots, 2n)$. [Reasoning]
5. Starting from the dyadic inequality, write a fully detailed proof of Lemma 8.1, including floors and ceilings. [Challenge]

6. Use a computer to graph $\pi(x)$, $x/\log x$, and the two Chebyshev bounds. Which statement is visible at moderate x, and which remains too coarse? [Exploration]

7. Show that Chebyshev's estimate implies the nth prime p_n has order of magnitude $n \log n$. [Synthesis]

Chapter 9

Capstone studio

Choose a thread, extend it, and explain it

The capstone is not a miniature research paper. It is a carefully explained mathematical journey: a motivating question, small examples, one substantial proof, and a clear account of how course ideas connect.

9.1 Project menu

A. Rational approximation and Pell equations

Choose a nonsquare d. Compute the continued fraction of $\sqrt{d}$, identify convergents that solve $x^2 - dy^2 = \pm 1$, and prove at least one recurring pattern. Include a number-line or lattice illustration.

B. Symmetry, orbits, and divisibility

Generalize necklace counting to length n. Classify possible orbit sizes for one composite n, explain stabilizers in concrete language, and derive a divisibility identity. Section 5.9 supplies a formula for every length. As a bonus extension, choose another poset from Section 5.10 and explain what its inverse counts.

C. Pascal's triangle modulo a prime

Visualize Pascal's triangle modulo p. Develop and prove a digit rule (Lucas's theorem is the natural destination), then explain how Kummer's carry rule and the visual pattern encode the same arithmetic.

D. Prime-counting laboratory

Compare $\pi(x)$ with $x/\log x$, $\mathrm{Li}(x)$, and the Chebyshev bounds over a range your computer can handle. Explain which plots are evidence and which claims are proved. Reconstruct one link in Chebyshev's proof in full detail.

9.2 Required deliverables

1. A four-to-six page mathematical exposition.
2. At least one original diagram or data visualization.
3. One theorem stated precisely and proved completely.
4. A one-page "proof map" naming the definitions and earlier results used.
5. A five-minute oral explanation, followed by questions.

9.3 Evaluation rubric

Criterion	Evidence of strong work	Weight
Mathematical correctness	Claims are true, hypotheses are visible, examples are checked, and the main proof closes every logical gap.	35%
Exposition	The question motivates the definitions; notation is stable; paragraphs explain why steps matter.	25%
Connection	At least two course strands are linked in a mathematically meaningful way.	15%
Visual or computational insight	A diagram or computation reveals structure and is interpreted rather than merely displayed.	15%
Revision and response	The final version addresses peer/instructor feedback and the oral defense shows ownership.	10%

Checkpoint

Before submitting, underline every sentence that makes a mathematical claim. For each underline, ask: Is it a definition, an observed example, a cited theorem, or a claim I have proved? Ambiguous status is often the real source of unclear exposition.

9.4 A final synthesis prompt

In 500–700 words, respond to this statement:

> Number theory repeatedly converts multiplication into something easier to organize: remainders, exponent vectors, additive logarithms, or equal-sized orbits.

Use one precise example from three different chapters. Your response should include at least one equation and explain not only what the conversion does, but why it helps a proof.

Appendix A

Two-by-two matrices and Möbius maps

A compact toolkit for continued fractions

This appendix develops only the matrix algebra used in Chapter 3. The central idea is simple: a matrix with determinant ± 1 can be reversed without ever leaving the integers.

A.1 Multiplication and determinants

A 2×2 matrix acts on a column vector by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

Matrix multiplication is defined so that composition works as expected: applying B and then A is the same as applying AB. Explicitly,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

The determinant of $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is $\det A = ad - bc$. It measures invertibility:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad \text{provided } ad - bc \neq 0.$$

The identity $\det(AB) = \det(A)\det(B)$ makes determinants especially useful for long products.

Definition A.1. The group $\mathrm{GL}_2(\mathbb{Z})$ consists of the integer matrices with determinant ± 1. Its subgroup $\mathrm{SL}_2(\mathbb{Z})$ consists of those with determinant $+1$.

Proposition A.2 (Integer reversibility)**.** *An integer 2×2 matrix has an inverse with integer entries if and only if its determinant is ± 1.*

Proof. If $\det A = \pm 1$, the displayed inverse formula has integer entries. Conversely, if both A and A^{-1} have integer entries, then $1 = \det(I) = \det(A)\det(A^{-1})$ is a product of two integers. Hence $\det(A) = \pm 1$. □

Big idea

The columns of a matrix in $\mathrm{GL}_2(\mathbb{Z})$ form an integer basis of $\mathbb{Z}^2$: every integer vector has a unique expression as an integer linear combination of those columns. This is the exact algebraic strength supplied by the determinant identity for consecutive convergents.

A.2 Matrices as fractional linear maps

Associate to an invertible matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, with $ad - bc \neq 0$, the Möbius map

$$T_A(t) = \frac{at + b}{ct + d}.$$

A direct substitution shows

$$T_A(T_B(t)) = T_{AB}(t).$$

Thus nested fractional expressions become matrix products. Multiplying A by any nonzero scalar leaves T_A unchanged, but retaining the specific integer matrix remembers determinant and divisibility information that the function alone would forget.

The inverse map is $T_{A^{-1}}$, so a change of variable transports fixed points by conjugation:

$$x = T_N(x), \quad y = T_M(x) \quad \Longrightarrow \quad y = T_{MNM^{-1}}(y).$$

Section 3.7 uses this identity to prove that adding a finite prefix to a periodic continued fraction preserves quadratic irrationality. Section 3.9 then proves the precise criterion for pure periodicity.

When $ct + d \neq 0$, the quotient rule gives

$$T_A'(t) = \frac{ad - bc}{(ct + d)^2}.$$

Therefore the determinant tells us whether the map is increasing or decreasing. This is the Möbius derivative trick used to locate alternating convergents.

A.3 Continued fractions as iterated products

One partial quotient corresponds to

$$D(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}, \qquad T_{D(a)}(t) = a + \frac{1}{t}.$$

Consequently,

$$T_{D(a_0)\cdots D(a_k)}(t) = [a_0; a_1, \ldots, a_k, t].$$

The convergent recurrences are equivalent to the product identity

$$D(a_0) \cdots D(a_k) = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix}.$$

Since every $D(a_i)$ has determinant -1, the product has determinant $(-1)^{k+1}$. Hence

$$p_k q_{k-1} - p_{k-1} q_k = (-1)^{k+1}.$$

This one product packages four facts:

- its first column is the current convergent (p_k, q_k);
- its second column is the preceding convergent (p_{k-1}, q_{k-1});
- its determinant says the two columns form an integer basis;
- its Möbius action inserts an arbitrary remaining tail.

Example A.3. For $[4; 2, 6, 7] = 415/93$,

$$D(4)D(2)D(6)D(7) = \begin{pmatrix} 415 & 58 \\ 93 & 13 \end{pmatrix}.$$

Its inverse is

$$\begin{pmatrix} 13 & -58 \\ -93 & 415 \end{pmatrix}.$$

The top row applied to the first column gives $13(415) - 58(93) = 1$. In the opposite direction, multiplying a vector by the inverse recovers its integer coordinates in the basis of consecutive convergents.

Exercises

1. Multiply three general matrices $D(a_0)D(a_1)D(a_2)$ and identify the two convergents in its columns. [Fluency]
2. Prove $\det(AB) = \det(A)\det(B)$ directly for 2×2 matrices. [Reasoning]
3. Find the inverse of $\left(\begin{smallmatrix} 17 & 7 \\ 12 & 5 \end{smallmatrix}\right)$ and use it to express $(1, 0)$ and $(0, 1)$ as integer combinations of its columns. [Fluency]
4. Verify directly that $T_A \circ T_B = T_{AB}$. Explain why multiplying a matrix by -1 does not change its Möbius map. [Reasoning]
5. Let $P = D(a_1) \cdots D(a_\ell)$. Show that a purely periodic tail $t = [\overline{a_1, \ldots, a_\ell}]$ is a fixed point of T_P, and derive its quadratic equation from the four entries of P. [Reasoning]

Appendix B

Streaming arithmetic with continued fractions

Gosper's homographic and bihomographic algorithms

Suppose a number is arriving one continued-fraction digit at a time. Can we transform it without first converting it to a decimal, and without waiting for the entire infinite input? Bill Gosper's answer is yes. The method reads just enough of the input to prove the next output digit, emits that digit, and then continues.

B.1 The problem—before the machinery

There are two versions.

One input. Given

$$x = [a_0; a_1, a_2, \dots]$$

as a stream, and a linear fractional transformation

$$L(x) = \frac{\alpha x + \beta}{\gamma x + \delta},$$

produce the continued fraction of $L(x)$ from left to right.

Two inputs. Given continued-fraction streams for x and y, produce the continued fraction of a two-variable fractional expression. In particular, $x + y$, $x - y$, xy, and x/y all have the form

$$\Phi(x, y) = \frac{axy + bx + cy + d}{exy + fx + gy + h}.$$

The intended computation is *exact*. It should never guess a digit from a floating-point approximation. Instead, every emitted digit comes with a proof that all inputs compatible with the digits seen so far would produce that same digit.

Big idea

A finite input prefix determines an *interval* of possible values. Map that interval through the desired transformation. If the whole image fits inside one continued-fraction digit cell $[k, k+1)$, the next output digit is certified to be k. If it crosses an integer boundary, read more input and shrink the interval.

The original algorithm appeared as Item 101B of the 1972 MIT memo *HAKMEM*. A modern refinement by Michael J. Collins allows exact intervals themselves to appear in the output stream, resolving boundary cases in which no next digit can be certified after finite time.

B.2 What the unread tail means

The subscript on a tail records the *next* digit, not the first digit of the original number. After the prefix $a_0, \ldots, a_{j-1}$ has been read, write

$$x = [a_0; a_1, \ldots, a_{j-1}, t_j], \qquad t_j = [a_j; a_{j+1}, \ldots]. \tag{B.1}$$

Here t_j is the complete unread remainder of the continued fraction, used as the last entry in the first expression. Before a_j is known, the only universal information is

$$1 \le t_j \le \infty$$

for a positive simple continued fraction. When the next digit $a_j = m$ arrives,

$$t_j = m + \frac{1}{t_{j+1}}, \qquad t_{j+1} = [a_{j+1}; a_{j+2}, \ldots] \in [1, \infty]. \tag{B.2}$$

Thus knowing the single digit m restricts the old tail to $t_j \in (m, m+1]$; absorbing m replaces it by the new unknown tail $t_{j+1} \in [1, \infty]$.

Known prefix versus unknown tail

At the start, $t_0 = x$ and no input digits have been absorbed. After a_0 is read, the state incorporates a_0 and the unknown variable is $t_1 = [a_1; a_2, \ldots]$. After a_1 is read, the state incorporates a_0, a_1 and the unknown variable is t_2, and so on. The state changes; the exact number represented does not.

Let

$$D(m) = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}, \qquad M_0 = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}.$$

After j input digits have been absorbed, set

$$M_j = M_0 D(a_0) D(a_1) \cdots D(a_{j-1}) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \tag{B.3}$$

Then the output value still waiting to be written is exactly

$$F_{M_j}(t_j) = \frac{At_j + B}{Ct_j + D}.$$

This is the invariant behind the algorithm.

B.3 The geometry: map an interval, then inspect its floor

For a general certified tail interval $I = [b_0, b_1]$, consider

$$F_M(t) = \frac{At + B}{Ct + D}.$$

Provided $Ct + D$ never vanishes on I, the derivative

$$F'_M(t) = \frac{AD - BC}{(Ct + D)^2} \tag{B.4}$$

has constant sign. Hence F_M is monotone, and the image $F_M(I)$ is simply the interval whose endpoints are $F_M(b_0)$ and $F_M(b_1)$, in the appropriate order. No point between them can escape those endpoint bounds.



If both endpoint images lie in the same floor cell and there is no pole between them, every possible output has floor k.

For the universal interval $[1, \infty]$, the two endpoint values are

$$F_M(1) = \frac{A + B}{C + D}, \qquad F_M(\infty) = \frac{A}{C}.$$

An endpoint at infinity may be open, and a finite continued fraction may end; exact implementations retain this endpoint information. The essential test is unchanged:

$$F_M(I) \subseteq [k, k+1) quad \Longrightarrow \quad \text{the next output digit is } k.$$

If the image crosses an integer boundary, or if a pole is possible, no digit is yet justified. We must refine the input interval.

B.4 The two algebraic moves

Absorb the next input digit. If $a_j = m$, substitute $t_j = m + 1/t_{j+1}$ into the current map. Clearing the factor t_{j+1} gives

$$F_M\left(m + \frac{1}{t_{j+1}}\right) = \frac{(Am + B)t_{j+1} + A}{(Cm + D)t_{j+1} + C}.$$

Thus the exact state update is

$$\boxed{\mathcal{I}_m(M) = MD(m) = \begin{pmatrix} Am + B & A \\ Cm + D & C \end{pmatrix}.} \tag{B.5}$$

Emit a certified output digit. If the image interval lies in $[k, k+1)$, write the digit k. The unread part of the output continued fraction is then $1/(F_M - k)$, represented by

$$\boxed{\mathcal{E}_k(M) = \begin{pmatrix} 0 & 1 \\ 1 & -k \end{pmatrix} M = \begin{pmatrix} C & D \\ A - kC & B - kD \end{pmatrix}.} \tag{B.6}$$

The homographic loop

Maintain the statement: "the unread output is $F_M(t_j)$ for every $t_j \in I$."

1. Map the current certified interval I through F_M by evaluating its endpoints.
2. If the image lies in one floor cell $[k, k+1)$, emit k, apply $M \leftarrow \mathcal{E}_k(M)$, and replace the output interval by $1/(F_M(I) - k)$.
3. Otherwise read $a_j = m$, use $t_j = m + 1/t_{j+1}$, apply $M \leftarrow \mathcal{I}_m(M)$, and return to the universal interval $t_{j+1} \in [1, \infty]$.
4. Repeat. A common nonzero factor may be removed from the four matrix entries without changing the map.

B.5 Worked stream: the first digits of $\pi/2$

Here $L(x) = x/2$, so $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, and the input is

$$\pi = [3; 7, 15, 1, 292, \dots].$$

After reading $a_0 = 3$ and $a_1 = 7$, the unknown variable is $t_2 = [15; 1, 292, \dots] \in [1, \infty]$, and

$$M_2 = M_0 D(3) D(7) = \begin{pmatrix} 22 & 3 \\ 14 & 2 \end{pmatrix}.$$

The first row below maps the endpoint tails 1 and ∞. Each following row applies (B.6); no further input digit is read.

event	current matrix M	image of the possible tail interval
read 3, 7	$\left(\begin{smallmatrix} 22 & 3 \\ 14 & 2 \end{smallmatrix}\right)$	[25/16, 11/7): emit 1
emit 1	$\left(\begin{smallmatrix} 14 & 2 \\ 8 & 1 \end{smallmatrix}\right)$	(7/4, 16/9]: emit 1
emit 1	$\left(\begin{smallmatrix} 8 & 1 \\ 6 & 1 \end{smallmatrix}\right)$	[9/7, 4/3): emit 1
emit 1	$\left(\begin{smallmatrix} 6 & 1 \\ 2 & 0 \end{smallmatrix}\right)$	(3, 7/2]: emit 3

After these four emissions the state is $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$: the unread output is $2t_2$. Only now do we read the next input digits, 15 and 1; absorbing them maps the new unknown tail into $[31, 32)$ and certifies 31. Therefore

$$\frac{\pi}{2} = [1; 1, 1, 3, 31, \dots].$$

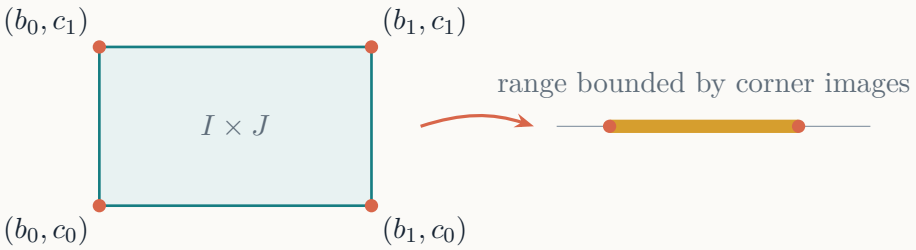
The calculation has used exact rational endpoints throughout. The decimal digits of π never entered.

B.6 Two inputs: rectangles replace intervals

Now suppose prefixes of two inputs have been read. Their unread tails, say u_j and v_ℓ, range independently over certified intervals I and J. The unread output has the form

$$B_{\mathcal{B}}(u,v) = \frac{auv + bu + cv + d}{euv + fu + gv + h}, \qquad \mathcal{B} = (a,b,c,d;\, e,f,g,h). \tag{B.7}$$

Thus the possible input is a rectangle $I \times J$, not a line segment.



Assume the denominator has no zero on the rectangle. For a fixed v, the map is linear fractional in u, so its extrema along that horizontal segment occur at the two ends. On either vertical edge it is linear fractional in v, so the remaining extrema occur at the edge endpoints. Consequently the entire range is bounded by the images of the four corners. In particular, for $I = J = [1, \infty]$ these values are

$$\frac{a}{e}, \qquad \frac{a+b}{e+f}, \qquad \frac{a+c}{e+g}, \qquad \frac{a+b+c+d}{e+f+g+h}, \tag{B.8}$$

with the usual interpretation of infinite and open endpoints.

The four basic operations differ only in their starting coefficients:

desired output	$(a,b,c,d;\, e,f,g,h)$
$x+y$	$(0,1,1,0;\, 0,0,0,1)$
$x-y$	$(0,1,-1,0;\, 0,0,0,1)$
xy	$(1,0,0,0;\, 0,0,0,1)$
x/y	$(0,1,0,0;\, 0,0,1,0)$

If the next u-digit is m, substitute $u = m + 1/u'$; if the next v-digit is n, substitute $v = n + 1/v'$. Collecting coefficients gives

$$\mathcal{I}^u_m(\mathcal{B}) = (ma + c,\ mb + d,\ a,\ b;\ me + g,\ mf + h,\ e,\ f), \tag{B.9}$$

$$\mathcal{I}^v_n(\mathcal{B}) = (na + b,\ a,\ nc + d,\ c;\ ne + f,\ e,\ ng + h,\ g). \tag{B.10}$$

When the four corner images all lie in one floor cell $[k, k+1)$, emit k and invert the fractional part:

$$\mathcal{E}_k(\mathcal{B}) = (e,f,g,h;\ a - ke, b - kf, c - kg, d - kh). \tag{B.11}$$

The geometry and the logic are exactly as before. If the image of the current rectangle crosses an integer boundary, absorb from one input and replace the rectangle by a thinner one; alternate inputs, or choose the one expected to shrink the image more. If the denominator may vanish inside the rectangle, no output is safe until further input or separate algebra resolves the issue.

Big idea

The four coefficients in the one-input problem, or eight in the two-input problem, are bookkeeping for a moving interval image. Matrix formulas make the updates fast, but the proof of correctness is geometric: every possible tail lies in the current interval or rectangle, and every emitted digit is shared by its entire image.

B.7 The boundary problem and the interval repair

The floor function jumps at every integer. This creates a subtle failure mode. Feed two *independent* copies of the continued fraction of $\sqrt{2}$ to the multiplication algorithm. The actual product is 2, but every finite pair of independent enclosures allows one copy to move a little below $\sqrt{2}$ and the other a little above it. The resulting product rectangle can straddle 2. Its image then meets both floor cells $[1, 2)$ and $[2, 3)$, so the rule cannot certify even the first digit.

This is an information problem, not numerical instability. The two stream variables have forgotten that they denote the same algebraic number. Retaining the relation $x = y$ reduces the expression to x^2, and the separate fact $x^2 = 2$ resolves the equality immediately.

Collins's interval-valued refinement distinguishes a continued fraction from its computational stream. A stream item may be either

- an explicit partial quotient, or
- an exact rational interval bounding the current unread value.

If no digit is forced, the algorithm may output a tighter certified interval instead of waiting silently. A consumer asks for an error tolerance ε and reads until the resulting value interval has width below ε. Exact endpoints and their open-or-closed status preserve the no-roundoff character of Gosper's method while guaranteeing useful progress at an integer boundary.

Digits and numerical accuracy are different requests

There may be no finite proof of the next continued-fraction digit even when a very accurate numerical enclosure is easy. A robust implementation therefore asks for a certified error tolerance, not an unconditional number of output digits. Away from boundaries, intervals quickly collapse into ordinary partial quotients and recover Gosper's original stream.

B.8 Exercises

Exercises

1. Suppose only the prefix $[2; 3]$ of x is known. Write $x = [2; 3, t_2]$ with $t_2 \in [1, \infty]$, evaluate the two endpoint values, and prove that $x \in [9/4, 7/3)$. [Fluency]

2. Prove the endpoint-image rule using (B.4). Apply it to $F(t) = (22t + 3)/(14t + 2)$ on $[1, \infty]$ and certify its floor. [Reasoning]

3. Derive the absorption and emission updates (B.5) and (B.6) directly by substitution. Express each as matrix multiplication. [Fluency]

4. Continue the $\pi/2$ calculation: from $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, absorb 15 and 1 and show that the new image lies in $[31, 32)$. [Reasoning]

5. Prove the four-corner range rule for a bihomographic map on a finite rectangle with no denominator zero. Why is checking four corners not enough if a pole lies inside? [Reasoning]

6. Derive (B.9) and (B.10). Verify that the four initial states really represent $+$, $-$, $\times$, and $/$. [Fluency]

7. Explain why independent intervals for two copies of $\sqrt{2}$ can straddle the boundary 2 forever when computing $\sqrt{2}\sqrt{2}$. State the extra algebraic information that resolves the equality. [Reasoning]

8. Design a lazy implementation whose output type is either `Digit(k)` or `Bound(lo,hi)`. State invariants that make every emitted item a certificate, then test all four operations on rational and quadratic inputs. [Project]

Appendix C

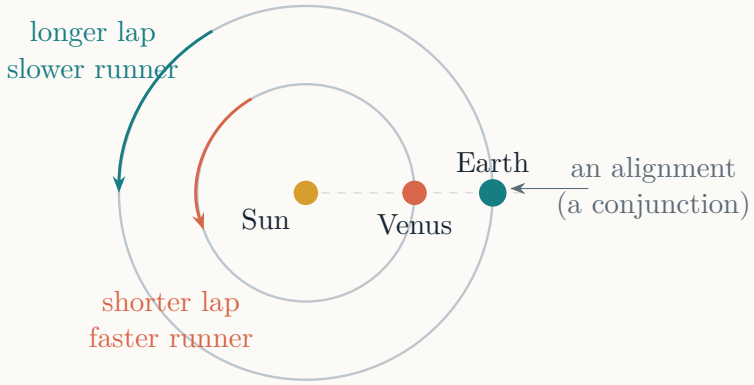
Venus writes Euclid in the sky

Three clocks, five alignments, and the Dresden Codex

Continued fractions are useful whenever two repeating motions almost return to their starting arrangement at the same time. Venus gives a particularly visible example. Small Fibonacci numbers describe a near-repeat of the modern planetary motions, while related integers organize an exact calendar table in the Dresden Codex. We will build the astronomical vocabulary from scratch and keep the approximate observation separate from the exact calendar arithmetic.

C.1 Begin with two runners on circular tracks

Imagine Earth and Venus as two runners circling the Sun. Venus uses the inner track and completes a lap sooner, so it repeatedly catches and passes Earth. At the instant shown below, the Sun, Venus, and Earth lie on the same ray.



Here are the three time intervals we need.

Symbol	Everyday meaning	Astronomical name and approximate value
E	time for Earth to make one Sun-centered lap	Earth's *sidereal period*: 365.256 days
V	time for Venus to make one Sun-centered lap	Venus's *sidereal period*: 224.7 days
S	time from one Earth–Venus alignment to the next	Venus's *synodic period*: 583.92 days

"Sidereal" means that a full lap is measured relative to the distant stars. A "conjunction" is an alignment; in the alignment drawn above, Venus lies between Earth and the Sun, so astronomers call it an *inferior conjunction*. "Synodic period" simply means the waiting time until the same kind of alignment occurs again.[1]

C.2 The overtaking clock

Earth completes $1/E$ laps per day, while Venus completes $1/V$. Because Venus is faster, it gains on Earth at the rate

$$\frac{1}{V} - \frac{1}{E} \quad \text{laps per day.}$$

The next alignment occurs when Venus has gained exactly one whole lap. Hence

$$\frac{1}{S} = \frac{1}{V} - \frac{1}{E}. \tag{C.1}$$

This is the same "lapping time" calculation used for two runners on a track.

Put $r = E/V$: it compares the length of an Earth lap in time with a Venus lap. Multiplying (C.1) by E gives

$$\frac{S}{E} = \frac{1}{r-1} =: T(r). \tag{C.2}$$

The measured ratio $E/V \approx 1.6255$ has the simple, very accurate fraction $13/8$ nearby. If we replace the measured ratio temporarily by this ideal one, then

$$T\left(\frac{13}{8}\right) = \frac{8}{5}.$$

In plain language:

- during about eight Earth laps, Venus makes about thirteen laps; and
- in that time Venus gains $13 - 8 = 5$ laps on Earth, producing about five repeat alignments.

None of these counts is exact for the physical planets. Their usefulness is that $13/8$ is an excellent approximation with a tiny denominator.

[1] For the modern mean values, see NASA, *2012 Transit of Venus* and NASA SP–461, *Pioneer Venus*. The real orbits are neither perfectly circular nor perfectly uniform; rounded mean periods are all we need here.

Why the golden ratio appears

The positive fixed point of the overtaking map $T(r) = 1/(r-1)$ satisfies

$$r = \frac{1}{r-1}, \qquad r^2 - r - 1 = 0,$$

so it is $\varphi = (1+\sqrt{5})/2$. More generally,

$$T\left(\frac{F_{n+1}}{F_n}\right) = \frac{F_n}{F_{n-1}}.$$

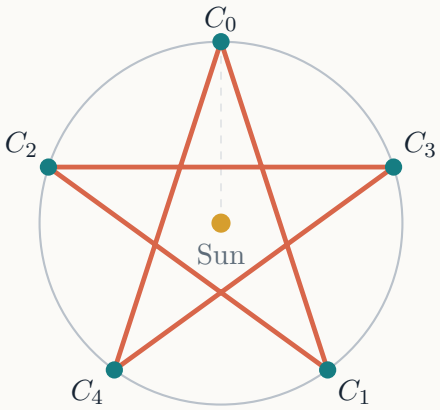
The calculation of one lapping time walks backward through consecutive Fibonacci ratios.

C.3 Why five alignments draw a star

During one alignment-to-alignment interval, Earth travels about $S/E \approx 8/5$ laps. One complete lap returns to the same direction, so only the leftover $3/5$ of a lap changes the direction of the alignment ray. The five successive directions are therefore approximately

$$0, \quad \frac{3}{5}, \quad \frac{1}{5}, \quad \frac{4}{5}, \quad \frac{2}{5} \pmod 1.$$

Connecting those directions in time order draws a five-pointed star.



Each C_j marks the direction of an alignment, not the path traveled by Venus. The actual fifth return misses the starting direction slightly.

The geometry of a perfectly regular pentagram contains the golden ratio exactly. The planetary star is only approximate: because 13/8 is not the exact period ratio, the alignment points drift and the whole pattern slowly turns over repeated eight-year cycles.[2]

[2]The European Southern Observatory's educational note *The Pentagram of Venus* illustrates the five alignments and the small miss after eight years.

C.4 The Euclidean algorithm inside $13 : 8$

The approximation contains the short Euclidean chain

$$\begin{aligned} 13 &= 1 \cdot 8 + 5, \quad 8 = 1 \cdot 5 + 3, \\ 5 &= 1 \cdot 3 + 2, \quad 3 = 1 \cdot 2 + 1, \\ 2 &= 2 \cdot 1. \end{aligned}$$

Consequently

$$\frac{13}{8} = [1; 1, 1, 1, 2] = [1; 1, 1, 1, 1, 1], \qquad \frac{8}{5} = [1; 1, 1, 2] = [1; 1, 1, 1, 1].$$

These are consecutive convergents of $\varphi = [1; \overline{1}]$. The orbit ratio 13/8, the difference 5, the step 3/5, and the pentagram are not independent coincidences: they are different consequences of one good rational approximation.

C.5 The Dresden Codex: an exact calendar model

The modern calculation above starts with measured periods and obtains approximations. The Maya Venus Table in the Dresden Codex instead uses chosen whole-day counts for calendrical bookkeeping. In particular:

- 584 days is the table's rounded alignment-to-alignment Venus cycle;
- 365 days is one *Haab*, the Maya solar-year calendar; and
- 260 days is one Tzolk'in, a repeating named-day cycle.

Five pages of the table present one 584-day Venus cycle apiece. The five cycles make an eight-Haab block because

$$5 \cdot 584 = 8 \cdot 365 = 2920. \tag{C.3}$$

This equality is exact arithmetic inside the calendar model, not an assertion that the physical Venus cycle is exactly 584 days.[3]

To make the 2920-day block agree again with the 260-day named-day cycle, take a least common multiple. Since $\gcd(2920, 260) = 20$,

$$\operatorname{lcm}(2920, 260) = \frac{2920 \cdot 260}{20} = 37960.$$

At that common return,

$$65 \cdot 584 = 104 \cdot 365 = 146 \cdot 260 = 13 \cdot 2920 = 37960. \tag{C.4}$$

[3]See the Saxon State and University Library Dresden's description of the Dresden Maya Codex, especially the Venus pages 24 and 46–50. For the page structure and cumulative day counts, see Akira Kaneko, "Identification of Postclassic Maya Constellations from the Venus Pages of the Dresden Codex," *Estudios de Cultura Maya* **35** (2010).

The same numbers are doing different jobs

In the modern orbital approximation, 13 counts Venus's Sun-centered laps in about eight Earth laps. In (C.4), 13 counts complete 2920-day calendar blocks in a 37960-day table. The identity $13 = 8 + 5$ is a delightful number-theoretic lens, but the manuscript alone does not show that Maya astronomers formulated their table using Fibonacci numbers or continued fractions.

Finally, rounding has a cost. The modern mean alignment cycle is about 583.92 days, so the calendar value 584 gains roughly

$$65(584 - 583.92) \approx 5.2 \text{ days}$$

over 65 cycles. Scholars study correction schemes in the table. Our exact identity (C.4) describes its integer framework, not a perfectly uniform planet.

Exercises

1. In your own words, distinguish a planet's Sun-centered lap time from the waiting time between two Earth–planet alignments. Which are E, V, and S? [Fluency]
2. Derive (C.1) by comparing lap rates. Then derive (C.2) and verify that $T(13/8) = 8/5$. [Reasoning]
3. Starting from a step of $3/5$ of a turn, list the first five positions modulo 1. Why does joining them chronologically make a star rather than the boundary of a pentagon? [Reasoning]
4. Run the Euclidean algorithm on $(13, 8)$. Read its quotients as the continued fraction of $13/8$, and explain why replacing the final 2 by $1, 1$ gives the same rational number. [Reasoning]
5. Factor 2920 and 260, compute their gcd and lcm, and recover every multiplier in (C.4). [Fluency]
6. Prove $T(F_{n+1}/F_n) = F_n/F_{n-1}$. Show that the positive fixed point of T is φ, and connect this with $[1; \overline{1}]$. [Reasoning]
7. Using 583.92 days as the modern mean alignment cycle, compare the elapsed time for 5, 65, and 650 cycles with the nominal 584-day count. Which error is negligible for a sketch but serious for a long calendar? [Exploration]

Appendix D

Finite fields and their arithmetic

Coordinates, classification, and applications

Big idea

Finite fields are the possible finite arithmetics in which every nonzero element can be divided by. Coordinates show why their sizes are prime powers. Polynomial constructions and cyclic multiplicative groups show that every prime-power size occurs and determines the arithmetic up to isomorphism. These tools then explain the Fibonacci periods first seen in Chapter 4. A final bonus proves that even dropping commutativity from finite division does not produce any new examples.

Read this after modular inverses and interpolation (Chapter 4), Fermat's theorem (Chapter 5), orders in finite groups (Chapter 6), and the prime binomial identity (Chapter 7). The first three sections develop the material through finite-fields Lecture 2. Cyclicity, matrix counts, and Fibonacci periods follow in the order of Lectures 3–4, with the finite-field classification inserted after cyclicity. Frobenius also describes all automorphisms and subfields. Section D.10 is independent bonus reading once dimensions and Lagrange's theorem are available; it supplies its own additional lemmas.

D.1 Rings, fields, and the prime field

Changing a modulus changes the rules of division. In $\mathbb{Z}/7\mathbb{Z}$ every nonzero residue has an inverse; in $\mathbb{Z}/8\mathbb{Z}$, $2 \cdot 4 = 0$ although neither factor is zero. We begin by naming the structures that distinguish these examples.

Definition D.1. A *ring* has an abelian group operation called addition, an associative multiplication with identity $1 \neq 0$, and both distributive laws

$$a(b+c) = ab + ac, \qquad (a+b)c = ac + bc.$$

Multiplication need not commute. A *unit* is an element with a two-sided multiplicative inverse. A *division ring* is a ring in which every nonzero element is a unit. A *field* is a commutative division ring. A nonzero element a is a *zero divisor* if $ab = 0$ or $ba = 0$ for some nonzero b.

The integers are a commutative ring, but not a field. The rational numbers are a field. Residues modulo n form a commutative ring whose units are exactly the classes coprime to n. Matrices

give familiar noncommutative rings: matrix multiplication distributes over addition but generally $AB \neq BA$. A unit cannot be a zero divisor, because multiplying by its inverse would force the other factor to be zero.

Proposition D.2. *A finite commutative ring with no zero divisors is a field. In particular, for $n \geq 2$, $\mathbb{Z}/n\mathbb{Z}$ is a field exactly when n is prime.*

Proof. Fix a nonzero a. If $ax = ay$, then $a(x - y) = 0$, so $x = y$. Multiplication by a is therefore injective on a finite set and hence surjective. In particular $ab = 1$ for some b, which is an inverse by commutativity. For prime p, Euclid's lemma excludes zero divisors modulo p; alternatively, Bézout gives each inverse. For composite $n = ab$, with $1 < a, b < n$, the nonzero residues of a, b have product zero. □

Definition D.3. The *characteristic* of a ring is the least positive integer m such that $m \cdot 1 = 0$, if such an integer exists; otherwise it is 0. Here $m \cdot 1$ means repeated addition, not exponentiation.

Proposition D.4. *Every finite field has prime characteristic p and contains a distinguished copy of $\mathbb{F}_p$, called its* prime field.

Proof. Finiteness makes two of $0, 1, 1 + 1, \ldots$ equal, so a positive multiple of 1 is zero. Let m be the least such multiple. If $m = ab$ with $1 < a, b < m$, then $(a \cdot 1)(b \cdot 1) = 0$ has two nonzero factors, impossible in a field. Also $m \neq 1$, so $m = p$ is prime. The map $\mathbb{Z} \to K$, $a \mapsto a \cdot 1$, respects both operations and has kernel $p\mathbb{Z}$. Thus its image is a subfield identified with $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$. □

D.2 Coordinates, exchange, and dimension

A vector space over a field k is an abelian group $(V, +)$ with scalar multiplication satisfying

$$a(u + v) = au + av, \quad (a + b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

The scalars belong to k, and the vectors belong to V. For example, in $\mathbb{F}_3^2$, $2(1, 2) = (2, 1)$. If a field K contains k, its own elements form a vector space over k: scalar multiplication is the field multiplication restricted to scalars in k.

Definition D.5. The *span* of $v_1, \ldots, v_r$ is the set of their linear combinations $\sum a_i v_i$. The list is *linearly independent* if $\sum a_i v_i = 0$ implies every $a_i = 0$. A *basis* is an independent spanning list. Its coordinates express every vector uniquely: spanning gives existence, and subtracting two expressions and using independence gives uniqueness.

An empty list spans $\{0\}$. If an independent list does not yet span, choose w outside its span. The enlarged list remains independent: in a relation $\sum a_i v_i + bw = 0$, a nonzero b would put w in the old span. Thus $b = 0$, and all the a_i also vanish. In a finite vector space this construction stops, because each span strictly enlarges. It produces a basis.

Before defining dimension we must know that different bases have the same length. Counting coordinates will prove this for finite fields, but the following argument works over any field.

Lemma D.6 (Exchange)**.** *If $v_1, \ldots, v_r$ are independent and $w_1, \ldots, w_m$ span V, then $r \leq m$.*

Proof. Insert the v_i one at a time into the spanning list. After k insertions, suppose the list consists of $v_1, \ldots, v_k$ and $m-k$ old vectors and still spans. Express v_{k+1} in this list. Some old vector has nonzero coefficient; otherwise v_{k+1} is in the span of the previous v_i, contradicting independence. Solve for that old vector by dividing by its coefficient. Replacing it by v_{k+1} preserves spanning, since the solved equation recovers the discarded vector. Previously inserted vectors are retained.

If $r > m$, after m insertions $v_1, \ldots, v_m$ would span, forcing v_{m+1} into their span, again contradicting independence. Thus $r \leq m$. If $m = 0$, $V = \{0\}$ has no nonempty independent list, so the conclusion also holds. □

Corollary D.7. *Any two finite bases of a vector space have the same length. This length is its* dimension *over the scalar field, denoted* $\dim_k V$.

Proof. Use each basis as the independent list and the other as the spanning list. The two inequalities force equality. □

Example D.8 (Coordinates over a prime field)**.** In $\mathbb{F}_3^2$, put $u = (1, 1)$ and $v = (1, 2)$. The equations $au + bv = 0$ give $a + b = 0$ and $a + 2b = 0$, hence $a = b = 0$. For an arbitrary (x, y), the same equations with right sides x, y give $b = y - x$ and $a = 2x - y$. Thus u, v are a basis, and $(2, 0) = u + v$ has coordinates $(1, 1)$ in this basis. The redundant list $u, v, u + v$ still spans but is not independent.

Theorem D.9 (The necessary size of a finite field)**.** *A d-dimensional vector space over a field with q elements has q^d elements. In particular, every finite field has p^d elements for a prime p and integer $d \geq 1$.*

Proof. Choose a basis. Each of its d coordinates has q choices, and the coordinate map $k^d \to V$ is bijective by existence and uniqueness. For a finite field K, use its prime field $\mathbb{F}_p$ as scalars. The basis-building procedure above stops because K is finite, so $|K| = p^d$. Since $1 \neq 0$, $d \geq 1$. □

Thus there is no field with 6 or 12 elements. A field with 81 elements must have characteristic 3 and dimension 4 over $\mathbb{F}_3$. At this point prime-power size is only a necessary condition. We will construct every prime-power size and prove uniqueness in Section D.5. First, a quadratic example makes the arithmetic concrete.

D.3 Build an extension before taking powers in it

The squares modulo 7 are $0, 1, 2, 4$. There is no element of $\mathbb{F}_7$ whose square is 5, so writing $\sqrt{5}$ does not yet give us a number we can calculate with in that field.

Introduce a symbol s with the rule $s^2 = 5$, and use pairs

$$a + bs, \qquad a, b \in \mathbb{F}_7.$$

Addition is coefficientwise. Multiply by distributing, then replace s^2 by 5:

$$(a + bs)(c + es) = (ac + 5be) + (ae + bc)s.$$

All coefficients are reduced modulo 7. For example,

$$(2 + s)(3 + 2s) = 16 + 7s = 2.$$

Formally, this is polynomial arithmetic modulo $X^2 - 5$:

$$K = \mathbb{F}_7[X]/(X^2 - 5).$$

The notation means that two polynomials represent the same element when their difference is a multiple of $X^2 - 5$. Polynomial division gives a unique representative $a + bX$, so there are exactly 49 elements. The usual arithmetic laws descend from polynomial addition and multiplication, just as they did for integer residues.

Proposition D.10 (Division in the new arithmetic)**.** *Every nonzero $a + bs$ has an inverse, given by*

$$(a + bs)^{-1} = \frac{a - bs}{a^2 - 5b^2}.$$

Consequently K is a field of 49 elements, denoted $\mathbb{F}_{49}$.

Proof. The product of $a + bs$ and $a - bs$ is $a^2 - 5b^2 \in \mathbb{F}_7$. This denominator is nonzero for a nonzero pair: if $b = 0$, it is a^2; if $b \neq 0$ and it vanished, then $(a/b)^2 = 5$, contradicting the list of squares. Its inverse exists in $\mathbb{F}_7$ and can be computed by extended Euclid. Multiplying gives the claimed inverse. □

For instance, $4 - 5 = -1 = 6$ has inverse 6 modulo 7, so

$$(2 + s)^{-1} = 6(2 - s) = 5 + s.$$

Indeed, $(2 + s)(5 + s) = 15 + 7s = 1$. Extended gcd has not disappeared: it now performs the division of coefficients inside the extension.

Common trap

$\mathbb{F}_{49}$ is not $\mathbb{Z}/49\mathbb{Z}$. In $\mathbb{F}_{49}$, seven copies of 1 sum to 0, and every nonzero element is invertible. In $\mathbb{Z}/49\mathbb{Z}$, the nonzero residue 7 satisfies $7 \cdot 7 = 0$ and has no inverse.

The construction works for any odd prime p and any nonsquare $d \in \mathbb{F}_p$: the pairs $a + bs$ with $s^2 = d$ form a field of p^2 elements. The same inverse formula uses $a^2 - db^2$ in the denominator. Equivalently, $X^2 - d$ has no root in $\mathbb{F}_p$, so, being quadratic, it is irreducible. If a chosen polynomial factors, its quotient need not be a field; irreducibility is doing the work.

D.4 The multiplicative group is cyclic

For a field K with q elements, its nonzero elements form a commutative group $K^\times$ of order $N = q - 1$. Lagrange gives $a^N = 1$ for every $a \neq 0$. The stronger claim is that some element has order exactly N.

Recall the polynomial root bound from interpolation: a nonzero polynomial of degree d over a field has at most d roots. Indeed, division by $X - a$ factors off a root a; another root $b \neq a$ must be a root of the remaining factor, because $b - a$ can be cancelled. Induction proves the bound. This cancellation is a field property.

Lemma D.11 (Combining coprime orders)**.** *If commuting elements a, b have relatively prime finite orders r, s, then ab has order rs.*

Proof. Certainly $(ab)^{rs} = 1$. If $(ab)^t = 1$, put $z = a^t = b^{-t}$. Then $z^r = z^s = 1$. Choose integers u, v with $ur + vs = 1$; inverses allow negative powers, so $z = (z^r)^u (z^s)^v = 1$. Consequently $r \mid t$ and $s \mid t$, hence $rs \mid t$. Thus rs is the least positive return exponent. □

Theorem D.12 (Cyclicity of finite-field multiplication)**.** *The multiplicative group of every finite field is cyclic.*

Proof. If $N = 1$, the group $\{1\}$ is cyclic. Otherwise, for each prime power ℓ^r occurring exactly in N, the polynomial $X^{N/\ell} - 1$ has fewer than N roots. Choose $a \in K^\times$ with $a^{N/\ell} \neq 1$ and put $g_\ell = a^{N/\ell^r}$. Then

$$g_\ell^{\ell^r} = a^N = 1, \qquad g_\ell^{\ell^{r-1}} = a^{N/\ell} \neq 1.$$

Its order divides ℓ^r but not ℓ^{r-1}, so it is exactly ℓ^r. The product of these g_ℓ has order the product of their pairwise coprime orders, by the lemma. This order is N, so its powers exhaust $K^\times$. □

A generator of $K^\times$ is called a *primitive element*. It also generates K as a field over its prime field, since every nonzero element is a power of it. These two kinds of generation are not equivalent in the opposite direction: in $\mathbb{F}_{49}$ the element s with $s^2 = 5$ generates the field over $\mathbb{F}_7$, but $s^{12} = 5^6 = 1$, so it does not generate the group of order 48.

Example D.13. In $\mathbb{F}_7^\times$, 6 has order 2 and 4 has order 3. Their product 3 therefore has order 6; its powers are $1, 3, 2, 6, 4, 5$. A primitive element exists, but it is not a distinguished choice of coordinate for the field.

D.5 Existence and uniqueness of Galois fields

Two fields are *isomorphic* if there is a bijection preserving addition, multiplication, and 1. Their elements may have different names or polynomial presentations while obeying the same arithmetic. We will prove that cardinality completely determines a finite field up to isomorphism. A finite field is also called a *Galois field*, and $\mathrm{GF}(q)$ is another notation for $\mathbb{F}_q$.

A polynomial construction supplies missing roots

A nonconstant polynomial over k is *irreducible* if it cannot be written as a product of two positive-degree polynomials over k. Every nonconstant polynomial has an irreducible factor: if it is reducible, factor it and continue with a factor of smaller positive degree. The decreasing degrees make this process terminate.

Lemma D.14 (Adjoin one root)**.** *For a monic irreducible polynomial $h \in k[X]$ of degree r, the polynomial residue ring $k[X]/(h)$ is a field containing k and a root of h. It has basis $1, t, \ldots, t^{r-1}$ over k, where t is the class of X. If $|k| = q$, it has q^r elements.*

Proof. Division by the monic polynomial h gives each residue a unique representative of degree less than r. Constants embed, and $h(t) = 0$. If a representative a is nonzero and has degree less than r, irreducibility gives $\gcd(a, h) = 1$. The polynomial Euclidean algorithm works over a field by successive divisions, just as for integers. Back substitution produces $u, v \in k[X]$ with $ua + vh = 1$. Reducing modulo h gives an inverse for a. Thus the residue ring is a field. Unique remainders give the stated basis and count. □

It follows that any polynomial over a finite field splits into linear factors in some finite extension. To see the termination explicitly, keep a factorization into known linear factors and a remaining factor. If that factor is nonconstant, choose an irreducible factor of it and adjoin a root by the lemma (or use a root already present). Divide out the resulting linear factor and repeat. Each step lowers the remaining degree by one, and every extension in the chain is finite. No prior construction of an algebraic closure is required. A field generated by all the roots of a polynomial is called its *splitting field*.

Theorem D.15 (Classification of finite fields)**.** *For every prime p and integer $n \geq 1$ there exists a field with p^n elements. Any two fields with that many elements are isomorphic by an isomorphism fixing their identified prime fields $\mathbb{F}_p$. Together with Theorem D.9, this describes all possible finite-field cardinalities.*

Proof. Existence. Set $q = p^n$ and choose a finite extension E of $\mathbb{F}_p$ in which $f(X) = X^q - X$ splits, using the preceding construction. The formal derivative is defined by $(\sum a_i X^i)' = \sum_{i \geq 1} i a_i X^{i-1}$, with coefficients read in the field; it obeys the product rule by direct multiplication. Here $f'(X) = -1$, since $q = 0$ in characteristic p. A repeated root would make both f and f' vanish there: differentiate $f = (X - a)^2 h$ to see this. Thus f has exactly q distinct roots in E.

Let $S = \{a \in E : a^q = a\}$. The prime binomial identity, iterated n times, gives $(a + b)^q = a^q + b^q$. Consequently S is closed under addition and additive inverses. It contains $0, 1$, and for $a, b \in S$,

$$(ab)^q = a^q b^q = ab, \qquad (a^{-1})^q = (a^q)^{-1} = a^{-1} \quad (a \neq 0).$$

So S is a field, and it has exactly q elements.

Uniqueness. Let K, L have q elements. Their characteristics are both p, by prime-power size and unique factorization. Identify their prime fields. Lagrange in either multiplicative group gives

$$X^q - X = \prod_{a \in K} (X - a) \quad \text{over } K, \qquad X^q - X = \prod_{b \in L} (X - b) \quad \text{over } L.$$

Choose a generator g of $K^\times$. There is a monic polynomial $m \in \mathbb{F}_p[X]$ of least positive degree with $m(g) = 0$, since $g^q = g$. It is irreducible: a factorization into smaller positive degrees would force one factor to vanish at g. Division with remainder also shows that every polynomial vanishing at g is divisible by m. In particular $m \mid X^q - X$.

Since $X^q - X$ splits over L, m has a root $b \in L$. The evaluation map

$$\mathbb{F}_p[X]/(m) \longrightarrow L, \qquad [a(X)] \longmapsto a(b)$$

is well defined and injective: its domain is a field, and if a nonzero element were sent to zero, multiplying by its inverse would send 1 to zero as well. The same evaluation at g identifies $\mathbb{F}_p[X]/(m)$ with $\mathbb{F}_p[g] = K$, because all nonzero elements of K are powers of g. We have therefore embedded K into L. Equal finite cardinalities make this embedding surjective, giving the required isomorphism. □

Common trap

Uniqueness means *up to isomorphism*, not that there is only one presentation or only one isomorphism. In $K = \mathbb{F}_7(s)$, $s^2 = 5$, both $s \mapsto s$ and $s \mapsto -s$ preserve the arithmetic. Likewise $\mathbb{F}_{p^n}$ is not the residue ring $\mathbb{Z}/p^n\mathbb{Z}$ when $n > 1$: their characteristics are p and p^n.

Example D.16 (Two presentations, one field)**.** Let $K = \mathbb{F}_3(s)$ with $s^2 = 2$. The polynomial $X^2 - X - 1$ is also irreducible over $\mathbb{F}_3$, since its values at $0, 1, 2$ are $2, 2, 1$. If $L = \mathbb{F}_3(t)$ with $t^2 = t + 1$, then

$$(t+1)^2 = t^2 + 2t + 1 = 2.$$

Thus $a + bs \mapsto a + b(t+1)$ is an isomorphism $K \to L$. The alternative root $-(t+1)$ gives another isomorphism.

D.6 Count matrices by choosing a basis

An invertible $d \times d$ matrix over $\mathbb{F}_q$ is the same thing as an ordered basis of $\mathbb{F}_q^d$: its columns give the images of the standard basis, and a basis gives unique solutions to every linear system. After r independent columns have been chosen, their span has q^r vectors. The next column can be any of the $q^d - q^r$ vectors outside that span. A full independent list of length d is a basis by exchange. Therefore, for $d \geq 1$,

$$\boxed{|\mathrm{GL}_d(\mathbb{F}_q)| = \prod_{r=0}^{d-1} (q^d - q^r).}$$

The factor for $r = 0$ excludes the zero vector; order of columns matters. For example, $|\mathrm{GL}_2(\mathbb{F}_2)| = (4-1)(4-2) = 6$.

For 2×2 matrices, Appendix A defines $\det\left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right) = ae - bc$ and proves invertibility when this is nonzero. The matrices with determinant 1 form $\mathrm{SL}_2(\mathbb{F}_q)$. Every $a \in \mathbb{F}_q^\times$ occurs as the determinant of $D_a = \mathrm{diag}(a, 1)$, and

$$S \longmapsto D_a S$$

is a bijection from determinant 1 matrices to determinant a matrices, with inverse multiplication by D_a^{-1}. There are $q - 1$ disjoint fibers, all equally large. Thus

$$\boxed{|\mathrm{SL}_2(\mathbb{F}_q)| = \frac{(q^2-1)(q^2-q)}{q-1} = q(q^2-1).}$$

The same fiber argument with the usual higher-dimensional determinant gives $|\mathrm{SL}_d(\mathbb{F}_q)| = |\mathrm{GL}_d(\mathbb{F}_q)|/(q-1)$. Neither matrix group is asserted to be cyclic; cyclicity above concerned the nonzero *field elements* under multiplication.

D.7 Frobenius moves the new elements

A field automorphism is a bijection from a field to itself preserving addition, multiplication, and 1. On a finite field K containing $\mathbb{F}_p$, define

$$\mathrm{Fr}(u) = u^p.$$

Theorem D.17 (Frobenius on a finite extension)**.** *The map* Fr *is an automorphism of K. Its fixed elements are exactly the embedded prime field $\mathbb{F}_p$.*

Proof. The coefficient argument in Chapter 7 gives $(u+v)^p = u^p + v^p$ in K, and $(uv)^p = u^p v^p$ follows from commutativity. Also $1^p = 1$. If $u^p = v^p$, then $(u-v)^p = 0$; a nonzero field element cannot have a zero power, so $u = v$. Thus the map is injective, and finiteness makes it surjective.

Fermat's theorem fixes every element of $\mathbb{F}_p$. There can be no additional fixed elements: the nonzero polynomial $X^p - X$ has degree p, so the root bound from interpolation allows at most p roots in K. The prime-field elements already supply all p of them. □

Example D.18 (Frobenius is conjugation in $\mathbb{F}_{49}$)**.** Since $s^2 = 5$,

$$s^7 = s(s^2)^3 = 5^3 s = 125s = -s.$$

For $a, b \in \mathbb{F}_7$, the sum identity and Fermat now give

$$(a+bs)^7 = a^7 + b^7 s^7 = a - bs.$$

Thus Frobenius exchanges s and $-s$, fixes the 7 constant elements, and applying it twice returns every element to itself. This is a nontrivial symmetry of the field, although it acts as the identity on the coefficients.

More generally, if $s^2 = d$ is nonsquare over $\mathbb{F}_p$ with p odd, then $(s^p)^2 = d^p = d$, so s^p is either s or $-s$. It cannot be s, since s is not in the fixed prime field. Therefore $(a+bs)^p = a - bs$ in this quadratic extension as well. Finiteness was essential to the surjectivity argument; no assertion about arbitrary infinite fields is needed here.

All automorphisms and all subfields

For $K = \mathbb{F}_{p^n}$, every element satisfies $a^{p^n} = a$, so Fr^n is the identity. No power Fr^r with $0 < r < n$ is the identity: otherwise the degree-p^r polynomial $X^{p^r} - X$ would have all p^n elements as roots. Thus Frobenius has order exactly n.

Proposition D.19. *The automorphisms of $\mathbb{F}_{p^n}$ are exactly $\mathrm{id}, \mathrm{Fr}, \ldots, \mathrm{Fr}^{n-1}$. For each positive divisor m of n, there is exactly one subfield with p^m elements, namely*

$$\{a \in \mathbb{F}_{p^n} : a^{p^m} = a\}.$$

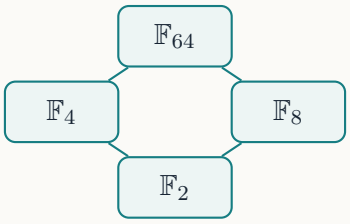
There are no other subfields.

Proof. Every field automorphism fixes 1 and hence $\mathbb{F}_p$. A multiplicative generator g generates the whole field, so its image determines the automorphism. Its minimal polynomial has degree n: evaluation gives an isomorphism $\mathbb{F}_p[X]/(m_g) \simeq K$, and comparison of cardinalities gives $\deg m_g = n$. The image of g must be one of its at most n roots. We have already found n distinct Frobenius powers, so these are all the automorphisms. Their group is cyclic of order n.

If $m \mid n$, then $p^m - 1 \mid p^n - 1$. In the cyclic group $K^\times$ of order $p^n - 1$, there are exactly $p^m - 1$ solutions of $a^{p^m - 1} = 1$: for a generator g, they are the powers whose exponents are multiples of $(p^n - 1)/(p^m - 1)$. Adjoining zero gives p^m solutions of $a^{p^m} = a$. As in the existence proof, these roots form a field. Any subfield with p^m elements consists entirely of such roots by Lagrange, so it must be this field.

Conversely, any subfield contains $\mathbb{F}_p$ and has size p^m for some m. Regard K as a vector space over it, with a finite basis of length r. Counting coordinates gives $p^n = (p^m)^r$, so $n = mr$. Thus $m \mid n$, as claimed. □

For example, the subfield sizes in $\mathbb{F}_{64}$ are $2, 4, 8, 64$, because the positive divisors of 6 are $1, 2, 3, 6$. The subfields of sizes 4 and 8 are not contained in one another. The two automorphisms of $\mathbb{F}_{49}$ are precisely the identity and the conjugation $a + bs \mapsto a - bs$.



The subfields of $\mathbb{F}_{64}$, with larger fields above smaller ones.

D.8 Return to the Fibonacci roots

Let $p \neq 2, 5$ be prime. If 5 is a square in $\mathbb{F}_p$, work in $K = \mathbb{F}_p$; otherwise construct $K = \mathbb{F}_p(s)$ with $s^2 = 5$ as above. In either case, K contains the two distinct nonzero roots

$$\alpha = \frac{1+s}{2}, \qquad \beta = \frac{1-s}{2}$$

of $X^2 - X - 1$. In particular, $\alpha + \beta = 1$, $\alpha\beta = -1$, and $\alpha - \beta = s \neq 0$.

The formula

$$F_n = \frac{\alpha^n - \beta^n}{\alpha - \beta} \qquad \text{in } K$$

now makes sense: all its divisions are by invertible field elements. It is proved by checking $n = 0, 1$ and observing that each of α^n, β^n satisfies the Fibonacci recurrence. The resulting values are the images of the ordinary Fibonacci numbers in the subfield $\mathbb{F}_p$.

Proposition D.20 (A period is a simultaneous multiplicative order)**.** *For $p \neq 2, 5$,*

$$\pi(p) = \operatorname{lcm}\big(\operatorname{ord}(\alpha), \operatorname{ord}(\beta)\big).$$

Proof. A return $F_t = 0$ first says $\alpha^t = \beta^t = c$. Then $F_{t+1} = c(\alpha - \beta)/(\alpha - \beta) = c$, so the full return condition $(F_t, F_{t+1}) = (0, 1)$ is equivalent to $\alpha^t = \beta^t = 1$. The least such positive t is the displayed least common multiple. □

When the roots are already in the prime field

If 5 is a square in $\mathbb{F}_p$, Fermat gives $\alpha^{p-1} = \beta^{p-1} = 1$, and hence

$$\pi(p) \mid p - 1.$$

For $p = 11$, choose $s = 4$. The roots are $\alpha = 8$, $\beta = 4$. Their orders are 10 and 5: $8^5 = -1$, $8^2 = 9 \neq 1$, while $4^5 = 1$ and $4 \neq 1$. Thus $\pi(11) = 10$. The ambient orders divide 10 by Fermat, which certifies the minimality claims in these power checks.

When Frobenius exchanges the roots

If 5 is nonsquare, Frobenius takes s to $-s$ and therefore

$$\alpha^p = \beta, \qquad \beta^p = \alpha.$$

Multiplying by the original root gives the sharper information

$$\alpha^{p+1} = \beta^{p+1} = \alpha\beta = -1.$$

Consequently

$$\pi(p) \mid 2(p+1), \qquad \pi(p) \nmid p+1.$$

The minus sign cannot be dropped: p is odd. Indeed, $F_{p+1} = 0$ but $F_{p+2} = -1$, so the state at $p+1$ is $(0,-1)$, not a restart at $(0,1)$.

Example D.21 (The period modulo 7)**.** In the field constructed earlier, $\alpha = 4+4s$ and $\beta = 4-4s$. Repeated squaring gives

$$\alpha^2 = 5+4s, \qquad \alpha^4 = 5s, \qquad \alpha^8 = -1, \qquad \alpha^{16} = 1.$$

Thus α has order 16: every proper divisor of 16 divides 8, and $\alpha^8 \neq 1$. Both roots have 16th power 1, so $\pi(7) = 16$. This computation uses the arithmetic of the extension explicitly; it does not treat the missing square root as an ordinary residue modulo 7.

D.9 From a bound to the exact period

The divisibilities above supply a candidate multiple, not always the least period. For instance, $\pi(29) = 14$, a proper divisor of 28.

We can decide which case applies without listing all squares. There are $(p-1)/2$ nonzero squares, since $x^2 = y^2$ exactly when $x = \pm y$. Fermat makes each such square a root of $X^{(p-1)/2} - 1$, and the root bound allows no others. Thus 5 is a square exactly when $5^{(p-1)/2} = 1 \pmod p$. Repeated squaring computes this test efficiently.

A practical procedure after finding a valid multiple B is:

1. Factor B and set $t = B$.

2. For each prime divisor q of t, test whether $(F_{t/q}, F_{t/q+1}) = (0,1) \pmod p$. If so, replace t by t/q and continue removing factors.

3. Stop when no prime factor can be removed. The remaining t is $\pi(p)$.

To justify the stopping rule, $\pi(p)$ divides the current t at every stage. If $t > \pi(p)$, some prime divides $t/\pi(p)$, and removing that prime still leaves a multiple of the true period. Thus a larger value cannot survive all the tests.

These tests can be made without computing all preceding Fibonacci numbers. The doubling identities are

$$F_{2n} = F_n(2F_{n+1} - F_n), \qquad F_{2n+1} = F_n^2 + F_{n+1}^2.$$

For a proof, the recurrence gives

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} - F_n & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

Square this matrix and compare its upper-right and lower-right entries. Thus a pair at index n gives the pair at $2n$; one additional recurrence step gives the pair at $2n+1$. Reading the binary digits of an index computes its pair by repeated doubling and reduction.

For $p = 29$, $11^2 = 5$ in $\mathbb{F}_{29}$, so start with $B = 28$. The pair at 14 is $(0, 1)$, reducing the candidate to 14. The pairs at 7 and 2 are $(13, 21)$ and $(1, 2)$, respectively, so neither prime factor of 14 can be removed. Hence $\pi(29) = 14$.

The exceptional primes

For $p = 2$, direct pair iteration gives $\pi(2) = 3$. For $p = 5$, the characteristic polynomial has a repeated root, so the distinct-root argument does not apply. Put $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Direct computation gives $A^5 = 3I \pmod 5$, hence $A^{20} = I$. But $A^{10} = 4I \neq I$, and A^4 has off-diagonal entry $F_4 = 3 \neq 0$. Every proper divisor of 20 divides 10 or 4, so the order of A is 20. The displayed matrix formula shows that $A^t = I$ exactly when $(F_t, F_{t+1}) = (0, 1)$; therefore $\pi(5) = 20$.

Checkpoint

Keep three questions separate. Reversible finite states prove that a period exists. Extension-field arithmetic and Frobenius give useful multiples of that period. Order tests identify the least period. For small moduli, simply following the pair orbit remains a complete method and requires none of the later field theory.

D.10 Bonus: every finite division ring is a field

The field axioms require multiplication to commute. Could a finite arithmetic allow division by every nonzero element while having $ab \neq ba$? Wedderburn's theorem says no. This is a bonus beyond the finite-fields lectures. The proof below uses the cyclotomic argument of Ernst Witt; see also André Weil, *Basic Number Theory*, Chapter I, §1. We develop the two extra ingredients before using them.

Conjugation partitions a group

For an element a of a group G, its *conjugacy class* is $\{gag^{-1} : g \in G\}$, and its *centralizer* is the subgroup

$$C_G(a) = \{g \in G : ga = ag\}.$$

Two elements g, h give the same conjugate exactly when $h^{-1}g \in C_G(a)$. Thus the class is in bijection with the left cosets of $C_G(a)$, so its size is $|G|/|C_G(a)|$. The *center* $Z(G)$ consists of elements commuting with all of G; these are exactly the one-element conjugacy classes. Since conjugacy classes partition G, choosing one a_i from each noncentral class gives the *class equation*

$$|G| = |Z(G)| + \sum_i \frac{|G|}{|C_G(a_i)|}.$$

This is another use of counting by equal-sized cosets.

The integer factor that will not fit

Among the complex nth roots of 1, call a root *primitive* if its multiplicative order is exactly n. They are the points $\exp(2\pi ik/n)$ with $\gcd(k,n)=1$, so there are $\varphi(n)$ of them. Define the cyclotomic polynomial

$$\Phi_n(X) = \prod_{\zeta \text{ primitive of order } n} (X-\zeta).$$

For example $\Phi_1 = X-1$, $\Phi_2 = X+1$, $\Phi_3 = X^2+X+1$, and $\Phi_4 = X^2+1$. Only the elementary geometry of roots of unity is needed here; irreducibility of cyclotomic polynomials is not needed.

Lemma D.22. *The polynomials Φ_n are monic with integer coefficients and*

$$X^n - 1 = \prod_{d \mid n} \Phi_d(X).$$

If d is a proper positive divisor of n, then

$$\Phi_n(X) \mid \frac{X^n-1}{X^d-1} \quad \text{in } \mathbb{Z}[X].$$

For integers $q \geq 2$ and $n > 1$, $|\Phi_n(q)| > q-1$.

Proof. Every nth root has a unique order $d \mid n$, giving the product identity over the complex numbers. To obtain integer coefficients, induct on n, starting with $\Phi_1 = X-1$. The product over proper divisors is monic with integer coefficients by induction. Dividing X^n-1 by it by monic long division produces an integer quotient and remainder. Over the complex numbers the division is exact with quotient Φ_n, so the remainder is zero and $\Phi_n \in \mathbb{Z}[X]$.

If $d \mid n$, cancel the factors indexed by divisors of d from the product for X^n-1. When $d < n$, the factor Φ_n remains, proving the polynomial divisibility and hence its integer-valued version at q.

For the inequality, each primitive nth root ζ is different from 1 when $n > 1$, and lies on the unit circle. Consequently

$$|q-\zeta|^2 = q^2 - 2q\,\mathrm{Re}(\zeta) + 1 > (q-1)^2.$$

Taking the product of these distances gives $|\Phi_n(q)| > (q-1)^{\varphi(n)} \geq q-1$, since $q-1 \geq 1$ and $\varphi(n) \geq 1$. □

Theorem D.23 (Wedderburn's little theorem)**.** *Every finite division ring is commutative, and therefore is a field.*

Proof. Let D be a finite division ring and let

$$Z = \{z \in D : zx = xz \text{ for every } x \in D\}$$

be its center. This is a field: it is closed under the ring operations, its multiplication commutes, and the inverse of a central nonzero element is central, as follows by multiplying $zx = xz$ by z^{-1} on both sides. Write $|Z| = q \geq 2$. Addition and multiplication by central scalars make D a vector space over Z; a finite basis exists by the earlier basis-building argument. If its dimension is n, then $|D| = q^n$. We will show $n = 1$.

For $a \in D$, put

$$C_D(a) = \{x \in D : xa = ax\}.$$

This set contains Z and is a vector subspace over Z, so it has q^d elements for some $d \geq 1$. It is closed under multiplication and under inverses of nonzero elements: if $xa = ax$, multiplying by x^{-1} shows $x^{-1}a = ax^{-1}$. Thus its nonzero elements form a subgroup of $D^\times$ of order $q^d - 1$. We do not assume that this centralizer is commutative. If a is noncentral, the subspace is proper and $d < n$.

Lagrange gives $q^d - 1 \mid q^n - 1$. This implies $d \mid n$: write $n = kd + r$, $0 \leq r < d$. Reduction modulo $q^d - 1$ gives $q^n - 1 \equiv q^r - 1$. If $r > 0$, this would be a positive multiple of $q^d - 1$ strictly smaller than $q^d - 1$, impossible. Thus $r = 0$.

Apply the class equation to $G = D^\times$. Its center is $Z \setminus \{0\}$, because commuting with every nonzero element is the same as commuting with all of D. Each group centralizer is $C_D(a) \setminus \{0\}$. Therefore

$$q^n - 1 = q - 1 + \sum_i \frac{q^n - 1}{q^{d_i} - 1}, \qquad d_i \mid n, \quad d_i < n,$$

where the sum runs over one representative of each noncentral conjugacy class.

Suppose $n > 1$. By Lemma D.22, the integer $\Phi_n(q)$ divides $q^n - 1$ and every term in this sum. Hence it divides $q - 1$. But $|\Phi_n(q)| > q - 1 > 0$, which is impossible for a divisor of $q - 1$. It follows that $n = 1$, so $D = Z$ is a field. □

Big idea

The contradiction joins linear algebra, group theory, and integer divisibility. Dimensions turn centralizers into powers of q; conjugation turns their sizes into a sum; cyclotomic factors divide every large term but are too large to divide the leftover $q - 1$.

Corollary D.24. *A finite ring with $1 \neq 0$ and no zero divisors is a field, even if commutativity was not assumed. Thus every finite division ring is, up to isomorphism, one of the Galois fields $\mathbb{F}_{p^n}$.*

Proof. For nonzero a, both maps $x \mapsto ax$ and $x \mapsto xa$ are injective and hence surjective. Choose b, c with $ab = 1$ and $ca = 1$. Then $c = c(ab) = (ca)b = b$, so each nonzero element has a two-sided inverse. The ring is a division ring, to which Wedderburn applies. The finite-field classification gives the last assertion. □

Finiteness matters. The real quaternions form an infinite division ring with $ij = k$ and $ji = -k$. Conversely, a finite matrix ring need not be a division ring: in $M_2(\mathbb{F}_2)$ the nonzero matrices $\mathrm{diag}(1, 0)$ and $\mathrm{diag}(0, 1)$ have product zero. Neither example contradicts the theorem.

Exercises

1. In $\mathbb{F}_7(s)$ with $s^2 = 5$, compute $(1 + 2s)(3 + s)$ and $(1 + 2s)^{-1}$. Check the inverse by multiplication. [Fluency]
2. Construct $\mathbb{F}_4$ using $t^2 = t + 1$ over $\mathbb{F}_2$. List its four elements and invert each nonzero element. Compute the square map and identify its fixed elements. [Reasoning]
3. In $\mathbb{F}_7(s)$, show that $u^7 = u$ exactly when $u \in \mathbb{F}_7$, and that $u^{49} = u$ for every element. Explain which calculation exhibits Frobenius as a nonidentity automorphism. [Reasoning]

4. In $\mathbb{F}_{29}$, verify that the roots of $X^2 - X - 1$ are 6 and 24, of orders 14 and 7. Deduce the period by their orders and compare with the factor-removal calculation. [Reasoning]

5. When 5 is nonsquare in $\mathbb{F}_p$ for odd p, prove directly that the Fibonacci pair at $p+1$ is $(0, -1)$. Use $p = 3$ to connect the result with the pair table in Chapter 4. [Synthesis]

6. Implement the doubling identities and compare the exact period procedure with direct pair iteration for several small primes. Treat 2 and 5 separately, and report candidate multiples as well as the least periods. [Exploration]

7. Find the units and characteristic of $\mathbb{Z}/9\mathbb{Z}$. Compare them with those of $\mathbb{F}_3(s)$, $s^2 = 2$. Why does the number of elements alone not identify a ring? [Fluency]

8. In $\mathbb{F}_5^2$, show $u = (1, 2), v = (2, 1)$ form a basis and find the coordinates of $(3, 4)$. Give a spanning list that is not a basis, and explain which coordinate property fails. [Fluency]

9. Could a field have 18, 27, or 64 elements? For each possible size, give the characteristic and dimension over the prime field, and identify the result that guarantees existence. [Reasoning]

10. Construct the field $\mathbb{F}_5(s)$ with $s^2 = 2$. Prove the nonvanishing needed for division, and find $(1 + s)^{-1}$. What fails if the relation is changed to $s^2 = 4$ on formal pairs? [Reasoning]

11. In $\mathbb{F}_9 = \mathbb{F}_3(s)$, $s^2 = 2$, show that s generates the field but not its multiplicative group. Show $1 + s$ has order 8 and find all primitive elements. [Reasoning]

12. The cubics $X^3 + X + 1$ and $X^3 + X^2 + 1$ over $\mathbb{F}_2$ give fields $K = \mathbb{F}_2(s)$ and $L = \mathbb{F}_2(t)$, respectively. Prove they are irreducible and exhibit an explicit isomorphism $K \to L$. *Hint:* use $s \mapsto t^{-1}$. [Synthesis]

13. Explain why $X^{p^n} - X$ has no repeated roots in any extension of $\mathbb{F}_p$. Contrast this with $X^p - 1$ in characteristic p. Why does the existence proof need the distinct-root assertion? [Reasoning]

14. Find every automorphism of $\mathbb{F}_9 = \mathbb{F}_3(s)$, $s^2 = 2$. Explain why uniqueness of a finite field up to isomorphism does not imply uniqueness of the isomorphism. [Reasoning]

15. List all subfield sizes in $\mathbb{F}_{4096}$ and determine the intersection of its subfields of sizes 16 and 64. Justify both the existence and the uniqueness used in your answer. [Synthesis]

16. Compute $|\mathrm{GL}_2(\mathbb{F}_3)|$, $|\mathrm{SL}_2(\mathbb{F}_3)|$, and $|\mathrm{GL}_3(\mathbb{F}_2)|$ by choosing columns. Why does every invertible matrix over $\mathbb{F}_2$ have determinant 1? [Fluency]

17. Compute $\Phi_6(X)$ by factoring $X^6 - 1$. In a hypothetical division ring of 64 elements with center $\mathbb{F}_2$, list the possible sizes of noncentral conjugacy classes from the proof. Show directly that its class equation cannot hold. [Synthesis]

18. Why cannot the finite-field cyclicity proof simply be applied to a finite division ring before proving Wedderburn? Locate the field property it uses. In the real quaternions, check that i, j, k are distinct roots of $X^2 + 1$. What familiar polynomial assertion fails? [Reasoning]

Appendix E

Selected hints and solutions

For checking arguments, not replacing them

Only selected exercises are treated here. A useful routine is: write a complete attempt, mark the exact step where you are stuck, then consult only the first relevant sentence below.

Chapter 1

1. Euclid gives

$$2024 = 2(748) + 528, \quad 748 = 528 + 220, \quad 528 = 2(220) + 88,$$

$$220 = 2(88) + 44, \quad 88 = 2(44),$$

so the gcd is 44.

2. The divisions are $527 = 341 + 186$, $341 = 186 + 155$, $186 = 155 + 31$, $155 = 5 \cdot 31$. Reverse:

$$31 = 186 - 155 = 2(186) - 341 = 2(527 - 341) - 341 = 2 \cdot 527 - 3 \cdot 341.$$

4. One solution is $35(-5) + 22(8) = 1$. All solutions are

$$x = -5 + 22t, \qquad y = 8 - 35t \qquad (t \in \mathbb{Z}).$$

To prove completeness, subtract two solutions and use $35(x - x') = -22(y - y')$ with $\gcd(35, 22) = 1$.

Chapter 2

3. Write $c = ar = bs$. Since $a \mid bs$ and $\gcd(a, b) = 1$, repeated Euclid's lemma (or Bézout) gives $a \mid s$. Thus $s = at$ and $c = abt$.

4. If $\sqrt{p} = a/b$ in lowest terms, then $a^2 = pb^2$. The exponent of p on the left is even; on the right it is odd. Unique factorization forbids this.

6. The primes at most 100 are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.$$

Every composite at most 100 has a prime factor at most $\sqrt{100} = 10$, so sieving by the primes $2, 3, 5, 7$ crosses out every composite.

7. No: $5! + 1 = 121 = 11^2$. Euclid's number only needs a prime divisor outside the proposed list; it need not itself be prime.

9. We already know that $a = \sqrt{2}$ is irrational. By the definition of the logarithm, $a^b = 3$. If $b = p/q$ were rational with $q > 0$, then $(\sqrt{2})^{p/q} = 3$; raising both sides to $2q$ would give $2^p = 3^{2q}$, contrary to unique factorization. Hence b is irrational. Unlike the proof by cases, this argument names a particular pair and proves that it works; it does not leave the witness dependent on an undecided alternative.

Chapter 3

1. Euclid gives

$$355/113 = [3; 7, 16],$$

with convergents 3, 22/7, and 355/113.

2. The convergents are

$$1,\ 2,\ \frac{3}{2},\ \frac{5}{3},\ \frac{8}{5},\ \frac{13}{8}.$$

Numerators and denominators are consecutive Fibonacci numbers; the recurrence with every $a_k = 1$ proves it.

4. Since $a_n = (a_n - 1) + 1/1$ at the bottom level,

$$[a_0; \dots, a_n] = [a_0; \dots, a_n - 1, 1].$$

Requiring the last partial quotient to exceed 1 selects a unique finite expansion.

6. With $\alpha \approx 0.2421988$, the continued fraction begins

$$\alpha = [0; 4, 7, 1, 3, 5, \dots],$$

so its first four nonzero convergents are

$$\frac{1}{4},\qquad \frac{7}{29},\qquad \frac{8}{33},\qquad \frac{31}{128}.$$

The Julian calendar uses $1/4$ leap day per year. A century-based cycle has $q = 100Q$, so we instead approximate

$$100\alpha \approx 24.21988 = [24; 4, 1, 1, 4, \dots].$$

Its first nonintegral convergent is $97/4$; dividing the denominator scale by 100 gives $97/400$, the Gregorian rate. The corresponding mean years are

$$365 + \frac{1}{4} = 365.25,\qquad 365 + \frac{97}{400} = 365.2425,\qquad 365 + \frac{31}{128} = 365.2421875.$$

Relative to the fixed approximation 365.2421988, their annual errors in days are about 7.80×10^{-3}, 3.01×10^{-4}, and -1.13×10^{-5}, respectively. Thus $31/128$ is numerically better, whereas $97/400$ fits a convenient rule organized by fours and centuries.

8. Euclid gives $527/341 = [1;1,1,5]$, whose convergents are $1, 2, 3/2, 17/11$. The penultimate convergent is $p_2/q_2 = 3/2$, so the remainder formula gives

$$\gcd(527, 341) = r_2 = 2 \cdot 527 - 3 \cdot 341 = 31.$$

9. The first six convergents of $\sqrt{3} = [1;\overline{1,2}]$ are

$$1, \quad 2, \quad \frac{5}{3}, \quad \frac{7}{4}, \quad \frac{19}{11}, \quad \frac{26}{15}.$$

They satisfy

$$1 < \frac{5}{3} < \frac{19}{11} < \sqrt{3} < \frac{26}{15} < \frac{7}{4} < 2.$$

For even k, $M_k'(t) < 0$, so the finite-tail value lies above $M_k(\infty) = c_k$; for odd k, the derivative is positive and the inequality reverses. Applying this once to x and once to c_{k+2} gives the two chains.

10. For $k = 0$ the formula is just $D(a_0) = \left(\begin{smallmatrix} p_0 & p_{-1} \\ q_0 & q_{-1} \end{smallmatrix}\right)$. If it holds at k, right multiplication by $D(a_{k+1})$ replaces the first column by $a_{k+1}(p_k, q_k)^T + (p_{k-1}, q_{k-1})^T$ and moves the old first column into second place. These are exactly the recurrences. The determinant is $(-1)^{k+1}$, so the inverse formula introduces no denominators.

11. For $1/7$, the remainders are $1, 3, 2, 6, 4, 5, 1, \ldots$, producing $0.\overline{142857}$. For $5/12$, they are $5, 2, 8, 8, \ldots$, producing $0.41\overline{6}$. The first repeated remainder marks the start and length of the repeating block.

15. Since

$$D(1)D(2) = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix},$$

the repeating tail obeys $t = (3t + 1)/(2t + 1)$. Hence $2t^2 - 2t - 1 = 0$, and positivity selects $t = (1 + \sqrt{3})/2$.

16. For $x = 1 + \sqrt{2}$, the conjugate $x' = 1 - \sqrt{2}$ lies in $(-1, 0)$; for $x = (1 + \sqrt{5})/2$, the conjugate is $(1 - \sqrt{5})/2$, again in $(-1, 0)$. Both numbers exceed 1, so Galois's criterion applies. Indeed their expansions are $[\overline{2}]$ and $[\overline{1}]$, respectively.

17. Factoring the difference of squares gives

$$\left|\sqrt{D} - p/q\right| = \frac{|D - p^2/q^2|}{\sqrt{D} + p/q} = \frac{1}{q^2(\sqrt{D} + p/q)}.$$

The denominator factor $\sqrt{D} + p/q$ exceeds 2, so Legendre's criterion forces p/q to be a convergent of $\sqrt{D}$.

18. We have $\sqrt{3} = [1;\overline{1,2}]$, of even period length 2; one period therefore gives the $+1$ solution $2^2 - 3(1)^2 = 1$. The first columns of the first three powers of $\left(\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right)$ are

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \begin{pmatrix} 7 \\ 4 \end{pmatrix}, \qquad \begin{pmatrix} 26 \\ 15 \end{pmatrix},$$

and each satisfies $p^2 - 3q^2 = 1$.

19. A finite computation verifies only the displayed finite averages; an asymptotic conclusion requires control of every later partial quotient. Khinchin's theorem excludes only a measure-zero

exceptional set and supplies no known test proving that a named constant such as π lies outside it. For e, the repeating pattern of positions gives

$$a_{3j-2} = 1, \qquad a_{3j-1} = 2j, \qquad a_{3j} = 1,$$

so $a_1 \cdots a_{3m} = 2^m m!$. Taking the $(3m)$th root and applying $m! \sim \sqrt{2\pi m}(m/e)^m$ yields

$$G_{3m}(e) = (2^m m!)^{1/(3m)} \sim (2m/e)^{1/3} \to \infty.$$

The denominator recurrence gives $q_{3m} \geq a_1 \cdots a_{3m} = 2^m m!$, so $q_{3m}^{2/(3m)} \to \infty$ too. Thus e is provably on the exceptional side of both laws.

20. The two matrices and their maps are

$$D(1)D(2) = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \quad t \longmapsto \frac{3t+1}{2t+1}; \qquad D(2)D(1) = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}, \quad t \longmapsto \frac{3t+2}{t+1}.$$

The rightmost matrix acts first. The fixed-point equations are $2t^2 - 2t - 1 = 0$ and $t^2 - 2t - 2 = 0$, with positive roots $(1+\sqrt{3})/2$ and $1+\sqrt{3}$, respectively.

21. For $j = 1$ and $j = 2$, respectively,

$$MNM^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}, \qquad MNM^{-1} = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}.$$

The positive fixed points are $\sqrt{3}$ and $1+\sqrt{3}$. Only the second is reduced, since $-\sqrt{3} < -1$ but $-1 < 1 - \sqrt{3} < 0$. Its pure period is $[\overline{2,1}]$.

22. A pole $ct + d = 0$ would make t rational if $c \neq 0$; if $c = 0$, invertibility implies $d \neq 0$. If $T_M(t) = r \in \mathbb{Q}$, then $(a - rc)t = rd - b$. The coefficient $a - rc$ cannot vanish: together with the equality it would force $ad - bc = 0$. Thus $t = (rd - b)/(a - rc)$ would be rational, a contradiction. The fixed-point assertion must exclude scalar matrices. For example, $T_I(t) = t$ for every t, including $\sqrt[3]{2}$, which is not quadratic: $X^3 - 2$ has no rational root and is an irreducible cubic over $\mathbb{Q}$.

23. Precisely $1+\sqrt{2}$, $(1+\sqrt{5})/2$, and $(1+\sqrt{3})/2$ are reduced, so those three are purely periodic. The conjugate of $\sqrt{2}$ is less than -1, and the last two numbers fail $x > 1$. All six are real quadratic irrationals, so Lagrange's theorem makes all their continued fractions eventually periodic, including the one with negative initial digit.

24. The successive triples are

$$(0,1,3),\ (3,4,1),\ (1,3,1),\ (2,3,1),\ (1,4,1),\ (3,1,6),\ (3,4,1).$$

Thus $\sqrt{13} = [3; \overline{1,1,1,1,6}]$. After the first step, $|P_n| \leq 3 < \sqrt{13}$ and $0 < Q_n \leq 4 < 2\sqrt{13}$; each corresponding surd exceeds 1 and has conjugate in $(-1, 0)$.

25. Here $-1/z' = 1 + \sqrt{3}$, whose floor is 2. The predecessor is $2 + 1/z = 1 + \sqrt{3}$. The bounds allow $P = -1, 0, 1$ and $Q = 1, 2, 3$, but the divisibility condition and both reduced inequalities leave only $(P, Q) = (1, 1), (1, 2)$. They represent $1+\sqrt{3}$ and $(1+\sqrt{3})/2$, and the algorithm interchanges them. A finite map can have an initial segment before its cycle: $0 \mapsto 1$, $1 \mapsto 1$. Unique reduced predecessors exclude that behavior in the continued-fraction argument.

26. The period matrix and equation are

$$N = \begin{pmatrix} 10 & 3 \\ 7 & 2 \end{pmatrix}, \qquad 7x^2 - 8x - 3 = 0, \qquad x = \frac{4 + \sqrt{37}}{7}.$$

Transposition gives $3z^2 - 8z - 7 = 0$, with positive root $z = (4 + \sqrt{37})/3$. Finally, $-1/x' = 7/(\sqrt{37} - 4) = (4 + \sqrt{37})/3$, as predicted.

27. For $D = 23$, the triples (P_n, Q_n, a_n) are

$$(0, 1, 4),\ (4, 7, 1),\ (3, 2, 3),\ (3, 7, 1),\ (4, 1, 8),\ (4, 7, 1).$$

Thus $L = 4$, the palindrome is $(1, 3, 1)$, and the final digit is $8 = 2a_0$. The first four convergents are $4, 5, 19/4, 24/5$, giving $24^2 - 23 \cdot 5^2 = 1$. Including the final 8 instead produces $211/44$, with remainder $211^2 - 23 \cdot 44^2 = -7$.

28. The period is $\sqrt{6} = [2; \overline{2, 4}]$. The four rows $(k, p_k/q_k, Q_{k+1}, p_k^2 - 6q_k^2)$ are

$$(0, 2/1, 2, -2),\ (1, 5/2, 1, 1),\ (2, 22/9, 2, -2),\ (3, 49/20, 1, 1).$$

Their last entries equal $(-1)^{k+1}Q_{k+1}$. Since $L = 2$, exactly the odd-indexed convergents solve $+1$, and none solves -1.

29. The least periods for $D = 2, 3, 5, 7, 13, 14$ are, respectively,

$$(2),\ (1, 2),\ (4),\ (1, 1, 1, 4),\ (1, 1, 1, 1, 6),\ (1, 2, 1, 6).$$

Only $D = 2, 5, 13$ have odd length. Their least positive solutions of the negative equation are $(1, 1), (2, 1), (18, 5)$. For $D = 3$, reducing $p^2 - 3q^2 = -1$ would give $p^2 \equiv 2 \pmod 3$, impossible because a square is 0 or 1 modulo 3.

30. Multiplication by $2 - \sqrt{3}$ sends

$$(p, q) \longmapsto (2p - 3q,\ 2q - p)$$

and preserves norm 1. For $q = 1$, positivity forces $p = 2$, and the image is $(1, 0)$. If $q > 1$, then $\sqrt{3} < p/q = \sqrt{3 + 1/q^2} < 2$. Hence $2p - 3q > 0$ and $0 < 2q - p < q$. Repeated descent strictly decreases the positive integer second coordinate until $(2, 1)$ and then $(1, 0)$ are reached. Reversing the steps proves the power formula.

31. The state algorithm gives

$$\sqrt{61} = [7; \overline{1, 4, 3, 1, 2, 2, 1, 3, 4, 1, 14}], \qquad L = 11.$$

The palindrome is the ten-digit block before 14. At indices 10 and 21, respectively, the convergents give

$$29718^2 - 61 \cdot 3805^2 = -1,$$

$$1766319049^2 - 61 \cdot 226153980^2 = 1.$$

The second also follows by squaring $29718 + 3805\sqrt{61}$. Theorem 3.22 excludes every smaller positive solution of either sign.

32. Clear denominators to check the seven equalities

$$6W = 5B + 6Y, \quad 20B = 9S + 20Y, \quad 42S = 13W + 42Y,$$
$$12w = 7(B + b), \quad 20b = 9(S + s),$$
$$42y = 13(W + w), \quad 30s = 11(Y + y).$$

The displayed table at $t = 1$ satisfies each equality and has total 50 389 082. The first three equations determine all bulls from W. The cow cycle has coefficient $(7/12)(9/20)(11/30)(13/42) \neq 1$, so it

determines all cows as well. Euclid gives gcd 4657 for the four bull counts and gcd 1 after including the cows. A Bézout combination of the scaled counts therefore equals t, proving $t \in \mathbb{Z}$ whenever the counts are integral. At $t = 1$, $W + B = 4 \cdot 4\,456\,749$ is not a square, so the full problem is not solved.

33. If $cr^2 = h(h+1)/2$, then $(2h+1)^2 = 8cr^2 + 1$. Conversely, an odd $X > 1$ gives the integer $h = (X-1)/2$ and $h(h+1)/2 = (X^2-1)/8 = cr^2$. Here $c = 51\,285\,802\,909\,803$. The identity $8c = 4\,729\,494 \cdot 9314^2$ makes $(X, r) \mapsto (X, Z) = (X, 9314r)$ a bijection onto the positive solutions of the smaller equation with $9314 \mid Z$. For the reverse map, use $r = Z/9314$; the equation then implies $X^2 \equiv 1 \pmod 8$, so X is odd.

34. The recurrence in Section 3.11 computes the two coefficients of ε^j modulo 9314 by induction. Its first vanishing second coefficient occurs at $j = 2329$, with first coefficient $9313 \equiv -1$. Because all positive Pell solutions are powers and their second coordinates increase, this first hit gives the smallest herd. Write any $j \geq 1$ as $j = 2329k + r$ with $0 \leq r < 2329$. Then $\varepsilon^j \equiv (-1)^k \varepsilon^r \pmod{9314}$, so its second coefficient vanishes exactly when $r = 0$. Finally, T is proportional to Z_j^2, while $Z_j = (\varepsilon^j - \varepsilon^{-j})/(2\sqrt{4\,729\,494})$. Squaring doubles the exponent: $2 \cdot 2329 = 4658$.

Chapter 4

1. Extended Euclid yields

$$1 = 11 \cdot 101 - 30 \cdot 37,$$

so $37^{-1} \equiv -30 \equiv 71 \pmod{101}$.

2. Divide $18x \equiv 30 \pmod{42}$ by 6 while reducing the modulus: $3x \equiv 5 \pmod 7$. Since $3^{-1} \equiv 5$, $x \equiv 4 \pmod 7$. Modulo 42 the six solutions are $4, 11, 18, 25, 32, 39$.

5. Here $N = 504$. One convenient construction is

$$x = 2(72)(4) + 3(63)(7) + 4(56)(5) = 3019 \equiv 499 \pmod{504}.$$

The multipliers $4, 7, 5$ invert $72, 63, 56$ modulo $7, 8, 9$, respectively.

8. The rows (r, s, t) are

$$(83, 1, 0),\ (29, 0, 1),\ (25, 1, -2),\ (4, -1, 3),\ (1, 7, -20),\ (0, -29, 83).$$

Thus $1 = 7 \cdot 83 - 20 \cdot 29$ and $29^{-1} \equiv 63 \pmod{83}$. Indeed, $29 \cdot 63 = 1827 = 22 \cdot 83 + 1$.

9. From $6 = 42 - 2 \cdot 18$, multiplication by 4 gives $24 = 4 \cdot 42 - 8 \cdot 18$. Hence $x \equiv -8 \equiv 6 \pmod 7$; the residues modulo 42 are $6, 13, 20, 27, 34, 41$. The gcd 6 does not divide 25, so the second congruence is impossible.

10. At every component modulus, e_i has value 0 or 1, exactly one selector has value 1, and distinct selectors cannot both have value 1. CRT transfers the three identities back modulo N. Consequently $(\sum a_i e_i)(\sum b_i e_i) = \sum a_i b_i e_i$ modulo N; addition is componentwise as well.

11. Put $x = 2 + 6t$. Then $6t \equiv 3 \pmod 9$, or $2t \equiv 1 \pmod 3$, so $t \equiv 2 \pmod 3$ and $x \equiv 14 \pmod{18}$. The second system prescribes different residues modulo $\gcd(6, 9) = 3$, so it is incompatible.

12. The selectors at $-1, 0, 2$ give

$$f(X) = 2\frac{X(X-2)}{3} - \frac{(X+1)(X-2)}{2} + 3\frac{X(X+1)}{6} = \frac{2X^2 - X + 3}{3}.$$

All solutions are $f(X) + (X+1)X(X-2)h(X)$ with $h \in \mathbb{Q}[X]$.

13. The denominators at $0, 1, 3$ are $3, 5, 6$ modulo 7. Their inverse certificates are

$$3 \cdot 5 - 7 \cdot 2 = 1, \quad 5 \cdot 3 - 7 \cdot 2 = 1, \quad 6 \cdot 6 - 7 \cdot 5 = 1.$$

Thus

$$f(X) = 5(X-1)(X-3) + 4 \cdot 3X(X-3) + 2 \cdot 6X(X-1) = X^2 + 2X + 1 \quad \text{in } \mathbb{F}_7[X].$$

Substitution gives the required values $1, 4, 2$.

14. In $f(2) - f(0)$ every term has a factor 2, so this difference cannot be 1 modulo 8. The coefficient ring is not a field; specifically, the difference between the two interpolation points is a nonunit.

15. Modulo 2, the orbit is $(0,1), (1,1), (1,0), (0,1)$, giving period 3. Modulo 3, it is

$$(0,1), (1,1), (1,2), (2,0), (0,2), (2,2), (2,1), (1,0), (0,1),$$

giving period 8. At the first positive zero, $n = 4$, the next residue is 2, not 1, so the recurrence has not restarted.

16. The periods are $\operatorname{lcm}(3,8) = 24$ and $\operatorname{lcm}(8,20) = 40$. A simultaneous return occurs exactly at an index divisible by both component periods. For modulus 15, the product $8 \cdot 20 = 160$ is a period but is four times the least one.

17. The orbit is $(1,0) \mapsto (0,0) \mapsto (0,0)$. Modulo 2 the rule is $(a,b) \mapsto (b,b)$, which forgets a; a state therefore does not determine a unique predecessor.

18. The squares modulo 7 are $0, 1, 2, 4$. A root x would satisfy $(2x-1)^2 = 5$, which is impossible. Binet's formula requires the missing roots (equivalently, a square root of 5); the extension in Appendix D supplies arithmetic for them.

Chapter 5

1. The binary length-3 words form four orbits:

$$\{000\}, \quad \{111\}, \quad \{001, 010, 100\}, \quad \{011, 110, 101\}.$$

3. If $ik \equiv jk \pmod{p}$, then $p \mid k(i-j)$. As $p \nmid k$, Euclid's lemma gives $p \mid i - j$. For $0 \le i, j < p$, this forces $i = j$.

5. Orbit sizes divide 6, and binary examples realize $1, 2, 3, 6$: 000000, 010101, 001001, and 000001.

6. Fermat gives $(a+b)^p = a + b = a^p + b^p$ in $\mathbb{F}_p$. Applying it repeatedly proves the corresponding identity for any finite sum. Fermat's statement about residues does not assert $u^p = u$ for new elements of an extension; the polynomial identity from Chapter 7 proves additivity there instead.

8. There are 2 constant words, $2^3 - 2 = 6$ words of least period 3, and $2^9 - 2^3 = 504$ of least period 9. The latter form $504/9 = 56$ necklaces.

9. The full-period words number $3^8 - 3^4 = 6480$, forming $6480/8 = 810$ necklaces. Thus $8 \mid 3^4(3^4 - 1)$. The factor $3^4 = 81$ is invertible modulo 8; the certificate $81 - 10 \cdot 8 = 1$ supplies its inverse 1. Cancellation gives $3^4 \equiv 1 \pmod{8}$.

10. Besides 0000 and 1111, remove 0101 and 1010. These are exactly the four words obtained by repeating a two-letter block. The count $16 - 2 = 14$ is not divisible by 4, whereas $16 - 4 = 12$ splits into three full-period orbits.

11. The values are

$$\varphi(72) = 4 \cdot 6 = 24, \qquad \varphi(100) = 2 \cdot 20 = 40, \qquad \varphi(81) = 27 \cdot 2 = 54.$$

The factors in $9 = 3 \cdot 3$ are not coprime: the pair of residues $(1, 2)$ modulo $3, 3$ cannot come from a single integer. There is no CRT bijection, and $\varphi(9) = 6 \neq 4 = \varphi(3)^2$.

12. The prime-power congruences are $5^4 \equiv 1 \pmod 8$ and $5^6 \equiv 1 \pmod 9$. Raise them to powers 6 and 4, respectively, then use CRT to obtain $5^{24} \equiv 1 \pmod{72}$. Since $2026 = 24 \cdot 84 + 10$, reduce to 5^{10}. Modulo 72, successive squares give $5^2 \equiv 25$, $5^4 \equiv 49$, and $5^8 \equiv 25$, so $5^{10} \equiv 25^2 \equiv 49$.

13. The factor 3^3 in $3^9 - 3^3 = 3^3(3^6 - 1)$ is not invertible modulo 9, so it cannot be cancelled. For an odd integer,

$$(2t + 1)^2 = 1 + 4t(t + 1) \equiv 1 \pmod 8,$$

because one of $t, t + 1$ is even. Euler's exponent need not be the least.

14. The proper divisors of 6 are $1, 2, 3$. Words whose periods divide 2 number a^2; those whose periods divide 3 number a^3. A word fixed by shifts of both 2 and 3 is fixed by their difference 1, so their intersection consists of the a constant words. Removing the union leaves $a^6 - a^3 - a^2 + a$ words, all in orbits of size 6. For p^k, all proper divisors divide p^{k-1}, so a single set contains every shorter-period word.

15. The functions n^2 and $\mathbf{1}$ are completely multiplicative. The other three are multiplicative but not completely: $\varphi(4) = 2 \neq \varphi(2)^2 = 1$, $\tau(4) = 3 \neq \tau(2)^2 = 4$, and $\mu(4) = 0 \neq \mu(2)^2 = 1$.

16. Since $360 = 2^3 \cdot 3^2 \cdot 5$, $\tau(360) = 4 \cdot 3 \cdot 2 = 24$ and $\sigma(360) = 15 \cdot 13 \cdot 6 = 1170$. Choose each prime exponent independently; for the sum, distribute the product of the three geometric sums.

17. For $d = 1, 2, 3, 6, 9, 18$, the classes with $\gcd(r, 18) = 18/d$ have sizes $1, 1, 2, 2, 6, 6$, totaling 18. The first class is $\{18\}$, accounting for $\varphi(1) = 1$.

18. In order from 1 to 12, the values are

$$1, -1, -1, 0, -1, 1, -1, 0, 0, 1, -1, 0.$$

The contributing divisors of 60 are $1, 2, 3, 5, 6, 10, 15, 30$. Their Möbius values sum to $1 - 1 - 1 - 1 + 1 + 1 + 1 - 1 = 0$, while

$$\varphi(60) = 60 - 30 - 20 - 12 + 10 + 6 + 4 - 2 = 16.$$

19. The first two inverses are $\mathbf{1}$ and id. The third is $\varphi + \varepsilon$, since $\mu * (\mathrm{id} + \mathbf{1}) = \varphi + \varepsilon$. Thus $f(1) = 2$ and $f(n) = \varphi(n)$ for $n > 1$. Inversion applies to every arithmetic function; multiplicativity is useful for products, not a hypothesis of the inversion theorem.

20. The divisors $1, 2, 4$ give $(\mu * \mu)(4) = 0 + 1 + 0 = 1 \neq \varepsilon(4)$. The cancellation lemma gives $\mu * \mathbf{1} = \varepsilon$.

21. Only $d = 1, 2, 3, 6$ contribute: $P_2(12) = 2^{12} - 2^6 - 2^4 + 2^2 = 4020$ primitive words and $4020/12 = 335$ primitive necklaces.

22. Substitute the primitive-word formula and write $d = me$:

$$\begin{aligned} nN_a(n) &= \sum_{d\mid n} \frac{n}{d} \sum_{m\mid d} \mu(d/m) a^m \\ &= \sum_{m\mid n} a^m \sum_{e\mid n/m} \mu(e) \frac{n/m}{e} = \sum_{m\mid n} a^m \varphi(n/m). \end{aligned}$$

Reindex to obtain the requested formula. For binary necklaces of length 12, the numerator is $4096 + 64 + 32 + 16 + 8 + 8 = 4224$, giving 352.

23. The exact values are $2, 5 - 2 = 3, 7 - 2 = 5$, and $16 - 5 - 7 + 2 = 6$. Their lower sums are $2, 5, 7, 16$.

24. Successive differences give $2, 5, 1, 12$. The coefficients from 0 are $1, -1, 0, 0$, so $\mu_P(0, 3) = 0$. The recurrence cancels the sum over the whole interval; it does not assign a sign merely from the distance between endpoints.

25. Upper totals count elements in every set indexed by S, permitting additional memberships. Inversion at $S = \varnothing$ gives $40 - 20 - 15 - 12 + 8 + 5 + 4 - 2 = 8$ in none. The union has 32 elements.

26. The coefficients are 1 at the bottom, -1 at each of the three middle partitions, and 2 at the top. Inversion gives $4^3 - 3 \cdot 4^2 + 2 \cdot 4 = 24$, agreeing with $4 \cdot 3 \cdot 2$.

27. At $d = 1, 2, 3, 4, 6, 12$, the coefficients are $1, -1, -1, 0, 1, 0$. The exponent interval for 6 has two one-step chains, giving $(-1)(-1) = 1$. The exponent of 2 in 12 spans two steps, whose chain coefficient is zero.

28. The interval $[x, y]$ has $y - x + 1$ elements, but there are infinitely many integers below y. Finite support for F, or the weaker condition that only finitely many $x \le y$ have $F(x) \neq 0$ for every y, makes each lower sum finite.

29. The samples are $(2, 2 - 2i, 2, 2 + 2i)$. The inverse four-point formulas give $(2, -1, 0, 1)$. The added polynomial $X^4 - 1$ vanishes at all four sample points; the degree bound is what singles out the representative $2 - X + X^3$.

30. Both polynomials have degree less than n and take the values δ_{jr} at the n distinct points ω^r. Their difference has n roots and degree below n, so it is zero.

31. The successive powers of 2 modulo 5 are $1, 2, 4, 3, 1$, so its order is 4. The samples are $(0, 4, 3, 2)$. The inverse formula uses $\omega^{-1} = 3$ and $4^{-1} = 4$:

$$c_k = 4 \sum_{j=0}^{3} y_j 3^{jk} \pmod 5,$$

giving $1, 2, 3, 4$ for $k = 0, 1, 2, 3$.

Chapter 6

1. $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$. Every nonidentity element squares to 1, so their orders are 2; the identity has order 1.

4. If $aH \cap bH$ contains x, write $x = ah_1 = bh_2$. Then $a = bh_2h_1^{-1}$, so every $ah \in aH$ equals $b(h_2h_1^{-1}h) \in bH$. Thus $aH \subseteq bH$; symmetry gives equality.

5. For $g \neq e$, the order of g divides the prime $|G|$, so it is $|G|$. Hence $G = \langle g \rangle$.

7. Reducing a product modulo m or n gives the product of the reductions. If $xy \equiv 1 \pmod{mn}$, its reductions also multiply to 1, so the CRT bijection preserves inverses. Counting its domain and codomain gives $\varphi(mn) = \varphi(m)\varphi(n)$.

Chapter 7

2. Legendre gives

$$v_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97,$$

$$v_5(100!) = 20 + 4 = 24.$$

Each trailing zero consumes one 2 and one 5, so there are 24.

4. Choose n objects from two disjoint groups of n. If k come from the first group, then $n - k$ come from the second. Summing over k gives

$$\sum_k \binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}.$$

7. Legendre gives

$$v_p\binom{2n}{n} = \sum_{j \geq 1} \left(\left\lfloor \frac{2n}{p^j} \right\rfloor - 2 \left\lfloor \frac{n}{p^j} \right\rfloor \right).$$

Each summand is 0 or 1, and there are at most $\left\lfloor \log_p(2n) \right\rfloor$ nonzero candidates. This is the exponent of p in the lcm, so the valuation of the binomial coefficient is no larger for every prime p.

8. Fermat makes $X^p - X$ zero at every point of $\mathbb{F}_p$, but its leading coefficient is 1, so it is not the zero polynomial. By contrast, expansion shows that every coefficient of $(X + Y)^p - X^p - Y^p$ vanishes modulo p. This is a polynomial identity and remains valid upon substitution in any field containing $\mathbb{F}_p$.

Chapter 8

2. The sum of the $2n + 1$ coefficients in row $2n$ is 4^n. The central coefficient is largest, so it is at least the average $4^n/(2n + 1)$ and strictly below the whole sum.

3. For $n < p \leq 2n$, the numerator $(n + 1) \cdots (2n)$ contains exactly one multiple of p, while $n!$ contains none. Because $p^2 > 2n$ for $p > n \geq 1$, the valuation is exactly one.

7 (hint). Insert $x = p_n$ into both Chebyshev bounds to compare n with $p_n/\log p_n$. First obtain coarse polynomial bounds on p_n so that $\log p_n$ can be compared with $\log n$, then substitute back.

Appendix B: continued-fraction arithmetic

1. With $t_2 \in [1, \infty]$,

$$x = 2 + \frac{1}{3 + 1/t_2}.$$

This expression increases with t_2. At $t_2 = 1$ it is $9/4$; as $t_2 \to \infty$ it approaches $7/3$. For an infinite continued fraction the second endpoint is not attained, so $x \in [9/4, 7/3)$.

2. The denominator is positive for $t \geq 1$, and

$$F'(t) = \frac{22 \cdot 2 - 3 \cdot 14}{(14t+2)^2} = \frac{2}{(14t+2)^2} > 0.$$

Thus the range is determined by the endpoint values $F(1) = 25/16$ and $F(\infty) = 11/7$. The whole interval lies in $[1, 2)$, so the certified output digit is 1.

3. Substituting $t = m + 1/u$ into $F_M(t) = (At + B)/(Ct + D)$ and clearing the factor u gives

$$\frac{(Am + B)u + A}{(Cm + D)u + C},$$

which is represented by $MD(m)$. After emitting k, the unread output is $1/(F_M - k)$, represented by $\left(\begin{smallmatrix} 0 & 1 \\ 1 & -k \end{smallmatrix}\right) M$.

4. From $M = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, absorption gives

$$MD(15) = \begin{pmatrix} 30 & 2 \\ 1 & 0 \end{pmatrix}, \qquad MD(15)D(1) = \begin{pmatrix} 32 & 30 \\ 1 & 1 \end{pmatrix}.$$

The latter map has range $[31, 32)$ on $[1, \infty)$, so the next output digit is 31.

5. For each fixed v, the map in u is linear fractional and hence monotone unless it is constant; its extrema on a horizontal slice occur on the two vertical edges. On either edge, the remaining function of v is again linear fractional, so its extrema occur at the two endpoints. Therefore all global extrema are corner values. If the denominator vanishes inside the rectangle, the map is not continuous there and can become unbounded, so corner values alone give no range certificate.

7. Treating the two copies as independent discards the relation $x = y$. Every finite continued-fraction prefix gives a nontrivial interval containing $\sqrt{2}$, so the product box permits values slightly below and slightly above 2. Since the floor jumps at 2, it is never certified from such boxes. Retaining the dependency $x = y$ changes the expression to x^2; using the known equation $x^2 = 2$ resolves it immediately. Without that extra algebraic relation, an interval-valued output can still certify arbitrarily small numerical error.

Appendix C: Venus and astronomical cycles

1. E is Earth's Sun-centered lap time and V is Venus's Sun-centered lap time. The quantity S measures something different: the waiting time between successive Earth–Venus alignments.

2. Venus and Earth move through $1/V$ and $1/E$ laps per day. Their relative phase therefore increases at $1/V - 1/E$ revolutions per day, so one relative revolution takes S days with $1/S = 1/V - 1/E$. If $r = E/V$, then

$$\frac{E}{S} = \frac{E}{V} - 1 = r - 1, \qquad \frac{S}{E} = \frac{1}{r-1}.$$

Substitution gives $T(13/8) = 1/(13/8 - 1) = 8/5$.

3. Repeated addition of $3/5$ modulo 1 gives

$$0,\ \frac{3}{5},\ \frac{6}{5} \equiv \frac{1}{5},\ \frac{4}{5},\ \frac{7}{5} \equiv \frac{2}{5},$$

then returns to 0. Each step skips one neighboring vertex of the regular pentagon, so the chronological edges are the diagonals and form a pentagram.

5. The factorizations are

$$2920 = 2^3 \cdot 5 \cdot 73, \qquad 260 = 2^2 \cdot 5 \cdot 13.$$

Thus the gcd is 20 and the lcm is $2^3 \cdot 5 \cdot 13 \cdot 73 = 37960$. Division by $584, 365, 260$, and 2920 gives the multipliers $65, 104, 146$, and 13.

6. The Fibonacci recurrence gives

$$T\left(\frac{F_{n+1}}{F_n}\right) = \frac{1}{F_{n+1}/F_n - 1} = \frac{F_n}{F_{n+1} - F_n} = \frac{F_n}{F_{n-1}}.$$

The positive solution of $r = 1/(r-1)$ is $r = (1+\sqrt{5})/2 = \varphi$. Its identity $\varphi = 1 + 1/\varphi$ is exactly the infinite continued fraction $[1; \overline{1}]$.

7. The nominal count gains 0.08 day per cycle, hence 0.4 day after 5 cycles, 5.2 days after 65, and 52 days after 650.

Appendix D: finite fields and division rings

1. The product is $13 + 7s = 6$. Since $1 - 5 \cdot 2^2 = -19 = 2$ in $\mathbb{F}_7$, its inverse is $4(1 - 2s) = 4 + 6s$. Multiplication gives $(1 + 2s)(4 + 6s) = 64 + 14s = 1$.

2. The elements are $0, 1, t, t+1$. The polynomial $X^2 + X + 1$ has no root in $\mathbb{F}_2$ and is irreducible. Directly, $t(t+1) = t^2 + t = 1$, so the nonzero inverses are $1, t+1, t$, respectively. Squaring fixes $0, 1$ and exchanges $t, t+1$.

3. Write $u = a + bs$. Then $u^7 = a - bs$, which equals u exactly when $2b = 0$, hence $b = 0$. Applying the map twice gives $u^{49} = u$. The example $s^7 = -s \neq s$ exhibits its nonidentity action.

4. Both 6 and 24 satisfy $X^2 - X - 1 = 0$ modulo 29. We have $6^7 = -1$, $6^{14} = 1$, and $6^2 = 7 \neq 1$, so 6 has order 14. Also $24^7 = 1$ and $24 \neq 1$, so 24 has order 7. Their least common multiple is 14, agreeing with factor removal.

5. Frobenius interchanges the roots, so $\alpha^{p+1} = \beta^{p+1} = -1$. Binet gives $F_{p+1} = 0$ and $F_{p+2} = (-\alpha + \beta)/(\alpha - \beta) = -1$. For $p = 3$ this is precisely the state $(0, 2)$ at index 4; the full return occurs at index 8.

6 (check values). For primes $3, 7, 11, 13, 29$, valid starting multiples are $8, 16, 10, 28, 28$ and the respective least periods are $8, 16, 10, 28, 14$. The exceptional primes $2, 5$ have periods $3, 20$. The modular arithmetic notebook gives implementations of direct pair iteration, fast doubling, and factor removal for comparison.

7. The units modulo 9 are $1, 2, 4, 5, 7, 8$, and the characteristic is 9. In $\mathbb{F}_3(s)$, $s^2 = 2$, the characteristic is 3 and all eight nonzero elements are units: 2 is nonsquare in $\mathbb{F}_3$, so the conjugate inverse formula applies. Both rings have nine elements, but different characteristics and different unit counts.

8. The equations $a + 2b = 0$, $2a + b = 0$ give $-3b = 0$ in $\mathbb{F}_5$, hence $b = a = 0$. For any target (x, y), solving these two equations gives unique coefficients, so the pair spans as well as being independent. In particular $(3, 4) = 4v$, with coordinates $(0, 4)$. The list $u, v, u + v$ spans but has the nontrivial relation $u + v - (u + v) = 0$ among its three listed vectors. Coordinates in that list are not unique: the third vector is also the sum of the first two.

9. The size 18 is impossible because it is not a prime power. Sizes $27 = 3^3$ and $64 = 2^6$ are possible by the existence part of Theorem D.15; their characteristics and dimensions are $(3, 3)$ and $(2, 6)$. The necessary-size theorem alone would not prove existence.

10. The squares in $\mathbb{F}_5$ are $0, 1, 4$. Thus for a nonzero pair (a, b), the scalar $a^2 - 2b^2$ cannot vanish: if $b \neq 0$, that would make 2 a square, and if $b = 0$, it would force $a = 0$. The conjugate formula gives $(1 + s)^{-1} = (1 - s)/(-1) = 4 + s$; indeed $(1 + s)(4 + s) = 1$. Under the alternative relation, $(s - 2)(s + 2) = 0$ with two nonzero formal pairs, so the ring is not a field.

11. Every field element is $a + bs$, so s generates the extension. But $s^2 = -1$ and $s^4 = 1$, with $s^2 \neq 1$, so its order is 4, not 8. For $g = 1 + s$, $g^2 = 2s$, $g^4 = 2 = -1$, and $g^8 = 1$. Its order is therefore 8. A power g^r generates precisely when $\gcd(r, 8) = 1$. The four primitive elements g, g^3, g^5, g^7 are, respectively, $1 + s, 1 + 2s, 2 + 2s, 2 + s$.

12. Each cubic has value 1 at both 0 and 1, so it has no root in $\mathbb{F}_2$. A reducible cubic over a field has a linear factor, so both are irreducible. In L, $t^3 + t^2 + 1 = 0$, whence $t^{-1} = t^2 + t$. Dividing the cubic relation by t^3 gives $(t^{-1})^3 + t^{-1} + 1 = 0$. Thus evaluation $s \mapsto t^{-1}$ defines a homomorphism $K \to L$. It preserves 1, so is injective since K is a field; both fields have eight elements, so it is an isomorphism.

13. The formal derivative of $X^{p^n} - X$ is -1, so a repeated root, which would also be a root of the derivative, is impossible. In contrast, $X^p - 1 = (X - 1)^p$ in characteristic p has only the root 1, with multiplicity p. Splitting a degree-p^n polynomial only counts p^n roots with multiplicity; distinctness is what gives a set of exactly p^n elements in the existence proof.

14. Every automorphism fixes $\mathbb{F}_3$ and sends s to a root of $X^2 - 2$, hence to s or $-s$. Both maps work: they are the identity and Frobenius $a + bs \mapsto a - bs$. They are distinct. The classification identifies an isomorphism class of fields, while the choice of isomorphism may still vary by an automorphism.

15. Since $4096 = 2^{12}$, the divisors $1, 2, 3, 4, 6, 12$ give the subfield sizes $2, 4, 8, 16, 64, 4096$, each occurring exactly once. Both the 16-element and the 64-element subfield contain the unique four-element subfield, by the subfield theorem applied inside each. Their intersection is a field of size 2^d with d dividing both 4 and 6. Thus $d \leq 2$, and the contained four-element field forces $d = 2$. Their intersection is $\mathbb{F}_4$.

16. The counts are $(9 - 1)(9 - 3) = 48$, $48/(3 - 1) = 24$, and $(8 - 1)(8 - 2)(8 - 4) = 168$. For the last count, the spans excluded at the three stages have sizes $1, 2, 4$. Over $\mathbb{F}_2$, the only nonzero determinant value is 1, so every invertible matrix has that determinant.

17. Dividing $X^6 - 1$ by $\Phi_1\Phi_2\Phi_3$ gives $\Phi_6 = X^2 - X + 1$. Here $q = 2, n = 6$. For a noncentral element, the centralizer dimension is a proper divisor $d = 1, 2, 3$ of 6, so its class has size $63, 21, 9$, respectively. All are divisible by $\Phi_6(2) = 3$. But the class equation would give $63 = 1+$ a sum of these multiples of 3, impossible modulo 3.

18. The cyclicity proof uses the polynomial root bound over a commutative field. That bound is not yet available for a division ring; factoring off one root and evaluating at another cannot use commutative polynomial arithmetic. In the quaternions, $i^2 = j^2 = k^2 = -1$, so the degree-two polynomial $X^2 + 1$ already has at least three distinct roots. Thus the degree bound can fail despite every nonzero element being invertible. Wedderburn establishes the commutativity needed for the finite-field argument.

Appendix F

Reference sheets

Definitions, algorithms, and proof moves

Core notation

Notation	Meaning
$a \mid b$	$b = ak$ for some integer k
$\gcd(a, b)$	greatest positive common divisor
$a \equiv b \pmod{n}$	$n \mid (a - b)$
$(\mathbb{Z}/n\mathbb{Z})^\times$	invertible residue classes modulo n
$\mathbb{F}_p$	the field $\mathbb{Z}/p\mathbb{Z}$ for prime p
$\mathbb{F}_q = \mathrm{GF}(q)$	the field of $q = p^n$ elements, unique up to isomorphism
$\varphi(n)$	number of units modulo n
$v_p(m)$	exponent of p in m
$\pi(x)$	number of primes at most x
$\vartheta(x)$	$\sum_{p \le x} \log p$
$\psi(x)$	$\sum_{p^k \le x} \log p$
$f(x) \asymp g(x)$	their ratio stays between positive constants eventually

Algorithms at a glance

Extended Euclid. Maintain $r_i = as_i + bt_i$, starting with $(a, 1, 0)$ and $(b, 0, 1)$. Each division quotient q updates all three columns by subtracting q times the current row from the preceding row. The last nonzero row certifies $\gcd(a, b) = as + bt$. For $au + nv = 1$, the coefficient u is $a^{-1} \pmod{n}$; reduce it and check the product.

Sieve of Eratosthenes. To find the primes through N, repeatedly circle the least uncrossed integer and cross out its larger multiples. Stop after the sieving primes $p \le \sqrt{N}$; a composite at most N must have a prime factor in that range.

Existence versus a witness. The classical dichotomy for $x = (\sqrt{2})^{\sqrt{2}}$ proves that one of $(\sqrt{2}, \sqrt{2})$ and $(x, \sqrt{2})$ has a rational power, but does not say which. The explicit choice $a = \sqrt{2}$, $b = \log_{\sqrt{2}} 3$ gives $a^b = 3$, and unique factorization proves that b is irrational.

Continued fraction. Repeatedly take the integer part, subtract it, and invert. Convergents satisfy

$$p_k = a_k p_{k-1} + p_{k-2}, \qquad q_k = a_k q_{k-1} + q_{k-2}.$$

The recurrences are the columns of the matrix product

$$\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix},$$

whose determinant is $(-1)^{k+1}$. Each factor lies in $\mathrm{GL}_2(\mathbb{Z})$; a product with an even number of factors lies in $\mathrm{SL}_2(\mathbb{Z})$. For $A/B = [a_0; \ldots, a_n]$, the Euclidean remainders obey

$$r_k = (-1)^k (q_k A - p_k B).$$

For irrational x, the zero-indexed convergents form alternating monotone bounds:

$$c_0 < c_2 < c_4 < \cdots < x < \cdots < c_5 < c_3 < c_1.$$

Legendre's test says that a reduced fraction satisfying $|x - p/q| < 1/(2q^2)$ must be a convergent. Lagrange's theorem says that a simple continued fraction is eventually periodic exactly for a quadratic irrational; Galois says it is purely periodic exactly when $x > 1$ and $-1 < x' < 0$.

Fixed points and prefixes. For $\det M \neq 0$, $T_M(t) = (at + b)/(ct + d)$ satisfies $T_M \circ T_N = T_{MN}$. An irrational fixed point of a nonscalar rational M obeys the genuine quadratic $ct^2 + (d - a)t - b = 0$. If $x = T_N(x)$ and $y = T_M(x)$, then

$$y = T_{MNM^{-1}}(y).$$

A finite continued-fraction prefix therefore preserves quadratic irrationality by conjugating the period matrix.

Reduced quadratic states. Write $x_n = (P_n + \sqrt{D})/Q_n$ with $Q_n \mid D - P_n^2$. With $a_n = \lfloor x_n \rfloor$, the exact update is

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = (D - P_{n+1}^2)/Q_n.$$

Every quadratic irrational eventually has a reduced tail. For reduced states, $|P_n| < \sqrt{D}$ and $0 < Q_n < 2\sqrt{D}$, so only finitely many states occur. A reduced successor z determines its reduced predecessor: $a = \lfloor -1/z' \rfloor$ and $u = a + 1/z$. Finiteness gives eventual repetition; unique predecessors make a reduced starting state purely periodic.

Typical information rate. For almost every x, the Khinchin–Lévy law gives

$$\lim_{n\to\infty} q_n^{1/n} = e^{\pi^2/(12 \log 2)} \approx 3.275823, \qquad \lim_{n\to\infty} |x - p_n/q_n|^{-1/n} = e^{\pi^2/(6 \log 2)} \approx 10.731.$$

So a continued-fraction term earns about 1.0306 decimal digits on average. For the first 100,000 partial quotients of π, computation gives $G_n(\pi) \approx 2.67956$ and $q_n^{2/n} \approx 10.67835$; neither limiting law has been proved for π. In contrast, $e = [2; 1, 2, 1, 1, 4, 1, 1, 6, \ldots]$ has $G_{3m}(e) \sim (2m/e)^{1/3} \to \infty$ and $q_{3m}^{2/(3m)} \to \infty$.

Gosper's streaming arithmetic. Represent a unary rational transformation by $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. Reading an input digit m and emitting a certified output digit k update the state by

$$M \longmapsto M \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}, \qquad M \longmapsto \begin{pmatrix} 0 & 1 \\ 1 & -k \end{pmatrix} M.$$

For two inputs use $(axy + bx + cy + d)/(exy + fx + gy + h)$; its eight integer coefficients are closed under the same absorb-or-emit operations. Exact interval bounds certify a digit when every possible value has the same floor. Interval-valued output is the safe fallback at integer boundaries where a digit-only process can stall.

Calendar design. If a tropical year is $365 + \alpha$ days, a cycle with p leap days in q years asks for $p/q \approx \alpha$. For $\alpha \approx 0.2421988$, the convergents begin

$$\frac{1}{4}, \quad \frac{7}{29}, \quad \frac{8}{33}, \quad \frac{31}{128}.$$

The Julian calendar uses $1/4$. Requiring $q = 100Q$ means approximating 100α; its convergent $97/4$ yields the Gregorian rate $97/400$.

Planetary lapping time. If an inner planet takes V days for one Sun-centered lap and Earth takes $E > V$ days, the waiting time S between successive alignments (the *synodic period*) satisfies

$$\frac{1}{S} = \frac{1}{V} - \frac{1}{E}, \qquad \frac{S}{E} = T(E/V), \qquad T(r) = \frac{1}{r-1}.$$

For Earth and Venus, $E/V \approx 13/8$ and $S/E \approx 8/5$; the five successive conjunction directions advance by about $3/5$ of a turn and draw a near-pentagram. The Dresden Codex uses the exact nominal calendar identities

$$5 \cdot 584 = 8 \cdot 365 = 2920, \qquad 65 \cdot 584 = 104 \cdot 365 = 146 \cdot 260 = 37960.$$

Square-root periods. For a nonsquare integer $D \geq 2$,

$$\sqrt{D} = [a_0; \overline{a_1, \ldots, a_{L-1}, 2a_0}], \qquad a_j = a_{L-j} \quad (1 \leq j < L),$$

where L is the least period length. The block before $2a_0$ is a palindrome. The complete quotients satisfy $Q_n = 1$ exactly when $L \mid n$, for $n \geq 1$.

Pell bridge. If $p^2 - Dq^2 = \pm 1$ with $p, q > 0$, then

$$\left|\sqrt{D} - p/q\right| = \frac{1}{q^2(\sqrt{D} + p/q)} < \frac{1}{2q^2},$$

so Legendre makes p/q a convergent. The exact identity

$$p_{n-1}^2 - Dq_{n-1}^2 = (-1)^n Q_n$$

shows that the Pell convergents have indices $mL - 1$ and sign $(-1)^{mL}$. The first $+1$ solution is at $L - 1$ for even L and $2L - 1$ for odd L; the -1 equation is solvable exactly when L is odd. Stop before the final $2a_0$. If (u, v) is the first $+1$ solution, every positive $+1$ solution is a power of $u + v\sqrt{D}$, or equivalently is generated by $\left(\begin{smallmatrix} u & Dv \\ v & u \end{smallmatrix}\right)$.

Cattle reduction. Archimedes' seven proportions give the total $50\,389\,082t$. The square condition forces $t = 4\,456\,749r^2$; the triangular condition is equivalent to

$$X^2 - 4\,729\,494Z^2 = 1, \qquad Z = 9314r.$$

The divisibility condition $9314 \mid Z$ must be retained. The first admissible power of the fundamental unit is 2329; the resulting smallest total has 206 545 digits. See Section 3.11 and Vardi's exposition.

Base-b expansion. Long division evolves by $br_n = qd_{n+1} + r_{n+1}$ with $0 \le r_{n+1} < q$. Repeated remainders make every rational expansion eventually periodic. A reduced fraction m/q is purely periodic exactly when $\gcd(q, b) = 1$; the reverse direction follows from Euler's theorem.

Linear congruence. Compute $au + nv = d = \gcd(a, n)$. There are solutions to $ax \equiv b \pmod{n}$ iff $d \mid b$. When it divides, $x_0 = u(b/d)$ is a solution, and all solutions are $x \equiv x_0 \pmod{n/d}$: exactly d residues modulo n.

CRT. For pairwise coprime moduli, put $N = \prod n_i$ and $N_i = N/n_i$. Extended Euclid gives $N_i M_i + n_i v_i = 1$. The selectors $e_i = N_i M_i$ are 1 at modulus n_i and 0 at the others; $x = \sum a_i e_i \pmod{N}$ reconstructs the prescribed remainders. For two arbitrary moduli m, n, compatibility means the prescribed remainders agree modulo $\gcd(m, n)$; uniqueness is modulo their lcm.

Totients and Euler. Counting multiples of p gives $\varphi(p^k) = p^{k-1}(p-1)$. CRT identifies units modulo coprime factors with pairs of units, so

$$\varphi(mn) = \varphi(m)\varphi(n) \quad (\gcd(m, n) = 1), \qquad \varphi(n) = n \prod_{p \mid n} (1 - 1/p).$$

For necklaces of length p^k, remove the $a^{p^{k-1}}$ words obtained by repeating a shorter block. Full rotation orbits give $p^k \mid a^{p^k} - a^{p^{k-1}}$. If $p \nmid a$, cancellation gives $a^{\varphi(p^k)} \equiv 1 \pmod{p^k}$. Raise to the common exponent $\varphi(n)$ and use CRT: $a^{\varphi(n)} \equiv 1 \pmod{n}$ when $n \ge 2$ and $\gcd(a, n) = 1$.

Multiplicative functions. Require $f(1) = 1$ and $f(mn) = f(m)f(n)$ for coprime inputs. Then $f(n) = \prod_i f(p_i^{k_i})$ for $n = \prod_i p_i^{k_i}$. Complete multiplicativity removes coprimality. The convolution $(f * g)(n) = \sum_{d \mid n} f(d)g(n/d)$ preserves multiplicativity and has identity $\varepsilon(1) = 1$, $\varepsilon(n) = 0$ for $n > 1$.

Möbius inversion. The function μ is $(-1)^r$ on products of r distinct primes (including $\mu(1) = 1$), and zero when a prime square divides the input. Since $\mu * \mathbf{1} = \varepsilon$,

$$F(n) = \sum_{d \mid n} f(d) \quad \iff \quad f(n) = \sum_{d \mid n} \mu(d) F(n/d).$$

No multiplicativity hypothesis is needed. In particular, $\sum_{d \mid n} \varphi(d) = n$ gives $\varphi(n) = \sum_{d \mid n} \mu(d) n/d$. Primitive length-$n$ words with a colors number $P_a(n) = \sum_{d \mid n} \mu(d) a^{n/d}$; divide by n for primitive necklaces. All necklaces number $n^{-1} \sum_{d \mid n} \varphi(d) a^{n/d}$.

Poset inversion. Define $\mu_P(x, x) = 1$ and $\mu_P(x, y) = -\sum_{x \le z < y} \mu_P(x, z)$. For finite sums,

$$G(y) = \sum_{x \le y} F(x) \quad \iff \quad F(y) = \sum_{x \le y} \mu_P(x, y) G(x).$$

For upper sums, $F(x) = \sum_{y \ge x} \mu_P(x, y) G(y)$. On divisibility intervals $\mu_P(r, s) = \mu(s/r)$; on subsets $\mu_P(S, T) = (-1)^{|T \setminus S|}$. Products of posets multiply their coefficients. In the incidence algebra, $\mu_P = \zeta^{-1}$.

Lagrange interpolation. For distinct points c_i in a field, values y_i determine the unique polynomial of degree less than r:

$$f(X) = \sum_{i=1}^{r} y_i \ell_i(X), \qquad \ell_i(X) = \prod_{j \ne i} \frac{X - c_j}{c_i - c_j}.$$

These are the CRT selectors for the polynomial moduli $X - c_i$. All polynomials with the prescribed values differ by a multiple of $\prod_i (X - c_i)$. Over $\mathbb{F}_p$, use extended gcd to invert denominators.

Fourier inversion is interpolation at roots of unity. For a primitive nth root ω in a field where n is invertible,

$$y_j = \sum_{k=0}^{n-1} c_k \omega^{jk}, \qquad \ell_j(X) = \frac{1}{n} \sum_{k=0}^{n-1} \omega^{-jk} X^k, \qquad c_k = \frac{1}{n} \sum_{j=0}^{n-1} y_j \omega^{-jk}.$$

The selectors satisfy $\ell_j(\omega^r) = \delta_{jr}$, reconstructing the unique polynomial of degree below n. Polynomial CRT modulo $X^n - 1$ turns cyclic convolution into pointwise multiplication of samples.

Fibonacci periodicity now. Track $(F_n, F_{n+1}) \pmod m$. The update $(a, b) \mapsto (b, a + b)$ has inverse $(u, v) \mapsto (v - u, u)$, so the orbit is periodic from its start. Its least positive return to $(0, 1)$ is the period $\pi(m) \le m^2 - 1$. For coprime m, n, $\pi(mn) = \operatorname{lcm}(\pi(m), \pi(n))$. Here $\pi(m)$ denotes a Fibonacci period, distinct from the prime-counting function $\pi(x)$ used in Chapter 8.

Finite fields. Every finite field has prime characteristic p and is a vector space over its prime field. Exchange makes dimension well defined, and d coordinates over $\mathbb{F}_q$ give q^d vectors. For every $q = p^n$ there is exactly one field up to isomorphism: the roots of $X^q - X$ in a splitting extension form such a field, and a minimal polynomial of a multiplicative generator gives an isomorphism between any two fields of size q. Its multiplicative group is cyclic of order $q - 1$. Uniqueness does not specify a unique isomorphism. The ring $\mathbb{Z}/p^n\mathbb{Z}$ is not $\mathbb{F}_{p^n}$ for $n > 1$.

Automorphisms, subfields, and matrices. Frobenius $a \mapsto a^p$ generates the cyclic automorphism group of $\mathbb{F}_{p^n}$, of order n. For each $m \mid n$, its unique subfield of size p^m is the root set of $X^{p^m} - X$; there are no others. Choosing independent columns gives

$$|\operatorname{GL}_d(\mathbb{F}_q)| = \prod_{r=0}^{d-1} (q^d - q^r), \qquad |\operatorname{SL}_2(\mathbb{F}_q)| = q(q^2 - 1).$$

Wedderburn's bonus theorem. Every finite division ring is a field. For a division ring with center of size q and dimension n over its center, the multiplicative class equation is

$$q^n - 1 = q - 1 + \sum_i \frac{q^n - 1}{q^{d_i} - 1}, \qquad d_i \mid n,\ d_i < n.$$

If $n > 1$, the integer $\Phi_n(q)$ divides the left side and every summand, so it divides $q - 1$; its absolute value is greater than $q - 1$, a contradiction. Thus the whole division ring equals its center.

Frobenius and period computation. On $\mathbb{F}_p$, Fermat makes $u \mapsto u^p$ the identity. The binomial coefficient proof of $(X + Y)^p = X^p + Y^p$ makes it an automorphism of every finite extension. In $\mathbb{F}_p(s)$ with $s^2 = d$ nonsquare and p odd, it sends $a + bs$ to $a - bs$. For Fibonacci and primes $p \neq 2, 5$, the roots of $X^2 - X - 1$ have simultaneous order $\pi(p)$. If 5 is a square modulo p, $\pi(p) \mid p - 1$; otherwise $\pi(p) \mid 2(p + 1)$ and $\pi(p) \nmid p + 1$. These are multiples, not always the least period. Remove factors while testing returns by fast doubling. The exceptional periods are $\pi(2) = 3$ and $\pi(5) = 20$.

Proof moves worth naming

- **Minimal counterexample / strong induction**: factor a composite object into smaller ones.
- **Bézout insertion**: replace 1 by $ax + by$ to transfer divisibility.

- **Exponent comparison**: use unique factorization or v_p to compare products and quotients.
- **Möbius monotonicity**: compute the derivative of $(at + b)/(ct + d)$ to determine how a continued-fraction tail moves its value.
- **Conjugation**: transport a fixed-point equation through a change of variable using MNM^{-1}.
- **Finite states with reversible steps**: bound integer states to force eventual repetition, then recover predecessors to remove any initial segment.
- **Equal-block partition**: cosets and rotation orbits turn a count into divisibility.
- **Invert a cumulative count**: choose Möbius coefficients, Lagrange selectors, or Fourier weights so that unwanted contributions cancel.
- **Count two ways**: binomial identities become transparent when both sides count the same set.
- **Logarithmic weighting**: products of primes become the sums ϑ and ψ.
- **Dyadic decomposition**: split a range into $(x/2, x], (x/4, x/2], \ldots$ and sum a geometric series.

Fermat and Euler: two counting routes

	Necklace proof	Group proof
Objects	a^p labeled colorings	nonzero residues modulo p
Structure	cyclic rotation orbits	group $(\mathbb{Z}/p\mathbb{Z})^\times$
Exceptional part	a constant colorings	identity/powers inside cyclic subgroups
Equal blocks	nonconstant orbits have size p	element orders divide $p - 1$
Conclusion	$a^p \equiv a \pmod{p}$	$a^{p-1} \equiv 1 \pmod{p}$ for $p \nmid a$
Euler	subtract all shorter periods at p^k, cancel, then use CRT	apply Lagrange to the $\varphi(n)$ units modulo n

Further reading

- G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*.
- I. Niven, H. Zuckerman, and H. Montgomery, *An Introduction to the Theory of Numbers*.
- J. Stillwell, *Elements of Number Theory*.
- R. P. Stanley, *Enumerative Combinatorics*, volume 1, second edition, Chapter 3, especially Sections 3.6–3.8 (incidence algebras, Möbius inversion, and products of posets).
- D. J. Kleitman, *The Finite Fourier Transform and the Fast Fourier Transform Algorithm*, MIT 18.310 lecture notes (polynomial evaluation at roots of unity).
- K. Conrad, *The Chinese Remainder Theorem* (including polynomial interpolation), and *Finite Fields* (constructions and Frobenius).
- J. S. Milne, *Fields and Galois Theory*, Section 4, "Finite fields" (existence, uniqueness, automorphisms, and subfields).
- E. Witt, "Über die Kommutativität endlicher Schiefkörper", *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* **8** (1931), 413 (the cyclotomic proof of Wedderburn's theorem).
- A. Weil, *Basic Number Theory*, Chapter I, §1 (finite fields and finite division rings).
- R. M. Corless, *Continued Fractions and Chaos*, section on periodic continued fractions (the distinction between Lagrange's theorem and Galois's criterion).
- G. Panti, *A General Lagrange Theorem* (2007; a broader perspective beyond simple continued fractions).
- I. Vardi, "Archimedes' Cattle Problem," *American Mathematical Monthly* **105**, no. 4 (1998), 305–319 (Pell equations, divisibility conditions, and a compact description of the enormous least solution; preprint).
- A. Ya. Khinchin, *Continued Fractions* (for the metric theory behind the "base 11" aside).
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- G. Lochs, "Vergleich der Genauigkeit von Dezimalbruch und Kettenbruch," *Abh. Math. Sem. Univ. Hamburg* **27** (1964), 142–144.
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- F. Eisenbrand, "Pope Gregory, the calendar, and continued fractions," in *Discrete Mathematics*, DOI: 10.4171/DMS/6/14.
- R. W. Gosper, "Continued Fraction Arithmetic," Item 101B in M. Beeler, R. W. Gosper, and

R. Schroeppel, *HAKMEM*, MIT Artificial Intelligence Memo 239 (1972), 39–44.

- M. J. Collins, "Arithmetic on Continued Fractions," *Qeios* **7** (2025), DOI: 10.32388/DD1PU3.2.
- NASA Goddard Space Flight Center, "2012 Transit of Venus" (orbital and synodic periods of Venus).
- European Southern Observatory, "The Pentagram of Venus".
- Saxon State and University Library Dresden, "The Dresden Maya Codex: Content".
- A. Kaneko, "Identification of Postclassic Maya Constellations from the Venus Pages of the Dresden Codex," *Estudios de Cultura Maya* **35** (2010), 33–74.
- J.-P. Serre, *A Course in Arithmetic* (for a more advanced next step).

Course packet designed as an adaptable teaching draft. Classical results and examples are attributed where appropriate; exposition, exercises, and diagrams prepared for this packet.